PEARSON EDEXCEL INTERNATIONAL A LEVEL BIOLOGY · WBI14

Unit 4: Energy, Environment, Microbiology and Immunity

Revision notes, spec point by spec point (Topics 5 and 6)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

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Topic 5 – Energy Flow, Ecosystems and the Environment

SPEC 5.1 Photosynthesis: the overall reaction

WHAT THIS SPEC POINT ASKS

Understand photosynthesis as using light energy to split the strong bonds in water, storing the hydrogen in a fuel (glucose) by combining it with carbon dioxide, and releasing oxygen.

5.1 The big picture

- Plants, algae and some bacteria are **autotrophs**: they make their own organic molecules from simple inorganic ones (CO_2 and water), using **light energy**. Animals and fungi are **heterotrophs**: they get organic molecules by eating or absorbing them.
- Photosynthesis turns **light energy** into **chemical energy** stored in the bonds of glucose. Respiration later releases that energy.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2 \text{ (light energy, chlorophyll)}$$

5.1 What happens to each reactant

- **Water**: the O–H bonds in water are strong. Light energy is needed to break them (**photolysis**). This gives hydrogen (as H^+ ions and electrons) and **oxygen**, which is released as a **waste product**.
- **Hydrogen** is the part worth keeping. It is passed on (via reduced NADP) and combined with CO_2 to make glucose. Glucose is the "fuel" that stores the hydrogen.
- **Carbon dioxide** supplies the **carbon** and the **oxygen** in glucose.

Atoms in glucose	Come from
Carbon	carbon dioxide
Oxygen	carbon dioxide
Hydrogen	water
The O_2 released	water

5.1 Two stages (details in 5.3 and 5.4)

Stage	Where	Needs	Makes
Light-dependent reactions	thylakoid membranes	light, water, ADP + Pi, NADP	ATP, reduced NADP, O_2 (waste)
Light-independent reactions (Calvin cycle)	stroma	CO_2, ATP, reduced NADP	GALP → glucose and other organic molecules; ADP, Pi and NADP go back to the thylakoids

KEY IDEA

Photosynthesis is a **redox** process. Water is **oxidised** (loses electrons and hydrogen). Carbon dioxide is **reduced** (gains hydrogen). Light supplies the energy to push the hydrogen "uphill" into glucose.

WORKED EXAMPLE: ENERGY FROM BOND ENERGIES

A question gives the O–H bond energy as 463 kJ mol^{-1} and says one O–H bond breaks in each water molecule in the equation $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$.
Energy needed = number of molecules × bonds broken per molecule × bond energy = 6 × 1 × 463 = **2778 kJ** per mole of glucose.
The skill is reading the stem: which bonds, how many per molecule, how many molecules. Bonds in glucose itself are C–H, C–O, C–C and O–H.

MARK-SCHEME LANGUAGE

- Sources of atoms: "C and O from carbon dioxide", "H from water". **Not** "O (in glucose) from water".
- Name the **molecules** (carbon dioxide, water), not just the elements.
- Why not all the sunlight is used: "not all wavelengths can be absorbed", "**green** light is reflected", "light is transmitted through the leaf". Shading and competition are ignored.
- Glucose is made in the **stroma**.

Often asked as: where the C, H and O in glucose come from; which stage uses CO_2 and which releases O_2 (MCQ); bond-energy calculations; why plants capture only a small share of the light energy.

COMMON MISTAKE

Saying the oxygen in glucose comes from water. The O_2 **released** comes from water; the O **in glucose** comes from CO_2.
"Light reflected" on its own: say **green** light is reflected.

BEYOND THE SPEC, BUT EXAMINED

- Bond-energy sums: pick the right bonds and multiply by the number of molecules. Candidates often missed which bond types (C–H, C–O, C–C) were needed to reach a total (WBI14 Oct 2020).
- Losses of light energy before photosynthesis even starts: reflected (green), transmitted, wrong wavelength, hits non-photosynthetic parts (WBI14 Oct 2024).

SPEC 5.2 ATP: photophosphorylation and hydrolysis

WHAT THIS SPEC POINT ASKS

Understand that making ATP from ADP (photophosphorylation, in photosynthesis) needs energy, and that hydrolysing ATP gives an immediate supply of energy for biological processes.

5.2 What ATP is

- **ATP (adenosine triphosphate)** is a phosphorylated nucleotide: the base **adenine**, the pentose sugar **ribose**, and **three phosphate groups**.
- Removing one phosphate gives **ADP** (adenosine diphosphate) and **inorganic phosphate (Pi)**.

5.2 Making ATP needs energy

$$ADP + Pi + energy \rightarrow ATP + H_2O$$

- Adding a phosphate to ADP is **phosphorylation**, a condensation reaction catalysed by **ATP synthase**.
- The energy can come from:
  - light, in the chloroplast: **photophosphorylation** (5.3);

  - the oxidation of respiratory substrates, in the mitochondrion: oxidative phosphorylation (Topic 7).
- In both, the energy first builds a **proton (H^+) gradient** across a membrane. H^+ flow back through ATP synthase, and that flow drives phosphorylation (**chemiosmosis**).

5.2 Hydrolysing ATP releases energy

$$ATP + H_2O \rightarrow ADP + Pi + energy$$

- Catalysed by **ATP hydrolase** (ATPase).
- Releases a **small, manageable** amount of energy (about 30 kJ mol^{-1}), so little is wasted as heat.
- It is **one reaction, one enzyme**, so energy is released **immediately**, exactly where it is needed. (Releasing energy from glucose takes dozens of steps.)
- ATP is **small and soluble**, so it moves easily around the cell.
- The released phosphate can be attached to another molecule (phosphorylating it), making it more reactive. This **couples** ATP hydrolysis to energy-requiring reactions.
- ATP is **not a store**: it is used within seconds and constantly remade from ADP + Pi. Glucose, starch and lipids are the stores.

Uses in plants: the **Calvin cycle** (reducing GP and regenerating RuBP), active transport of mineral ions into roots, protein synthesis, phloem loading.

KEY IDEA

Plants trap light energy, but light energy cannot drive reactions directly. It is first transferred to **ATP** (and reduced NADP), which then supply the energy for the Calvin cycle.

MARK-SCHEME LANGUAGE

- "ATP is the {usable source of energy / energy currency}"; "light energy cannot be used (directly)"; "ATP is needed in the Calvin cycle to convert GP into GALP".
- Products of hydrolysis: "ADP and (inorganic) phosphate / Pi". **Not** "P" or "phosphorus". "Dephosphorylation" is accepted.
- Phosphorylation: "hydrogen ions pass through ATP synthase, releasing energy for phosphorylation of ADP".
- MCQ logic: ADP is **phosphorylated** and this **needs** energy; ATP is **hydrolysed** and this **releases** energy.

Often asked as: why plants need ATP even though they trap light; products of ATP hydrolysis; fill-in blanks on a photophosphorylation diagram (ADP, Pi, ATP); MCQs.

COMMON MISTAKE

"ATP makes energy" or "ATP is energy". Energy can't be made. ATP is a **source** or **carrier** of energy.
Writing "P" for phosphate. If unsure of the symbol, write the word **phosphate**.
Answering "how light is converted into ATP" when the question asks **why** plants need ATP.

BEYOND THE SPEC, BUT EXAMINED

- **ATP synthase** and **chemiosmosis** appear in mark schemes for the light-dependent reactions (WBI14 Jan 2021).
- Glucose made in photosynthesis is respired to give the ATP needed for protein synthesis (WBI14 Jan 2020).

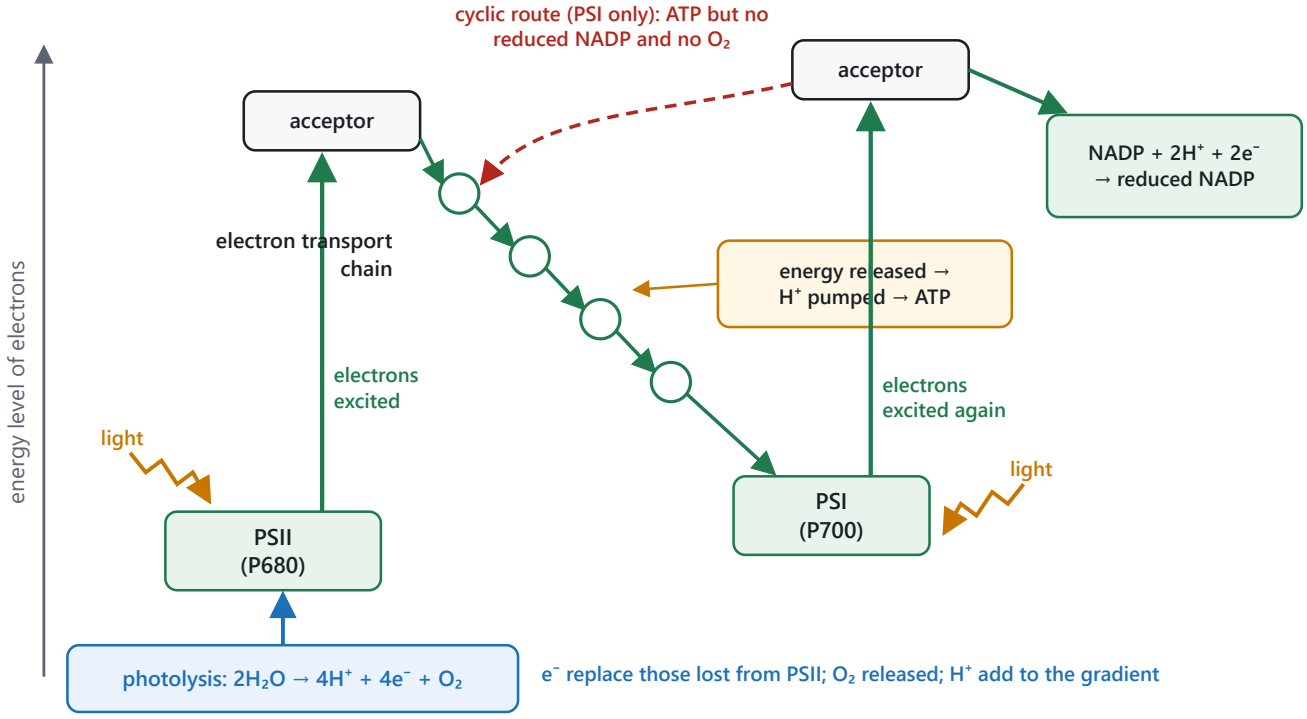
SPEC 5.3 The light-dependent reactions

WHAT THIS SPEC POINT ASKS

Understand how light energy excites electrons in chlorophyll, how those electrons drive ATP production and NADP reduction in cyclic and non-cyclic photophosphorylation, and how photolysis of water produces oxygen.

5.3 Where and what

- On the **thylakoid membranes** of the chloroplast.
- Chlorophyll and accessory pigments sit in protein complexes called **photosystems**. Each has a **reaction centre** chlorophyll a molecule. **PSII** absorbs best at 680 nm (P680); **PSI** at 700 nm (P700). (They are numbered in order of discovery, not the order electrons pass through them.)
- Products: **ATP**, **reduced NADP**, and **oxygen** (waste).



Non-cyclic route (solid green): water → PSII → electron transport chain → PSI → NADP. Cyclic route (dashed red): electrons from PSI go back down the chain to PSI, making ATP only.

5.3 Non-cyclic photophosphorylation, step by step

1. Light is **absorbed** by chlorophyll in **PSII**. Its energy **excites electrons** to a higher energy level, and they leave the chlorophyll (the chlorophyll is oxidised).
2. The electrons are taken up by an **electron acceptor** and passed along a chain of **electron carriers** (the **electron transport chain**) in a series of redox reactions.
3. As they move down the chain, the electrons **lose energy**. This energy pumps **H^+ from the stroma into the thylakoid space**.
4. H^+ build up in the thylakoid space (a **proton gradient**). They diffuse back into the stroma through **ATP synthase**, which uses the energy to **phosphorylate ADP → ATP** (**chemiosmosis**).
5. The electrons reach **PSI**. Light absorbed by PSI **excites them again**.
6. The electrons pass to **NADP**. Together with **H^+** it becomes **reduced NADP**.
7. **Photolysis**: PSII has lost electrons, so it must get them back. Light energy splits water:

$$2H_2O \rightarrow 4H^+ + 4e^- + O_2$$

- The **electrons** replace those lost from PSII.
- The **H^+** add to the proton gradient and are used to reduce NADP.
- The **oxygen** diffuses out of the leaf as a by-product (or is used in respiration).

The electron path is one-way, from water to NADP, which is why it is called **non-cyclic**.

5.3 Cyclic photophosphorylation

- Uses **PSI only**. Excited electrons from PSI pass to the electron transport chain and then **return to PSI**.
- Their energy still pumps H^+, so **ATP is made**.
- **No** reduced NADP, **no** photolysis, **no** oxygen.

	Non-cyclic	Cyclic
Photosystems	PSII and PSI	PSI only
Electron source	water (photolysis)	PSI (electrons go round)
Final electron acceptor	NADP	PSI
ATP	yes	yes
Reduced NADP	yes	no
O_2 released	yes	no

EXAM TIP

For "describe how light energy is used to make ATP", cover **both** routes: ATP is made in cyclic **and** non-cyclic photophosphorylation. Very few candidates say this, and the 4-mark version needs it.

MARK-SCHEME LANGUAGE

- "Light is **absorbed** by {chlorophyll / photosystems}, which **excites electrons**" (or "releases high-energy electrons").
- "Electrons are passed along a series of electron carriers, releasing energy to phosphorylate ADP into ATP."
- Photolysis "{replaces electrons lost by chlorophyll / provides protons for formation of reduced NADP}".
- Purpose of the light-dependent reactions: "ATP and reduced NADP for the Calvin cycle".
- The product made only in non-cyclic is **reduced NADP**. NADPH is accepted; NADP, NADH or reduced NAD are not.
- Location: thylakoid **membranes**; H^+ accumulate in the **thylakoid space**.
- Magnesium: "a component of chlorophyll, which absorbs light".

Often asked as: describe how light leads to ATP (4 marks); role of photolysis; role of chlorophyll; which product is only in non-cyclic; location MCQs; why a shortage of water or Mg^{2+} slows photosynthesis.

COMMON MISTAKE

Light "hits" or "strikes" chlorophyll. The word that scores is **absorbed**. Examiners have flagged this in report after report.
Saying NADP is "made" in the light-dependent reactions. It is **reduced**.
Mixing up cyclic and non-cyclic: oxygen and reduced NADP come only from **non-cyclic**.
Saying the light-dependent reactions happen in the stroma, or "in the grana" without the membrane.

BEYOND THE SPEC, BUT EXAMINED

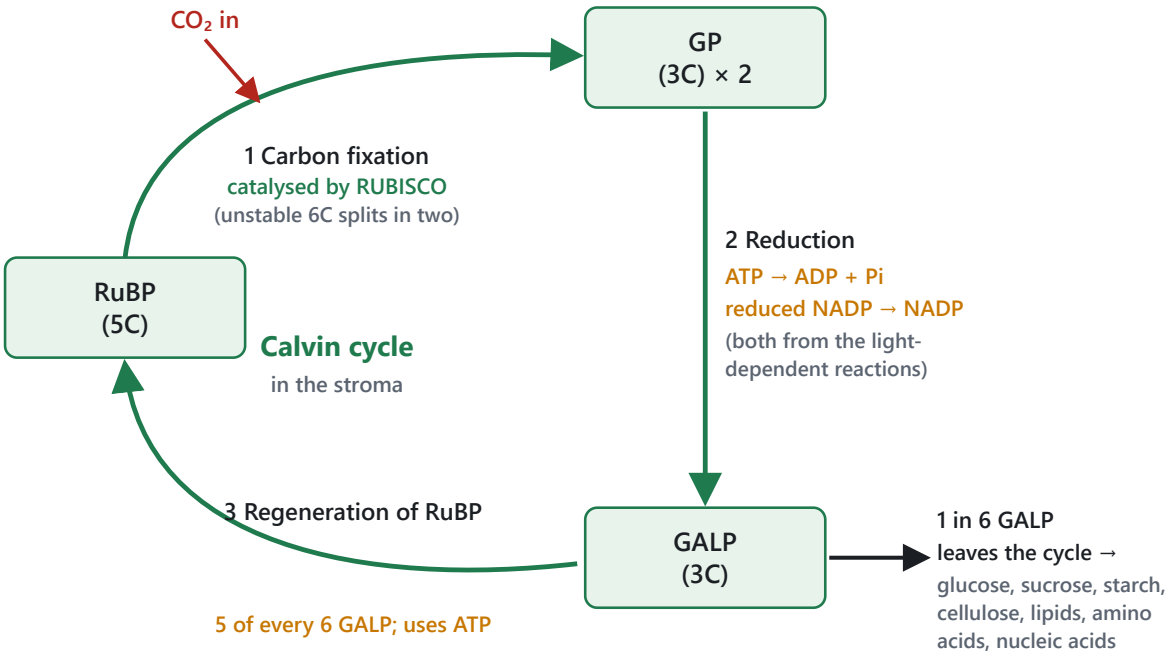
- **Mg^{2+}** is part of the chlorophyll molecule; Mg^{2+} uptake by roots needs active transport, so ATP and oxygen for aerobic respiration (WBI14 Jun 2023).
- **Cyanobacteria** have no chloroplasts: their light-dependent reactions happen on internal membranes (WBI14 Jan 2023).
- Less water reaching the leaf means less photolysis, so fewer electrons and H^+ for the light-dependent reactions (WBI14 Jun 2024).
- Why cyclic exists: the Calvin cycle uses 3 ATP for every 2 reduced NADP. Non-cyclic alone gives too little ATP, so cyclic tops it up.

SPEC 5.4 The light-independent reactions (Calvin cycle)

WHAT THIS SPEC POINT ASKS

(i) Understand the light-independent reactions as the reduction of CO_2 using ATP and reduced NADP: carbon fixation in the Calvin cycle and the roles of GP, GALP, RuBP and RUBISCO. (ii) Know that the simple sugars made are used for respiration and to build polysaccharides, amino acids, proteins, lipids and nucleic acids.

5.4 The cycle in three steps (in the stroma)



Carbon fixation, reduction and regeneration. One GALP in every six leaves the cycle to make useful molecules.

1. **Carbon fixation.** CO_2 diffuses into the stroma and combines with **RuBP** (ribulose bisphosphate, a **5-carbon** compound). The enzyme is **RUBISCO**. The 6-carbon product is unstable and splits at once into **two molecules of GP** (glycerate 3-phosphate, **3-carbon**).
2. **Reduction.** GP is reduced to **GALP** (glyceraldehyde 3-phosphate, 3-carbon; some books call it triose phosphate, TP, or G3P).
   - **Reduced NADP** supplies the hydrogen (it is oxidised back to NADP).
   - **ATP** supplies the energy (hydrolysed to ADP + Pi).
   - The NADP, ADP and Pi return to the thylakoids to be used again.
3. **Regeneration of RuBP.** Most GALP (**5 in every 6**) is rearranged back into RuBP. This uses **more ATP**. Without regeneration the cell would run out of RuBP and **carbon fixation would stop**.

5.4 Counting carbons

WORKED EXAMPLE: SIX TURNS OF THE CYCLE

6 CO_2 + 6 RuBP (6 × 5C = 30C) → 12 GP (36C)
12 GP → 12 GALP, using **12 ATP** and **12 reduced NADP**
10 GALP (30C) → 6 RuBP (30C), using **6 ATP**
2 GALP (6C) left over → **1 glucose** (6C)
Totals per glucose: **18 ATP** and **12 reduced NADP**, so ATP : reduced NADP = 18 : 12 = **1.5 : 1**.
Per CO_2 fixed: 3 ATP and 2 reduced NADP.

5.4 What happens if light or CO_2 changes

Change	GP	RuBP	Why
Light reduced (or switched off)	rises	falls	less ATP and reduced NADP, so GP isn't reduced to GALP and builds up; RuBP is still used in fixation but is not regenerated
CO_2 reduced	falls	rises	less fixation, so less GP made; RuBP is still regenerated from GALP but not used up
Temperature lowered	both processes slow		RUBISCO and other Calvin cycle enzymes have less kinetic energy, so fewer enzyme–substrate complexes form

The Calvin cycle does not need light directly, but it **stops within seconds in the dark** because it runs out of ATP and reduced NADP. Don't call it the "dark reaction".

5.4 What the products are used for

Product	How it is made	Extra element needed
Glucose (hexose)	two GALP join	none
Sucrose (transport in phloem)	glucose + fructose	none
Starch (storage), **cellulose** (cell walls)	polymers of glucose	none
Lipids	glycerol from GALP; fatty acids from GALP via acetyl coenzyme A	none
Amino acids → proteins	from sugars + nitrogen; joined by **peptide bonds** in condensation reactions at ribosomes	**nitrogen** (from nitrate); some need **sulfur** (from sulfate)
Nucleotides → nucleic acids (and ATP)	sugar + base + phosphate	**nitrogen** (bases) and **phosphorus** (from phosphate)

- Some glucose is **respired** to make the ATP these syntheses need.
- Animals, fungi and decomposers get these molecules by eating plants (or eating what ate them), so the sugars made here fuel almost every food chain.

EXAM TIP

"How does GALP become protein / root biomass?" wants a short chain, not the whole Calvin cycle: GALP → glucose → (with fructose) **sucrose** → **transported in phloem** → converted to amino acids **using nitrates** → joined by peptide bonds → protein; some glucose respired for ATP.

MARK-SCHEME LANGUAGE

- Fixation: "carbon dioxide {binds to / reacts with} RuBP", "catalysed by RUBISCO", "forming GP".
- Regeneration: "(otherwise) the cell will run out of RuBP", so "no carbon fixation".
- "GALP converted into {hexose / glucose}"; "amino acids made from glucose and nitrates"; "amino acids joined by peptide bonds / condensation"; "glucose respired to produce ATP".
- Why glucose alone can't make amino acids: "amino acids contain **nitrogen**", "some contain **sulfur**", "nitrogen from nitrates, sulfur from sulfates". "Amino acids have an amino group" alone is too vague.
- Tick grids: protein needs nitrate; nucleic acid needs nitrate **and** phosphate; triglyceride needs neither.
- Transport: "glucose and fructose used to produce sucrose", "sucrose / amino acids moved in the phloem". **Not** "glucose transported".
- GP and GALP have accepted alternative names (triose phosphate, G3P). Carbon numbers are not required but help.

Often asked as: describe carbon fixation; why most GALP stays in the cycle; ATP : reduced NADP ratio; how GALP becomes protein, root biomass or another product; why a plant needs nitrate or sulfate; temperature and RUBISCO.

COMMON MISTAKE

Writing out the whole Calvin cycle when only "GALP plus nitrate → amino acids" was wanted.
Forgetting **nitrate** when explaining how plants make proteins.
Saying glucose is transported in the phloem. It is **sucrose**.
Saying the Calvin cycle "happens at night".

BEYOND THE SPEC, BUT EXAMINED

- Calculating the ATP : reduced NADP ratio for a number of turns (WBI14 Jan 2025).
- Phosphate is needed for ATP, nucleotides and phospholipids; nitrate for amino acids and the bases in nucleotides (WBI14 Jun 2023).
- Mineral ions such as phosphate are taken up by **active transport** and travel in the **xylem** (WBI14 Jan 2023).
- Mutualistic fungi (mycorrhizae) supply plants with nitrogen compounds (WBI14 Jun 2024).

SPEC 5.5

Chloroplast structure

WHAT THIS SPEC POINT ASKS

Understand how the structure of a chloroplast suits it to photosynthesis.

5.5 The parts and why they help

Feature	Description	How it helps photosynthesis
Envelope	double membrane (inner and outer)	controls what enters and leaves; inner membrane has transport proteins
Thylakoids	flattened membrane sacs	membranes hold photosystems, electron carriers and ATP synthase: the light-dependent reactions
Grana (singular granum)	stacks of thylakoids	huge membrane **surface area** for pigments and light absorption
Intergranal lamellae	thylakoid membranes joining grana	also carry PSI; link the grana
Thylakoid space (lumen)	inside each thylakoid	small volume, so a **proton gradient** builds quickly for ATP synthesis
Stroma	fluid matrix around the thylakoids	contains **RUBISCO** and the other Calvin cycle enzymes; close to the thylakoids, so ATP and reduced NADP reach it quickly
Starch grains	in the stroma	store the glucose made
Circular DNA and **70S ribosomes**	in the stroma	code for and make some chloroplast proteins (e.g. part of RUBISCO)
Lipid droplets	in the stroma	lipid store (not credited as a "feature" label)

- Chloroplasts are about **2–10 µm** long, so you need an electron microscope to see the grana.
- Not every plant cell has them: root cells and phloem sieve tubes don't.

WORKED EXAMPLE: ACTUAL SIZE OF A CHLOROPLAST

On an electron micrograph at ×8000 a chloroplast measures 58 mm.
Actual size = image size ÷ magnification. Convert first: 58 mm = 58 000 µm.
58 000 ÷ 8000 = 7.25 µm = **7.3 µm** (2 s.f.).
In standard form, keep **one non-zero digit before the decimal point**: 7.3×10^{-3} mm.

MARK-SCHEME LANGUAGE

- Labels that score (beyond stroma and thylakoid): "DNA (loop)", "starch grain", "{envelope / inner membrane / outer membrane}", "grana", "intergranal lamellae", "(70S) ribosome". Lipid droplets are ignored; "linear DNA" and "stoma" are wrong.
- Chloroplast v mitochondrion: **similarities** "both have (circular) DNA", "both have (70S) ribosomes", "both have a double membrane"; **differences** "stroma v matrix", "thylakoids / grana v cristae", "starch grains only in chloroplasts".
- Thylakoid membrane v cell surface membrane: "both have a phospholipid bilayer"; thylakoid "has chlorophyll / photosystems, ATP synthase, electron carriers".
- Why lots of thylakoids: comparative, "**more** light absorbed / **more** chlorophyll / **more** ATP and reduced NADP".

Often asked as: draw and label three features; compare with a mitochondrion; compare membranes; label MCQs; magnification and actual size; why many thylakoids.

COMMON MISTAKE

Compare questions answered with two separate descriptions. Pair each point: "**both** have…", "chloroplasts have… **whereas** mitochondria have…". Separate points are not pieced together.
Describing what the membranes **do** when the question asks about **structure**.
Calling a chloroplast "a cell", or labelling "inner part" instead of a named structure.
Standard form with two digits before the decimal point (73×10^{-4}).

BEYOND THE SPEC, BUT EXAMINED

- Chloroplasts and mitochondria both have **70S ribosomes and circular DNA**, like bacteria (WBI14 Jun 2021).
- Chloroplast DNA replicates in the light, independently of mitosis (WBI14 Oct 2021).
- Size conversions between nm, µm and mm for ribosomes, thylakoids and starch grains (WBI14 Jan 2022).
- Amyloplasts and vacuoles contain no DNA (WBI14 Jun 2025).

SPEC 5.6

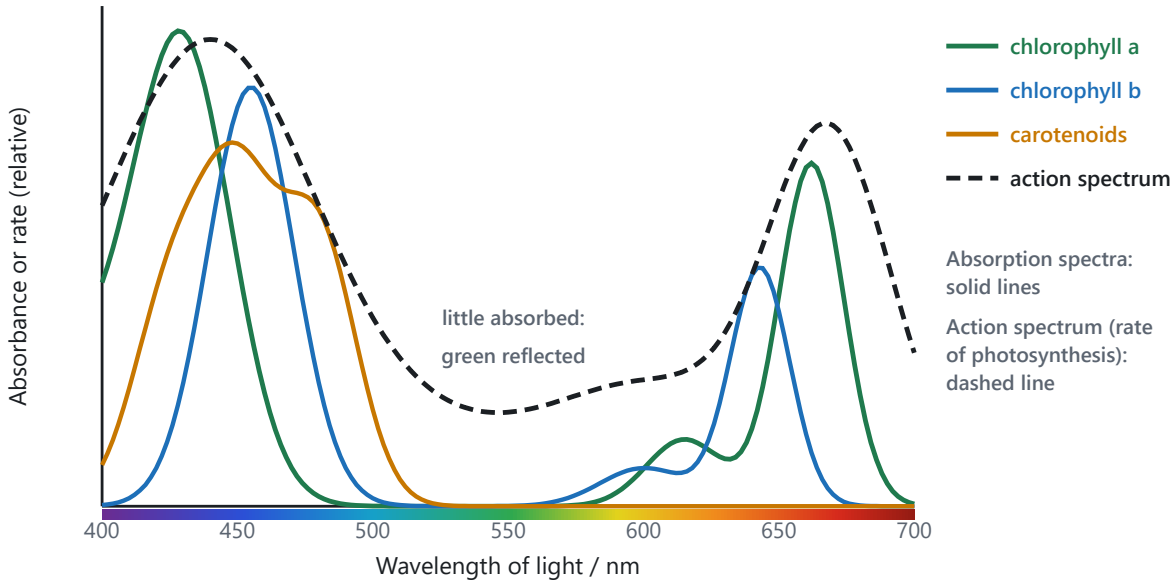
Absorption and action spectra

WHAT THIS SPEC POINT ASKS

Understand what an absorption spectrum and an action spectrum are.

5.6 Definitions

- **Absorption spectrum**: how much light **a pigment absorbs** at each wavelength.
- **Action spectrum**: the **rate of photosynthesis** at each wavelength of light.



Sketch of the absorption spectra (solid) and the action spectrum (dashed). Peaks are in blue-violet (about 430–480 nm) and red (about 640–680 nm). The action spectrum follows the combined absorption of all the pigments.

5.6 The pigments

Pigment	Colour	Absorbs mainly
Chlorophyll a	blue-green	blue-violet (~430 nm) and red (~660 nm); the reaction-centre pigment
Chlorophyll b	yellow-green	blue (~450 nm) and orange-red (~640 nm)
Carotenoids (β-carotene, xanthophylls)	orange, yellow	blue-violet (400–500 nm)

- Chlorophylls absorb little **green** light: it is **reflected** (or transmitted), so leaves look green.
- The **accessory pigments** (chlorophyll b, carotenoids) absorb wavelengths chlorophyll a absorbs poorly, and pass the energy on to the reaction centre. Having several pigments **widens the range of wavelengths** a plant can use.

5.6 Reading the two together

- The action spectrum **closely follows** the combined absorption spectra. That is the evidence that the light absorbed by these pigments is what drives photosynthesis.
- The action spectrum **doesn't fall to zero** in the green: some green light is absorbed by accessory pigments, and light bouncing around inside a thick leaf gets more chances to be absorbed.
- To measure an action spectrum: shine light of one wavelength at a time (coloured filters or LEDs) of the **same intensity** onto an aquatic plant, and measure the rate of oxygen production (as in CP10, 5.8).

MARK-SCHEME LANGUAGE

- Action spectrum: "**rate of photosynthesis** at different wavelengths of light".
- Sketching an action spectrum from an absorption spectrum: "a line that roughly follows the contours of the absorption spectrum" and "does not drop to zero"; the blue peak is higher than the red peak.
- Why several pigments: "so that light can be absorbed at different wavelengths", "so that the rate of photosynthesis is maximised".
- Least absorbed: **green** (around 500–560 nm).
- Ratios of peak heights: give as a ratio (e.g. 1.2 : 1), not a percentage.

Often asked as: define action spectrum; draw it on an absorption graph; why plants have different pigments; read absorbance values; which colour of light boosts growth.

COMMON MISTAKE

Swapping the two definitions. Absorption = **pigment absorbs**; action = **rate of photosynthesis**. Long answers to a 2-mark question. "To maximise light absorption" is usually the key phrase, not a description of the graph in words.

BEYOND THE SPEC, BUT EXAMINED

- **Seaweeds at different depths**: different seaweeds have different pigment mixtures, so each absorbs the wavelengths that reach its depth (red light penetrates least far). Living at different depths also reduces competition for light (WBI14 Jun 2024).
- Photosynthetic bacteria have absorption peaks at different wavelengths from plants (about 380 nm and 880 nm), so they can use light plants don't (WBI14 Jan 2023).
- Blue and red light give the highest growth rates; amber the lowest (WBI14 Oct 2024).

SPEC 5.7

Chromatography of chloroplast pigments

WHAT THIS SPEC POINT ASKS

Understand that chloroplast pigments can be separated by chromatography and identified by their Rf values.

5.7 How it works

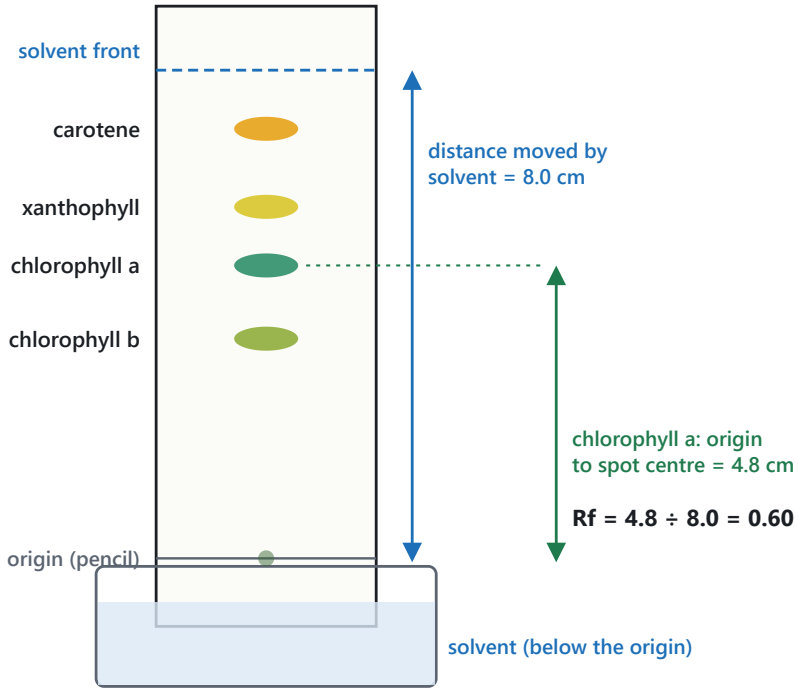
- A **stationary phase** (chromatography paper or a thin-layer plate) and a **mobile phase** (a non-polar solvent).
- Each pigment has its own balance between **solubility in the solvent** and **attraction to the stationary phase**. More soluble pigments are carried further. So the pigments separate into bands.

PIGMENT CHROMATOGRAPHY: METHOD IN BRIEF

1. Grind a **fresh, green** leaf with a little sand and **propanone** (acetone). Grinding breaks cells and membranes; the solvent dissolves the pigments (they are not water-soluble).
2. Draw an **origin line in pencil** about 2 cm from the bottom of the paper or TLC plate. (Ink would dissolve and run.)
3. Use a fine capillary tube to put a **small, concentrated spot** of extract on the origin. Let it dry, then add more on the same spot. Repeat 5–10 times.
4. Stand the paper in a little solvent in a covered tank, with the **solvent level below the origin** (otherwise the pigments dissolve into the solvent pool).

5. Remove it when the solvent is near the top. **Mark the solvent front at once** in pencil, before it evaporates.
6. Measure from the origin to the **centre of each spot** and from the origin to the solvent front. Work out each Rf.

Safety: the solvents are flammable and harmful; no flames, good ventilation or a fume cupboard.



A typical order of bands, top to bottom: carotene, xanthophyll, chlorophyll a, chlorophyll b. The exact Rf values depend on the solvent.

$$\text{Rf} = \text{distance moved by the pigment} \div \text{distance moved by the solvent front}$$

- Rf has **no units** and is always **between 0 and 1**.
- In one solvent, at one temperature, on one type of paper, each pigment has a **constant Rf**. Compare your Rf with **reference values for the same solvent**, or run known pigments alongside on the same paper in the same tank.

WORKED EXAMPLE: IDENTIFYING A PIGMENT

The solvent front moved 8.0 cm from the origin. A blue-green spot moved 4.8 cm (to its centre).
Rf = 4.8 ÷ 8.0 = **0.60**.
The reference table for this solvent gives chlorophyll a = 0.60, so the spot is chlorophyll a. Quote Rf to 2 or 3 decimal places.

MARK-SCHEME LANGUAGE

- Rf: "distance moved by pigment ÷ distance moved by solvent front", measured "from the origin" to the "{centre / leading edge}" of the spot.
- Identify: "calculate the Rf value", "compare with Rf values of known pigments **in the same solvent**", or "run known pigments in the same solvent in the same tank".
- Why Rf identifies β-carotene: it is soluble in that solvent, has a specific Rf in that solvent, and is coloured so it can be seen.
- Why two methods give different Rf: different solvents, different solubilities, different stationary phase.
- Technique points: pencil origin, small concentrated spot (dried between applications), solvent below the origin, mark the solvent front straight away.

Often asked as: which technique separates pigments; calculate Rf from a diagram; use Rf to identify a pigment; describe how to measure Rf; the same method applied to antibiotics or proteins.

COMMON MISTAKE

Missing the words "**same solvent**". The examiners say this is the key mark.
Describing how to set up a chromatogram when the question asks how to **identify** a pigment.
Measuring to the top of the paper instead of the solvent front.

BEYOND THE SPEC, BUT EXAMINED

- The same method separates non-pigments (antibiotics, proteins): the most soluble or smallest molecules travel furthest (WBI14 Jun 2023, Oct 2022).
- Comparing pigment bands from spinach and seaweeds to explain their colours (WBI14 Jun 2024).
- Different solvents extract pigments differently (WBI14 Oct 2021).

SPEC 5.8

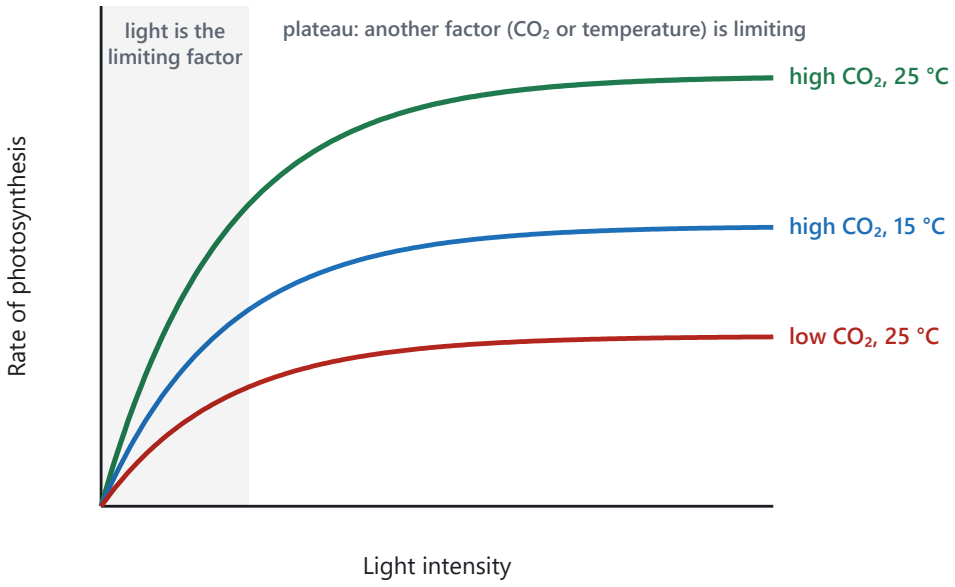
Core Practical 10: factors affecting the rate of photosynthesis

WHAT THIS SPEC POINT ASKS

Investigate how light intensity, light wavelength, temperature and CO_2 availability affect the rate of photosynthesis in an aquatic plant.

5.8 Limiting factors

- The rate of photosynthesis is limited by whichever factor is in **shortest supply** (the **limiting factor**): light intensity, CO_2 concentration or temperature.
- Raising the limiting factor increases the rate, until something else becomes limiting and the graph **levels off**.



On the rising part, light is limiting. On the plateau, another factor is limiting: raising CO_2 or temperature lifts the plateau.

- **Light**: provides energy for the light-dependent reactions (exciting electrons, photolysis). More light → more ATP and reduced NADP.
- **CO_2**: the substrate for RUBISCO. More CO_2 → more fixation → more GP.
- **Temperature**: affects the **enzymes** (RUBISCO, ATP synthase). Warmer → more kinetic energy → more enzyme–substrate collisions, up to the optimum; then enzymes denature and the rate falls. Temperature has little effect at low light intensity, because light is limiting there.

CORE PRACTICAL 10: METHOD IN BRIEF

1. Cut a 5–10 cm piece of pondweed (*Elodea* or *Cabomba*) underwater and place it, cut end up, in a boiling tube of **sodium hydrogencarbonate** solution (a CO_2 source; use excess so CO_2 isn't limiting).
2. Stand the tube in a **water bath** (or a beaker of water) to keep the temperature constant. Darken the room.
3. Place the lamp (an LED gives off little heat) at a set distance, e.g. 10 cm, measured with a ruler. Leave the plant **5 minutes to acclimatise**.
4. Collect the gas for a set time (e.g. 5 minutes) in a **photosynthometer** (capillary tube + syringe): draw the bubble into the capillary and measure its **length**. Or count bubbles per minute, or use a gas syringe or an oxygen probe.
5. Repeat at least 3 times at each distance and calculate a mean. Then move the lamp to 20, 30, 40 cm…

Changing the other factors:

- **Wavelength**: coloured filters or coloured LEDs at the same distance and intensity.
- **CO_2**: different concentrations of sodium hydrogencarbonate, other factors fixed.
- **Temperature**: water baths at e.g. 10, 20, 30, 40 °C, with the light fixed.

Variables: IV = the factor you change. DV = volume of oxygen per unit time. Control: the other three factors, the same plant (same mass and length), acclimatisation time, collection time.

5.8 Processing the results

- Volume of gas in a capillary = $\pi r^2 l$ (r = radius of the bore, l = length of the bubble).
- Rate = volume ÷ time.

- Light intensity is proportional to **$1/d^2$** (d = distance from the lamp): the **inverse square law**. Doubling the distance cuts intensity to a quarter. Plot rate against $1/d^2$ for a direct measure of intensity.

WORKED EXAMPLE: RATE FROM A PHOTOSYNTHOMETER

Capillary bore radius r = 0.5 mm. Collection time 5 min.

Distance / cm	$1/d^2$ / cm^{-2}	Bubble length / mm (mean of 3)	Volume / mm^3	Rate / $mm^3\ min^{-1}$
10	0.0100	24.0	18.8	3.77
20	0.0025	8.0	6.3	1.26
40	0.000625	2.2	1.7	0.35

At 10 cm: $V = \pi \times 0.5^2 \times 24.0 = 18.85\ mm^3$ (18.8 to 3 s.f.); rate = 18.85 ÷ 5 = **3.77 $mm^3\ min^{-1}$**.
Moving from 20 cm to 10 cm makes the light **4 times** as intense (1/100 ÷ 1/400) and the rate rises 3.0 times. Rate is not exactly proportional, so another factor is starting to limit at high intensity.

5.8 Errors and improvements

Problem	Why it matters	Fix
Lamp heats the water	temperature changes as well as light	LED, heat shield (a tank of water), or water bath
Bubbles vary in size	counting bubbles is inaccurate	measure volume in a photosynthometer or gas syringe
Some O_2 dissolves in the water; some is used in respiration	you measure **net** O_2 release, which underestimates the true rate	treat results as comparative; same conditions throughout
Gas is not pure O_2	some N_2 and CO_2 diffuse in	still fine for comparing
Background light	uncontrolled light intensity	darkened room
CO_2 used up over time	CO_2 becomes limiting	excess sodium hydrogencarbonate; fresh solution

MARK-SCHEME LANGUAGE

- Units for rate: e.g. "mg CO_2 **used** $mm^{-2}\ hr^{-1}$" (per area per time), or volume of O_2 per unit time.
- Tangent for a rate from a curve: "tangent correctly drawn, touching the curve at the stated time", answer to a sensible number of decimal places.
- Temperature effect: "more collisions", "RUBISCO works faster", "ATP synthase works faster".
- Light no longer limiting: "light (intensity) is not rate limiting" once extra pigment widens the range of wavelengths absorbed.

Often asked as: the method itself hasn't been set in WBI14 papers 2020–2026, but its ideas have: rate against temperature for two species, a tangent on an O_2 curve, wavelength and biomass, rate units. Learn the method anyway; the spec names it.

COMMON MISTAKE

Saying temperature affects the light-dependent reactions "by giving more light". It affects **enzymes**.
Reading two lines' crossing point carelessly. Read graph values to the nearest small square.
A tangent that doesn't touch the curve at the time asked.

BEYOND THE SPEC, BUT EXAMINED

- **C3 and C4 plants**: C4 plants (e.g. maize) have a higher optimum temperature (about 34–36 °C, against 26–28 °C for C3 plants) (WBI14 Jan 2024).
- Different species have different optimum temperatures for photosynthesis, which helps explain where they grow (WBI14 Jan 2022).
- Temperature, RUBISCO and ATP synthase as the explanation of a rate-against-temperature curve (WBI14 Jan 2026).

SPEC 5.9 GPP, NPP and plant respiration

WHAT THIS SPEC POINT ASKS

(i) Understand how gross primary productivity (GPP), net primary productivity (NPP) and plant respiration (R) are related. (ii) Be able to calculate NPP.

5.9 Definitions

- **Primary productivity** is the rate at which producers (plants, algae) make new organic matter by photosynthesis. It is measured as energy (or biomass) **per unit area per unit time**: **kJ m^{-2} yr^{-1}** (or g m^{-2} yr^{-1}; per m^3 in water).
- **GPP (gross primary productivity)**: the **total** rate at which light energy is converted into chemical energy in organic molecules by photosynthesis.
- **R (plant respiration)**: the energy the plant itself releases by respiring some of those molecules. Much of it is lost as **heat**.
- **NPP (net primary productivity)**: what is left as **new plant biomass**. This is the energy available to **primary consumers** and **decomposers**.

$$\text{NPP} = \text{GPP} - \text{R}$$

5.9 Where the light energy goes

- Only about **1–3%** of the light energy reaching a plant ends up as GPP, because:
  - some light is **reflected** (especially green) or **transmitted** through the leaf;
  - some wavelengths can't be absorbed by the pigments;
  - some light misses chloroplasts or falls on non-photosynthetic parts;
  - other factors (CO_2, temperature, water) limit the rate.
- Plants then respire a large share of GPP (often 20–50% or more), leaving the rest as NPP.

WORKED EXAMPLE: NPP, AND REARRANGING

1. A grassland has GPP = 24 600 kJ m^{-2} yr^{-1} and R = 15 900 kJ m^{-2} yr^{-1}.
   NPP = 24 600 − 15 900 = **8 700 kJ m^{-2} yr^{-1}**.
2. A forest has NPP = 3 400 kJ m^{-2} yr^{-1}. Respiration uses 60% of GPP. Find GPP.
   NPP is 40% of GPP, so GPP = 3 400 ÷ 0.40 = **8 500 kJ m^{-2} yr^{-1}** (and R = 5 100).
3. Incident light 1.2 × 10^6 kJ m^{-2} yr^{-1}; GPP 1.8 × 10^4 kJ m^{-2} yr^{-1}.
   Percentage captured = 1.8 × 10^4 ÷ 1.2 × 10^6 × 100 = **1.5%**.

5.9 Measuring it

- Biomass is measured as **dry mass** (water content varies a lot): dry samples in an oven at about 80 °C until the mass stops changing (**constant mass**). Take samples from a known area at intervals: the gain in dry

mass per area per year estimates NPP.

- Energy content can be found by burning dry samples in a calorimeter.

MARK-SCHEME LANGUAGE

- Labels on a flow diagram: "**gross primary productivity**" (or GPP) going in, "(plant) **respiration**" (or R) leaving, NPP left over.
- "Energy lost **by** respiration". **Not** "energy lost **for** respiration".
- Units: "**kJ m^{-2} yr^{-1}**" (energy per area per time).
- NPP that ends up below ground: "Calvin cycle produces GALP" → "sucrose" and "amino acids" → "transported in the phloem" → "used to synthesise biomass".

Often asked as: label a GPP–R–NPP diagram; MCQ on NPP = GPP − R; units of NPP; calculate NPP or a percentage of it; compare ecosystems; 6-mark on factors affecting root biomass.

COMMON MISTAKE

Using NPP for **animals**. GPP and NPP are **producer** (trophic level 1) terms.
Adding reasons inside a diagram box ("energy lost for respiration") and losing the mark.
Leaving out the units, or writing them as kJ/m^2/yr^{-1}. Use kJ m^{-2} yr^{-1}.

BEYOND THE SPEC, BUT EXAMINED

- Factors that affect NPP and root biomass in a 6-mark answer: temperature (enzymes, RUBISCO), soil moisture (photolysis, transport), tree density (shading), soil nitrogen (amino acids), soil depth (WBI14 Oct 2022).
- Biomass expressed as carbon, in gigatonnes (Gt C) (WBI14 Oct 2021).

SPEC 5.10

Efficiency of energy and biomass transfer

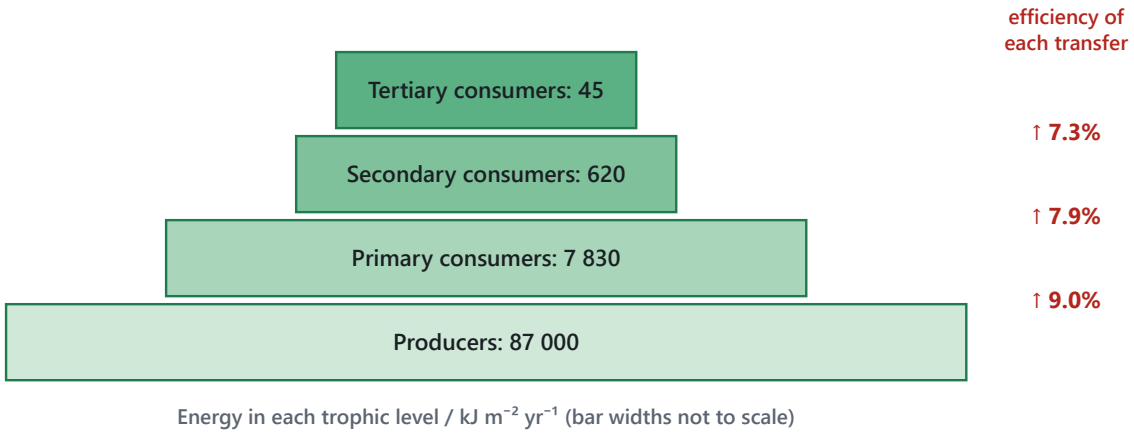
WHAT THIS SPEC POINT ASKS

Know how to calculate the efficiency of biomass and energy transfers between trophic levels.

5.10 Where energy goes at each level

- **Producers** lose energy as light not absorbed (5.9) and in respiration.
- **Consumers** don't get all the energy in the level below:
  - parts are **not eaten** (roots, bones, shells, fur);
  - parts are eaten but **not digested**: lost in **faeces** (egestion), e.g. cellulose;
  - some is lost in **urine** (excretion of urea);
  - a lot is used in **respiration** and lost as **heat** (movement, keeping warm).
- Only the energy built into **new biomass** (growth and reproduction) is available to the next level. That is typically about **10%** (often 5–20%), but not a fixed rule: calculate it from the data.

$$\text{efficiency (\%)} = \text{energy (or biomass) transferred to the higher trophic level} \div \text{energy (or biomass) in the lower trophic level} \times 100$$



A pyramid of energy. Each level holds less energy than the one below, so the pyramid can never be inverted.

WORKED EXAMPLE: EFFICIENCY AND RATIOS

Using the pyramid: producers 87 000, primary consumers 7 830 kJ m^{-2} yr^{-1}.
Efficiency = 7 830 ÷ 87 000 × 100 = **9.0%**.
Secondary consumers: 620 ÷ 7 830 × 100 = **7.9%**.
Ratio of energy in producers : primary : secondary = 87 000 : 7 830 : 620. Divide by the smallest: **140 : 13 : 1**. One side must be 1.
Trap: (87 000 − 7 830) ÷ 87 000 × 100 = 91% is the share **lost**, not the efficiency.

5.10 Consequences

- **Food chains are short** (rarely more than 4–5 levels): after a few transfers there isn't enough energy left to support another population.
- **Endotherms** (birds, mammals) are less efficient than **ectotherms** (reptiles, insects) because they respire more to keep warm.
- Herbivores lose more in faeces than carnivores (cellulose is hard to digest).
- Eating **lower in the food chain** (more plants) feeds more people per hectare: fewer transfers, less energy lost.
- **Farming**: keeping animals indoors, warm and with limited movement means less energy is respired and more goes into growth (with welfare trade-offs).

5.10 Pyramids

Pyramid	Shows	Can it be inverted?
Numbers	number of organisms at each level	yes (one oak tree supports thousands of insects)
Biomass	dry mass per area at one moment	occasionally (phytoplankton turn over fast, so their standing mass is small)
Energy	energy flow per area per year	**never**: energy is always lost between levels

MARK-SCHEME LANGUAGE

- No fourth trophic level: "not enough energy (in trophic level 3) to sustain a fourth trophic level", "energy is lost between trophic levels". "No energy" is ignored.
- Ratio: one value must be **1** (e.g. "98 : 10 : 1"; 100 : 10 : 1 accepted).
- Plant-based diet: "fewer trophic levels", "less energy lost from the food chain".
- Energy flow diagrams: arrows from each level to "respiration / heat", and the total lost.

Often asked as: efficiency calculation from a diagram; ratio of energy at three levels; why there is no fourth level; complete an energy-flow diagram; trophic level of a human diet; why a meat-eating diet needs more land.

COMMON MISTAKE

Subtracting first, which gives the percentage **lost**. Divide the higher level by the lower one.
Rounding badly: give percentages to a sensible precision (e.g. 1–2 d.p.), no recurring decimals.
Choosing values that make no biological sense from a crowded diagram. Check which two numbers are the energy entering each level.

BEYOND THE SPEC, BUT EXAMINED

- **Trophic level number (TLN)** of a diet: a mean value, so humans can be at 2.3 or 2.8; TLN rises with more meat (WBI14 Oct 2023).
- Reading proportional diagrams (e.g. Voronoi diagrams of biomass, where area ∝ biomass) (WBI14 Oct 2021).
- Biomass is used to compare organisms because they differ in size and can't all be counted (WBI14 Oct 2021).

SPEC 5.11

Population, community, habitat and ecosystem

WHAT THIS SPEC POINT ASKS

Understand the terms population, community, habitat and ecosystem.

Term	Meaning	Example (a rainforest)
Habitat	the place where an organism lives	the forest canopy where a slender loris lives
Population	all the organisms of **one species** living (and breeding) in a particular area at the same time	all the slender lorises in that forest
Community	all the populations of **different species** living and **interacting** in an area at the same time	lorises, tigers, trees, insects, fungi in that forest
Ecosystem	the community **interacting with each other and with the abiotic (physical and chemical) environment**	the forest's organisms plus its soil, water, light, temperature

- Smallest to largest: population → community → ecosystem. A **niche** (5.13) is a species' role, not a place.
- An ecosystem is a **dynamic** system: **energy flows** through it (in as light, out as heat) and **nutrients cycle** within it.
- **Microhabitat**: a small part of a habitat with its own conditions (e.g. under a log).

EXAM TIP

These definitions are usually 2 marks each: one for the definition, one for **using the organism and place named in the question**. "A population is all the slender lorises living in this forest" beats a textbook sentence.

MARK-SCHEME LANGUAGE

- Ecosystem: "organisms **interacting** with each other **and** with their (physical / chemical) environment". "Biotic and abiotic factors" accepted; "habitat" ignored.
- Habitat: "a place where organisms live" + the named place.
- Population: "organisms of **one species** in a particular area" + the named species.
- Community: "organisms of **different species** interacting in a particular area" + named organisms.

Often asked as: define all four using a food web or species list; define ecosystem; define population "as used in this context"; say what "sustainable" means in context.

COMMON MISTAKE

Missing the word **interact** from the ecosystem definition. Listing biotic and abiotic factors without saying they interact loses the mark.
Calling a niche a habitat.

BEYOND THE SPEC, BUT EXAMINED

- "**Sustainable**" in context: e.g. producing enough food without damaging the environment for the future (WBI14 Jan 2022). A general definition without the context doesn't score (WBI14 Jun 2024).

SPEC 5.12

Biotic and abiotic factors

WHAT THIS SPEC POINT ASKS

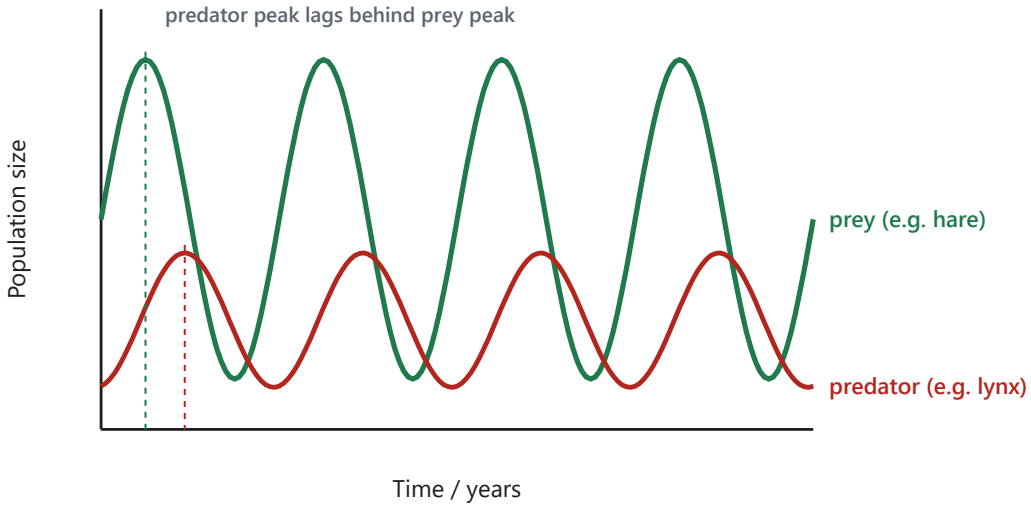
Understand that the numbers (abundance) and distribution of organisms in a habitat are controlled by biotic and abiotic factors.

- **Abundance**: how many individuals there are. **Distribution**: where they are found.
- **Abiotic factors**: non-living, physical and chemical.
- **Biotic factors**: caused by living organisms.

Abiotic factor	How it affects organisms
Light intensity (and wavelength)	photosynthesis rate, so plant growth; shade-tolerant v sun plants
Temperature	enzyme activity, growth, development, breeding times
Water / humidity	photolysis, turgor, transpiration; desiccation risk
Soil (edaphic): pH, mineral ions, depth, drainage	nutrient uptake (nitrate, phosphate), root growth
Oxygen and CO_2	respiration (O_2 is low in waterlogged soil and warm water); photosynthesis
Salinity, wind, wave action	water potential and osmosis; physical damage

Biotic factor	How it affects organisms
Intraspecific competition (same species)	for food, mates, territory, light: limits population size
Interspecific competition (different species)	for the same resource: one species may outcompete the other
Predation	predator and prey numbers rise and fall in linked cycles
Food supply	more food → more survive and breed
Parasitism and disease	spread faster in dense populations
Mutualism, pollinators, grazing, human activity	e.g. cutting grass favours fast-growing species

5.12 Predator–prey cycles



Prey rise first; predators then rise because there is more food; prey fall as more are eaten; predators then starve and fall; prey recover.

- The **predator peak lags behind** the prey peak, and predator numbers are usually lower.
- Real cycles are also driven by the prey's own food supply, disease and other predators.

5.12 Density-dependent and independent

- **Density-dependent** factors have a bigger effect in crowded populations: competition, disease, predation (mostly biotic).
- **Density-independent** factors affect a population whatever its size: floods, fire, frost (mostly abiotic).
- Together these keep a population around the **carrying capacity**: the largest population the habitat can support.

MARK-SCHEME LANGUAGE

- **Competition must say what for**: "competition for {light / water / mineral ions / space}". Cutting a field: "the grass is faster growing, reducing the light available to other plants".
- Predator–prey: "lynx decrease due to starvation when the number of hares decreases"; "hares increase when they reproduce"; "hares decrease as they are hunted by more lynx"; "changes in numbers are **staggered**".
- Other biotic factors in a scavenging study: decomposers (microorganisms, insects), other scavengers, predators, pathogens, people.
- Classify: grass cutting = biotic (human activity); salt and wave action = abiotic.

Often asked as: classify factors; explain differences between habitats; other biotic factors that could affect results; sketch or explain a predator–prey graph; which region suits a species given temperature data.

COMMON MISTAKE

Writing about abiotic factors when the question asks for **biotic** ones (and the reverse).
"Competition" with no resource named.
Sketching predator–prey curves carelessly: the predator peaks must come **after** the prey peaks and be offset by the same amount each time.

BEYOND THE SPEC, BUT EXAMINED

- Predator–prey oscillations and the lag between the peaks (WBI14 Jan 2025).
- Biotic factors that could confound a field study (WBI14 Jun 2022).

SPEC 5.13

Niche

WHAT THIS SPEC POINT ASKS

Understand how the idea of a niche explains the distribution and abundance of organisms in a habitat.

5.13 What a niche is

- A **niche** is the **role** a species plays in its habitat: what it eats, what eats it, when and where it feeds, the conditions it tolerates, and how it interacts with other species and the abiotic environment.
- Habitat = address; niche = job.

5.13 How niche explains distribution and abundance

- **Distribution**: a species is only found where the biotic and abiotic conditions match its adaptations (its tolerance range).
- **Abundance**: the more resources its niche gives access to, and the less competition, the larger the population.
- **Competitive exclusion**: two species **cannot occupy exactly the same niche** in the same habitat for long. The better-adapted one outcompetes the other, which dies out locally or shifts.
- Similar species can coexist when their niches **differ slightly** (**resource partitioning**): e.g. warblers feeding in different parts of the same tree, or two barnacle species at different heights on a rocky shore.

WORKED EXAMPLE: TWO BARNACLES

Species A is found only on the upper shore; species B on the middle and lower shore.

- A tolerates **drying out** (desiccation) at low tide better, so it survives higher up.
- B grows faster and **outcompetes** A for space lower down, but can't survive the dry upper shore.
- So each occupies a different niche, which **reduces competition**, and both persist.

MARK-SCHEME LANGUAGE

- "Niche is the **role** that an organism plays in its habitat" (with an example from the question).
- Distribution: "the two species are found on different parts of the rock", "a stated reason (food, temperature tolerance, desiccation)", "which **reduces competition**".
- Predator linked to prey: "the loris provides food for the tigers", "therefore tigers are found where lorises are found".
- Six-mark evolution of a new niche: use the data, mention variation in **polygenic** features, selection pressure, changed allele frequency, reduced gene flow, reproductive isolation.

Often asked as: explain how a distribution shows different niches; niche MCQs; define niche with an example; how a species came to occupy a new niche (6-mark).

COMMON MISTAKE

Choosing "habitat" in a niche MCQ. A niche is a **role**, not a place.
Using **abundance** and **distribution** as if they mean the same.

BEYOND THE SPEC, BUT EXAMINED

- Adaptations in three types: **anatomical** (structure), **physiological** (body processes, e.g. producing metabolic water), **behavioural** (WBI14 Jan 2022, Jan 2020).
- **Fundamental niche** (everything a species could use) v **realised niche** (what it actually uses because of competition), as in the barnacle example.

SPEC 5.14

Core Practical 11: ecology of a habitat

WHAT THIS SPEC POINT ASKS

Study the ecology of a habitat, e.g. using quadrats and transects to find the distribution and abundance of organisms, and measuring the abiotic factors that matter there.

5.14 Choosing a sampling method

- You can't count everything, so you **sample** and scale up.
- **Random sampling**: for a fairly uniform area, or to compare two areas. Lay two tapes at right angles as axes; use **random numbers** to pick coordinates for each quadrat. This **avoids bias**.
- **Systematic sampling** along a **transect**: for a **gradient** (shore, edge of a wood, path, sand dunes). Place quadrats at regular intervals along a line.
  - **Line transect**: record what touches the line at intervals.
  - **Belt transect**: quadrats placed along the line, continuously or at intervals (**interrupted belt**).

5.14 Measuring abundance

Measure	How	Use for
Density	count individuals per quadrat	large, separate plants (daisies) or slow animals (limpets)
Percentage cover	estimate % of the quadrat covered (a 10 × 10 gridded quadrat: each square = 1%)	grasses and mosses where individuals can't be told apart
Frequency	% of quadrats in which the species is present	quick surveys
ACFOR scale	abundant, common, frequent, occasional, rare	quick but **subjective**
Point quadrat	frame of pins; % cover = pins touching the species ÷ total pins × 100	dense, low vegetation

CORE PRACTICAL 11: METHOD IN BRIEF (BELT TRANSECT PLUS ABIOTIC FACTORS)

1. Choose a habitat with a gradient (e.g. from open grassland into woodland). Lay a tape measure along it.
2. Place a 0.5 m × 0.5 m quadrat every 2 m along the tape (interrupted belt transect).
3. In each quadrat, record the **abundance** of each species (percentage cover or count). Use a key to identify species.
4. At each quadrat, measure the **abiotic factors**: light intensity (light meter at the same height and time of day), soil moisture (moisture probe, or mass lost on drying), soil pH (probe or indicator), soil and air temperature (thermometer or probe). In water: oxygen probe, flow rate, turbidity.
5. Repeat the transect at least 3 times (parallel lines) and calculate mean values.
6. Plot a **kite diagram** (width of the kite = abundance at each point) to show distribution, and scatter graphs of abundance against each abiotic factor.
7. Test for a correlation using **Spearman's rank** (r between −1 and +1). Compare the calculated value with the **critical value at p = 0.05**: if it is larger, reject the null hypothesis (no correlation).

Safety and ethics: check tides and weather; wash hands after handling soil; return organisms and stones as you found them; trample as little as possible.

5.14 Enough samples?

- Plot a **running mean** of species number (or abundance) against number of quadrats. When it **levels off**, you have enough.
- More quadrats and repeats → a more representative mean and a smaller standard deviation.

WORKED EXAMPLE: ESTIMATING A POPULATION AND A DIVERSITY INDEX

Population. Ten random 0.25 m^2 quadrats in a 1 200 m^2 field contain a mean of 6.4 dandelions each. Density = 6.4 ÷ 0.25 = 25.6 per m^2. Population ≈ 25.6 × 1 200 = 30 720 ≈ **31 000**.

Simpson's diversity index (the formula is printed on the paper): D = N(N − 1) ÷ Σn(n − 1), where N = total number of organisms, n = number of each species.
Counts: species A 10, B 6, C 4. N = 20.
Σn(n − 1) = (10 × 9) + (6 × 5) + (4 × 3) = 90 + 30 + 12 = 132.
N(N − 1) = 20 × 19 = 380. D = 380 ÷ 132 = **2.88**. A **higher** value means a **more diverse** community.

MARK-SCHEME LANGUAGE

- Design (lab choice experiment): "all surfaces set up in the same container / water", "same surface area", "stated number (at least 100) of larvae", "leave for a period of time", "calculate number per unit area", "repeat (at least 3 times) and calculate mean and standard deviation".
- Diversity: "count the number of different species", "find the abundance of each species", "use the (bio)diversity index"; a successful restoration shows a **higher** index. Hardy–Weinberg is not accepted.
- Reliability with error bars: smaller error bars relative to the mean = more reliable; overlapping bars → difference may not be significant.

Often asked as: the quadrat or transect method itself hasn't been set in WBI14 2020–2026. What has: designing a sampling or choice experiment, estimating a population from survey counts, percentage calculations from survey data, using a diversity index to judge a restoration.

COMMON MISTAKE

Putting each treatment in a **separate** container in a choice experiment. The organisms must be able to choose, so all options go in the **same** container.
Not naming how a variable is controlled. "Keep it the same" scores less than "measure at the same time of day".
Using the t-test or chi-squared for a **correlation** question. Use Spearman's rank (or Pearson).

BEYOND THE SPEC, BUT EXAMINED

- **Simpson's diversity index**: know what N and n are and that a higher value = more diverse (WBI14 Jun 2025, Oct 2025).
- Estimating a total population by scaling up survey counts (WBI14 Oct 2020).
- Mean and standard deviation in a design answer (WBI14 Jan 2020).

SPEC 5.15

Succession

WHAT THIS SPEC POINT ASKS

Understand the stages of succession, from colonisation of bare ground to a climax community.

5.15 What succession is

- **Succession** is the **change in the species** that make up a community over time.
- **Primary succession** starts on ground with **no soil**: bare rock after a volcanic eruption, sand dunes, ground left by a retreating glacier.
- **Secondary succession** starts where soil already exists but the community has been removed (fire, felling, ploughing). It is **faster**: soil, seeds and roots are already there.

5.15 The stages

Stage	Organisms	How they change the conditions
Pioneer (colonisation)	lichens, mosses, algae, marram grass on dunes	break down rock (weathering); when they die, **decomposers** turn them into **humus**, starting a thin soil
Intermediate	grasses, ferns, small flowering plants, then shrubs	deeper soil, more organic matter, **holds more water and mineral ions**; more shelter and shade; pioneers are **outcompeted**
Climax community	trees (in most temperate climates), a few dominant species	stable, self-sustaining; little further change while the climate stays the same

Each community **modifies the environment** so that it becomes **more suitable for the next species** (and often less suitable for itself). Conditions become less extreme. Biomass rises and, usually, so does biodiversity, although it may fall slightly at the climax as a few large species dominate.

5.15 Pioneer species are adapted to harsh conditions

- Tolerate **extremes**: drought, lack of shade, wind, temperature swings.
- **Low requirement for minerals and soil**; some fix nitrogen.
- **Fast life cycles**, lots of seeds or spores, **wide dispersal** (by wind), often asexual reproduction.

5.15 Abiotic changes through a sand dune or glacier succession

- Soil **depth** and **organic matter (humus)** increase as dead material is decomposed.
- **Water** retention and **mineral ion** content increase.
- **pH** usually falls as humus builds and rain leaches out minerals.
- **Bare ground** decreases.
- **Carbon : nitrogen ratio**: rises as decomposition adds carbon; falls as nitrogen is added by leaf litter and **nitrogen-fixing** and nitrifying bacteria.

5.15 Plagioclimax

- Human activity (grazing, mowing, burning) can **stop** succession at an earlier stage: a **plagioclimax** (e.g. grazed grassland, heather moorland). Conservation sometimes does this on purpose, to keep a habitat and its species.

MARK-SCHEME LANGUAGE

- Succession: "a (series of) changes in the (types of) **species** (within a habitat) with time".
- Pioneer characteristics: "withstand {harsh / extreme} environments", "{desiccation / lack of shade}", "low requirement for {minerals / soil}", "{fast life cycle / grow fast / reproduce asexually / lots of seeds / wide dispersal}".
- Why stages: "changes to the habitat have to take place before a different organism can survive there", plus the named change and the organism that follows.
- Dune 6-mark (description + explanation): soil depth and organic matter rise "because these are older dunes", humus builds up, microorganisms decompose; pH falls; bare ground falls; species number rises as the soil becomes fertile.
- Insects on a carcass: the first species is the pioneer; later species arrive and the first "is outcompeted".

Often asked as: 6-mark data questions on sand dunes, glacier retreat or forest regrowth; three characteristics of pioneers; why succession happens in stages; define succession; regrowth after fire.

COMMON MISTAKE

Describing **what pioneers do** when asked for their **characteristics**.

Saying succession is a change in the soil or habitat. It is a change in the **species**; the habitat changes cause it.

"Competition" without the resource (light, water, mineral ions, space).

Writing so much general description that the data parts (biodiversity, C : N) get no marks. Plan, and cover every part.

BEYOND THE SPEC, BUT EXAMINED

- **Carbon : nitrogen ratio** in soils during succession, needed for full marks (WBI14 Jan 2024).
- Secondary succession after fire or an eruption (WBI14 Oct 2022).
- A climax community is "not yet reached" while species numbers are still changing (WBI14 Oct 2024).
- pH is a log scale: a difference of 2 pH units is a **100-fold** difference in H^+ concentration (WBI14 Jun 2021).

SPEC 5.16

Evidence for climate change

WHAT THIS SPEC POINT ASKS

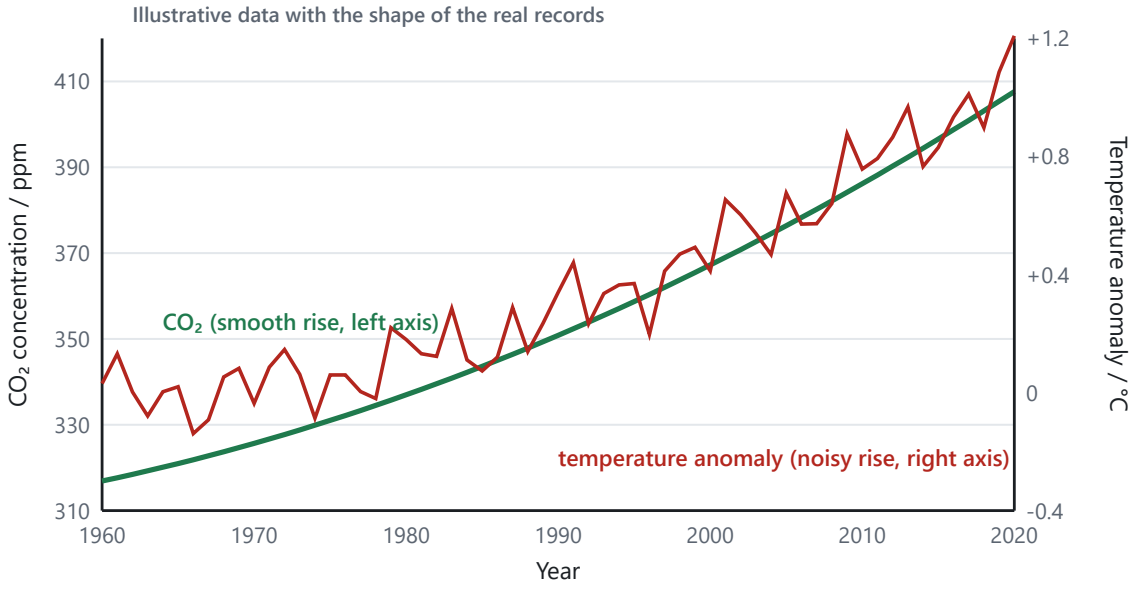
Understand the types of evidence for climate change (CO_2 records, temperature records, pollen in peat bogs, dendrochronology) and its causes, telling correlation apart from cause.

- **Weather** is day-to-day conditions. **Climate** is the long-term average pattern of weather (temperature, rainfall) in a region.
- **Climate change** is a significant, long-term change in those averages.

5.16 The four main kinds of evidence

Evidence	What it measures	How it works	Limitations
Temperature records	air and sea temperature	thermometer readings worldwide since about 1850	few stations early on, mostly in Europe and North America; methods and sites have changed
CO_2 records	atmospheric CO_2 concentration	direct measurement since 1958 (Mauna Loa, Hawaii); before that, air **bubbles trapped in ice cores** from Antarctica and Greenland (deeper = older; back about 800 000 years)	ice-core dating gets less precise with depth; one site may not represent the whole globe
Pollen in peat bogs	which plants grew in the past, so the climate then	peat is **waterlogged, acidic and anaerobic**, so decomposition is very slow and pollen is preserved; layers build up, deeper = older; each species' pollen has a distinctive shape and each species grows in a known climate range	pollen can blow in from far away; some species make more pollen than others; plants may have adapted; layers can be disturbed
Dendrochronology (tree rings)	climate year by year	one ring grows each year; **wide ring = good growing conditions** (warm, wet), narrow = poor; cores from living trees and old timber are overlapped (cross-dated) to extend the record back	ring width also depends on rainfall, disease, competition, pests, soil; only works where there are distinct seasons

5.16 Correlation and cause



CO_2 and temperature both rise over the same period: a positive correlation. On its own this doesn't prove CO_2 causes the warming.

- A **correlation** means two variables change together. It does **not** show that one causes the other: a third factor could cause both, or it could be chance.
- To claim **cause** you need a **mechanism** (here, the greenhouse effect, 5.17), controlled experiments where possible, and consistent evidence from many independent sources.

EXAM TIP

For "describe the data": state the trend **with values and years** ("CO_2 rose from about 317 ppm in 1960 to about 414 ppm in 2020"), mention any fluctuations, then compare the two variables. Don't explain unless asked. Don't extrapolate from a single peak or trough of a noisy graph.

WORKED EXAMPLE: TREE AGE AND GROWTH RATE

A core taken from the **base of the trunk** shows 69 rings. The tree is about **69 years** old (one ring per year).
The trunk's radius at the base is 23 cm. Mean growth rate = 23 ÷ 69 = **0.33 cm yr^{-1}** (radius).
A ring narrower than its neighbours: a cooler (or drier) year → enzymes such as RUBISCO work more slowly → less photosynthesis, less GALP → less new xylem and biomass.

MARK-SCHEME LANGUAGE

- Ring width: "**narrower** because (mean) temperatures were **cooler**", "{enzymes / RUBISCO} working more slowly", "photosynthesis slower / less GALP produced", "less {biomass / xylem} produced". The converse is accepted; "more water, wider rings" too.
- Tree age: sample "from the {trunk / base of the tree}" (with an **increment borer**), "count the rings", "one ring is made each year". Cutting the tree down or using branches is not accepted. Quote the diagram ("69 rings = 69 years").
- Cores are matched by aligning rings; an overlapping ring is counted **once**.
- Pollen: "type of plant present when conditions are favourable"; a change in pollen type means "conditions changed"; competition for named resources.
- Bog preservation: "conditions not suitable for decomposition", "cold, so (microbial) enzymes work slowly", "low pH", "waterlogged, so anaerobic, so not enough ATP", "protected from insects".
- Correlation: "it is only a correlation", "no evidence of cause".

Often asked as: name dendrochronology; why rings differ; how to age a tree; growth rate from a core; pollen diagrams over centuries; describe a temperature graph; correlation v cause.

COMMON MISTAKE

Saying rings are "different" without saying **narrower** or **wider** and why, comparatively.
Measuring the diameter "of the tree" without saying **at the base**.
Reading a pollen diagram upside down: the **deepest** layer is the **oldest**.
Writing about succession when the question is about climate change.

BEYOND THE SPEC, BUT EXAMINED

- Increment borer and radius / circumference calculations from tree cores (WBI14 Oct 2021).
- Why bodies are preserved in bogs (WBI14 Jan 2025).
- Radiocarbon dating gets only 1 mark when asked how to age a tree; counting rings is the expected answer (WBI14 Jan 2024).
- Ice-core oxygen isotope ratios (^{18}O : ^{16}O) are another temperature record.

SPEC 5.17 Anthropogenic climate change and the greenhouse effect

WHAT THIS SPEC POINT ASKS

Understand the causes of climate change due to human activity, including how greenhouse gases cause the greenhouse effect.

5.17 The greenhouse effect

1. **Short-wavelength** radiation from the Sun (UV and visible light) passes **through** the atmosphere and reaches the Earth's surface.
2. The surface absorbs it, warms up, and re-emits energy as **long-wavelength infrared (IR)** radiation.
3. **Greenhouse gases** in the atmosphere **absorb** IR and re-radiate it, some back towards the surface. Heat is **trapped** and the lower atmosphere and surface warm.

- This **natural** greenhouse effect keeps Earth about 30 °C warmer than it would be: liquid water and life depend on it.
- **Anthropogenic climate change**: human activities add more greenhouse gases, so more IR is trapped (the **enhanced** greenhouse effect), causing **long-term changes in mean temperature and rainfall patterns**.

5.17 The main greenhouse gases

Gas	Human sources	Notes
Carbon dioxide	**burning** fossil fuels (power, transport, industry); **deforestation** (burning, and fewer trees to absorb CO_2); draining and burning peat	largest total effect; long-lived
Methane	**cattle and other ruminants** (gut microorganisms); **rice paddies**; **landfill** (anaerobic decay); leaks from gas and coal; thawing permafrost (a positive feedback)	traps far more heat **per molecule** than CO_2 but is at lower concentration and breaks down faster
Nitrous oxide	nitrogen fertilisers	very high warming effect per molecule
Water vapour	(mostly natural)	warmer air holds more, amplifying warming

Oxygen and nitrogen are **not** greenhouse gases.

MARK-SCHEME LANGUAGE

- Anthropogenic climate change: "caused by the effects of **humans**" **and** "changes to (mean) {temperature / rainfall}" ("long-term change in weather patterns" accepted). "Global warming" and "weather" alone are ignored.
- Greenhouse effect: "greenhouse gases {accumulate / increase} in the (upper) **atmosphere**"; "allow {UV / short wavelength / visible} light **through** to the Earth's surface"; "but {trap / absorb} **IR** (long wavelength) radiation"; "causing the Earth's {atmosphere / surface} to **heat up**".
- Human sources must say **what is done**: "**burning** (fossil) fuels", "farming cattle", "rice paddies", "landfill", "draining peat bogs", "deforestation".
- Volcanoes: ash and sulfur dioxide **block** incoming light, so less IR is re-emitted and temperatures **fall**; as the particles disperse, temperatures rise again. One eruption's CO_2 is tiny next to yearly human emissions.

Often asked as: which gases are greenhouse gases; UV and IR through the atmosphere; define anthropogenic climate change; explain the greenhouse effect; name human sources; volcano v human activity; global warming potential.

COMMON MISTAKE

"Fossil fuels" without "**burning**".
Not saying **where** the gases build up (the atmosphere) or **what** heats up (Earth's surface / atmosphere).
Mixing up short and long wavelengths. Safer: **UV / visible in, IR trapped**.
Defining climate change only as "happens over a long time": say what changes (mean temperature, rainfall).
Plotting a variable against "global warming" instead of against **temperature**.

BEYOND THE SPEC, BUT EXAMINED

- **Global warming potential** and atmospheric lifetime of methane and nitrous oxide compared with CO_2 (WBI14 Jan 2025).
- Volcano data: sulfur dioxide and ash cool the Earth; anthropogenic CO_2 (about 3.5×10^{10} t a year) equals hundreds of eruptions (WBI14 Jun 2022).
- Using bacteria to remove methane could release other greenhouse gases (WBI14 Jan 2025).

SPEC 5.18

The carbon cycle and reducing atmospheric CO_2

WHAT THIS SPEC POINT ASKS

Understand how knowledge of the carbon cycle can be used to design ways of reducing atmospheric CO_2.

5.18 Stores (sinks) and flows (fluxes)

Store	Form of carbon
Atmosphere	CO_2 (and methane)
Oceans	dissolved CO_2 and **hydrogencarbonate** ions; carbonate in shells
Living organisms	carbohydrates, lipids, proteins (biomass)
Soil and peat	dead organic matter, humus
Fossil fuels	coal, oil, natural gas
Sedimentary rocks	limestone and chalk (calcium carbonate)

Process	Effect on atmospheric CO_2
Photosynthesis (plants, algae, phytoplankton)	**removes** CO_2: it is fixed into organic molecules
Respiration (plants, animals, **decomposers**)	**returns** CO_2
Decomposition (6.16)	returns CO_2 from dead material via microbial respiration
Combustion of fossil fuels and biomass	returns CO_2 quickly
Feeding	moves carbon between organisms
Ocean exchange	CO_2 dissolves in cold water and comes out of warm water
Fossilisation, sedimentation	lock carbon away for millions of years

- Human activity has **sped up** the flows into the atmosphere (combustion, deforestation, draining peat) and **reduced** the flows out (fewer forests).

5.18 Using the cycle to cut CO_2

Method	How it works	Limitation
Burn **less fossil fuel**: renewables (solar, wind, tidal, geothermal), public transport, lift sharing, insulation	less CO_2 **released** by combustion	cost; weather-dependent; land use
Reforestation and **stopping deforestation**	more photosynthesis, so more CO_2 **absorbed** and fixed; carbon **stored** in wood	takes decades; trees can burn; needs land
Biofuels (5.26)	CO_2 released was recently absorbed by the crop	land, fertiliser and processing emissions
Protect **peat bogs and soils**	waterlogged peat decomposes very slowly, keeping carbon locked up	drained peat releases CO_2
Protect oceans, mangroves, seagrass, kelp	phytoplankton and coastal plants fix carbon; dead material sinks	warming and acidification reduce uptake
Fewer cattle, less rice in paddies, capture landfill gas	less **methane** released	food supply and income
Carbon capture and storage	CO_2 from power stations pumped underground	expensive, unproven at scale

WORKED EXAMPLE: CARBON UNITS

1 Pg (petagram) = 10^{15} g = 10^{12} kg. So 61 Pg = 61 × 10^{12} kg = **6.1×10^{13} kg**.
1 Gt (gigatonne) = 10^{9} t = 10^{12} kg. So 11.3 Gt = **1.13×10^{13} kg**.
Check standard form: **one non-zero digit** before the decimal point.

MARK-SCHEME LANGUAGE

- Reducing CO_2 (method **plus** reason): "named way to reduce burning fossil fuels (lift sharing, public transport, solar power)" → "less carbon dioxide released"; "reforestation" → "more plants absorb more CO_2 for photosynthesis"; "reduce the number of cattle" → "less methane released".
- Return of carbon: "as **carbon dioxide** from **respiration**" of plants, animals or decomposers. CO, Co or C alone are wrong.
- Soil carbon: the ecosystem with the slowest decomposition keeps "more carbon" and releases "less CO_2".
- Whales: "a lot of carbon stored in the body", "dead organisms (on the sea floor) decompose very slowly" → "carbon sink".

Often asked as: carbon unit conversions; how carbon moves between air and ocean; how carbon returns to the air; methods to reduce CO_2 (including a 6-mark); carbon storage in soils.

COMMON MISTAKE

Giving a method without the **reason** it lowers CO_2.
Saying soil carbon is used by plants for photosynthesis. Plants fix CO_2 from the **air**.
Replanting answers not linked back to climate change when the question is about climate.

BEYOND THE SPEC, BUT EXAMINED

- Decomposition of tea bags: dry mass found by removing water, and mass lost linked to CO_2 released by microbial respiration (WBI14 Oct 2021).
- Marine snow and coccolithophores carrying carbon to the deep ocean (WBI14 Oct 2024).
- Biofuel crops regrown "remove CO_2 for the Calvin cycle"; the phrase "carbon neutral" alone is ignored (WBI14 Jan 2026).

SPEC 5.19

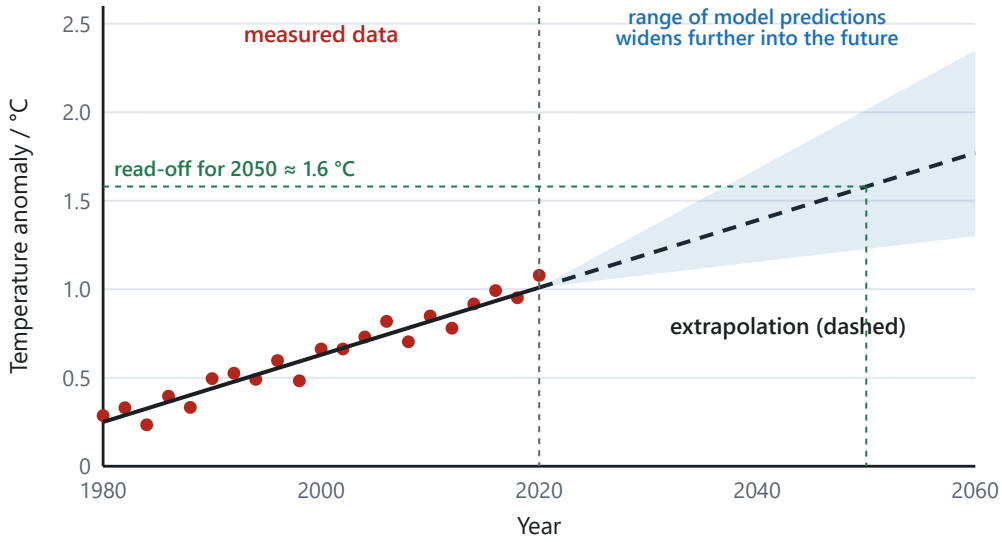
Extrapolation and climate models

WHAT THIS SPEC POINT ASKS

(i) Understand that data can be extrapolated to make predictions, and that these are used in models of future climate change. (ii) Understand that climate models have limitations.

5.19 Extrapolation

- **Extrapolation** means **extending a trend line beyond the range of the data** to predict a value.
- Draw a line of best fit through the whole data set (not through peaks or troughs), extend it, and read off the value.



Extending the best-fit line from the measured data gives a prediction for 2050 of about 1.6 °C. The further from the data, the less certain the prediction.

5.19 Models

- A climate **model** is a **prediction** of what might happen, **based on data and current understanding** (often a computer simulation). It runs scenarios: e.g. emissions cut quickly v "business as usual".
- Models are tested by running them on **past** data and checking they reproduce the climate we actually measured.
- Predictions are used to plan: flood defences, crop choices, energy policy, international agreements.

5.19 Why models have limitations

- Future **emissions** depend on human choices, technology and population: unknown.
- The climate system is **complex** and not fully understood: clouds, ocean currents, ice melt.
- **Feedback loops** and **tipping points** (e.g. thawing permafrost releasing methane; melting ice reflecting less light) could speed change up suddenly.
- Unpredictable natural events: **volcanic eruptions**, changes in solar output.
- Extrapolation **assumes the trend continues** in the same way. It may speed up, slow or level off (as sea-ice loss flattened for a while).
- The further ahead, the **wider the uncertainty**.

MARK-SCHEME LANGUAGE

- Method: "extrapolation (of the line / graph)" / "extending the line (to 2050)". "Draw a line of best fit" alone is ignored.
- Model: "making predictions", "based on {knowledge / data}", "a visual representation (of what might happen)".
- Why a prediction might be revised: "climate change is happening faster than thought", "more ice melts", "scientists may use other data".
- Answers read off a graph must fall in the accepted range and have a sensible number of decimal places.

Often asked as: extrapolate to a future year or find the year a value is reached; what is a model; why a prediction may change or fail.

COMMON MISTAKE

Extrapolating from the last peak or trough instead of from the overall trend.
Giving estimates to unrealistic precision (e.g. 1.6372 °C). Match the precision of the data.
Describing the diagram in words when asked what a model **is**.

BEYOND THE SPEC, BUT EXAMINED

- A falling trend (sea ice) can **flatten**, so a straight-line extrapolation fails (WBI14 Jan 2026).
- Extrapolation in other contexts, e.g. predicting disease deaths, and how a vaccine would change the estimate (WBI14 Jun 2022).

SPEC 5.20

Effects of climate change on plants and animals

WHAT THIS SPEC POINT ASKS

Understand how climate change (changing rainfall patterns and seasonal cycles) affects plants and animals: species distribution, development and life cycles.

5.20 The changes

- Higher mean temperatures; **changed rainfall** (more droughts in some places, more floods in others); more extreme weather; **earlier springs and later autumns**; melting ice and rising sea levels; ocean warming and acidification.

5.20 Effects on species

Effect	Example and explanation
Distribution shifts	species move **towards the poles** or **uphill** to stay within their temperature range. Species that can't move fast enough, or have nowhere to go (mountain tops, islands), decline or go extinct locally
Development speeds up	higher temperature → more **kinetic energy** → more enzyme–substrate collisions → faster respiration, mitosis, DNA replication → insects and other ectotherms develop faster and may fit in more generations a year
Life cycles and timing	earlier flowering, leaf burst, egg-laying and migration. Species that respond to **temperature** shift, but those that respond to **day length** don't, causing a **mismatch** (e.g. chicks hatching after the caterpillar peak)
Rainfall	drought: less water for photolysis and turgor, stomata close, less CO_2 in, less photosynthesis; flooding: waterlogged roots short of O_2
Sex ratios	in reptiles with **temperature-dependent sex determination**, warmer nests skew the ratio (e.g. more female turtles)
Pests and disease	vectors such as mosquitoes spread to new regions; parasites survive mild winters

WORKED EXAMPLE: BUILDING A CAUSE–EFFECT CHAIN (TICKS AND MOOSE)

Warmer winters → **less snow** → fewer ticks die when they drop off in early spring, and warmer conditions **shorten the tick life cycle** → more ticks per moose → more blood removed, fur lost, less insulation, wounds infected → moose weaker and die → **moose population falls**.
Top answers link every step from climate to the change in numbers.

MARK-SCHEME LANGUAGE

- Species moving north: "distributed more towards the north", "related to a described change in climate", "**effect explained**" (enzymes less effective, prey moved, dehydration, plants die).
- Faster life cycle: "increase in temperature increases the **kinetic energy** … more collisions between enzymes and substrates", then a named process (respiration, mitosis, DNA polymerase).
- Polar bears: "berries and grass provide less energy than seal fat"; "energy used to swim further is greater than the energy gained".
- Sea ice from sea water melts faster: "higher salt concentration, so less heat needed to melt".

Often asked as: species moving north (3 marks); fertility or development against temperature; moose and ticks (6-mark); temperature-dependent sex determination; polar bears; coral reefs.

COMMON MISTAKE

"Habitat destruction" with no link to temperature or rainfall. Say **how** the climate change affects the organism.
Only saying a species "moved north" for 3 marks: add the reason (its enzymes, food, water).
Not reading the axes before describing a correlation.

BEYOND THE SPEC, BUT EXAMINED

- **Temperature-dependent sex determination**: turtles are male at lower temperatures, tuatara male at higher, crocodiles male at middle temperatures; the **pivotal temperature** gives 50% males. The switch is **epigenetic** (DNA methylation, histone modification) (WBI14 Jun 2025).
- Longer (older) lizards have shorter **telomeres** because their cells have divided more (WBI14 Oct 2023).
- Coral: sexual reproduction gives variation; asexual keeps alleles that suit current conditions (WBI14 Oct 2025).

SPEC 5.21

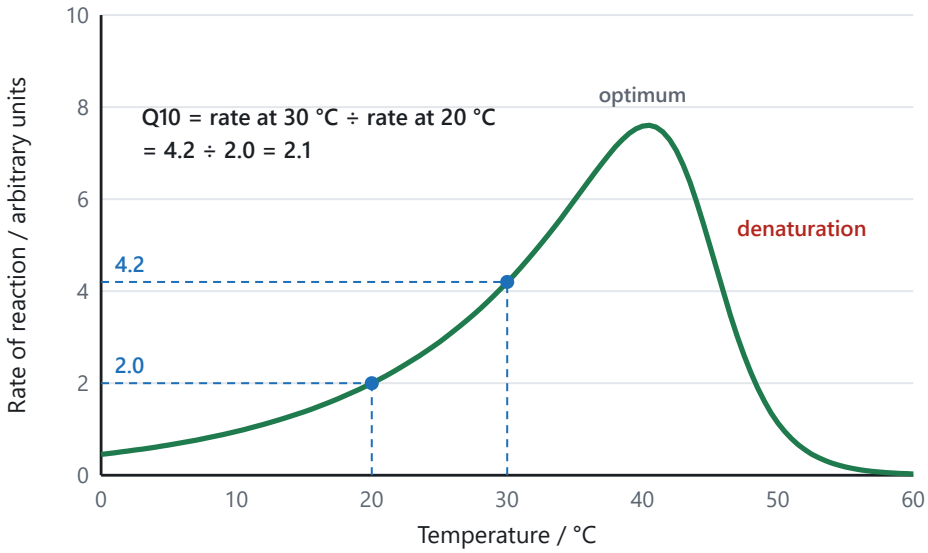
Temperature, enzyme activity and Q10

WHAT THIS SPEC POINT ASKS

Understand how temperature affects the rate of enzyme activity, and the effect on plants, animals and microorganisms, including the temperature coefficient Q10.

5.21 Why rate changes with temperature

- **Below the optimum**: raising the temperature gives enzyme and substrate molecules more **kinetic energy**, so they move faster → **more frequent collisions** → **more enzyme–substrate complexes** form per second → faster rate. At low temperature the enzyme is not damaged; it just works slowly.
- **Above the optimum**: atoms in the enzyme vibrate so much that **hydrogen bonds and ionic bonds** holding the tertiary structure **break**. The **active site changes shape**, so the substrate no longer fits (it is no longer complementary): the enzyme is **denatured**. This is usually permanent.



Read the rate at two temperatures 10 °C apart, then divide the higher-temperature rate by the lower one.

5.21 Q10, the temperature coefficient

$$Q10 = \text{rate of reaction at } (T + 10)\ °C \div \text{rate of reaction at } T\ °C$$

- Q10 tells you how many times faster a reaction goes for a 10 °C rise. For most enzyme reactions between about 0 and 40 °C, **Q10 ≈ 2–3**: the rate roughly doubles.
- Q10 has **no units**.
- Q10 ≈ 1 means temperature has little effect (e.g. a physical process like diffusion, or near the optimum).

WORKED EXAMPLE: Q10 FROM A GRAPH

1. From the graph: rate at 20 °C = 2.0 units; rate at 30 °C = 4.2 units.
   Q10 = 4.2 ÷ 2.0 = **2.1** (no units).
2. From a table: oxygen production by algae is 6.0 at 10 °C and 21.5 at 20 °C.
   Q10 = 21.5 ÷ 6.0 = **3.6**.

5.21 Effects on organisms

- **Ectotherms** (most plants, invertebrates, reptiles, fish, microorganisms) have body temperature close to the environment's, so their **metabolism, growth and development rates** change directly with temperature.
- **Endotherms** (mammals, birds) keep their body temperature roughly constant, but their food plants and prey are affected.
- **Plants**: photosynthesis (RUBISCO) and respiration both speed up with warmth until enzymes denature; very high temperatures also close stomata.
- **Microorganisms**: decomposition is faster when warm (more CO_2 released), and pathogens grow faster.
- Each species has its own optimum and **zone of tolerance**; enzymes of species from hot places (C4 plants, thermophilic bacteria) have higher optima.

MARK-SCHEME LANGUAGE

- Q10 = "rate at (T + 10 °C) ÷ rate at T". To describe it: "read off the rate at two temperatures **10 °C apart**", "divide the rate at the **higher** temperature by that at the lower".
- Calculation marks: "values for (e.g.) 20 °C and 30 °C read from the graph", "substituted into the equation", answer "with **no units**".
- Explanation: "more **kinetic energy**", "more frequent / energetic **collisions**", "more **enzyme–substrate complexes**"; high temperature: "enzymes **denature** / active site changes shape". Low temperature: "not enough kinetic energy".
- Sketch: rises to an **optimum** (about 40 °C for many enzymes), then falls steeply.
- Enzyme stored for longer loses activity: "active site changes shape", "more vibration between R groups".

Often asked as: calculate Q10 from a graph (up to 4 marks); describe how to find Q10; sketch rate against temperature; explain photosynthesis or development rates at different temperatures; why decomposition is faster when warm.

COMMON MISTAKE

Reading the rate at the lower temperature and **adding 10 to the rate**. You add 10 to the **temperature** and read a second rate.
"Enzymes are killed", "the active site denatures", "enzymes start to denature". The **enzyme** denatures; it was never alive.
Too many decimal places, or units on Q10.
Drawing tangents and subtracting them: Q10 uses rates read at two temperatures.

BEYOND THE SPEC, BUT EXAMINED

- Optimum temperatures differ between species (C3 v C4 plants; *Spartina* about 35 °C v *Leucopoa* about 23 °C) and this affects where they grow (WBI14 Jan 2024, Jan 2022).
- **Zone of tolerance** of a fish linked to its enzymes (WBI14 Jan 2023).
- If the two temperatures are not 10 °C apart: Q10 = $(R_2 \div R_1)^{(10 \div (T_2 - T_1))}$. Not usually needed, but useful for CP12 data.

SPEC **5.22**

Core Practical 12: temperature and development

WHAT THIS SPEC POINT ASKS

Investigate how temperature affects the development of organisms (e.g. seedling growth rate or brine shrimp hatching rate), using organisms ethically.

5.22 Why it matters

- Development depends on many enzyme-controlled reactions (respiration, protein synthesis, mitosis). Warmer conditions speed these up (to an optimum), so climate change alters how fast organisms grow and hatch (5.20, 5.21).

CORE PRACTICAL 12: METHOD IN BRIEF

Brine shrimp hatching

1. Make up salt water of the same concentration (e.g. 35 g dm^{-3}) in identical containers, same volume each.
2. Set the containers in **water baths** or incubators at a range of temperatures, e.g. 15, 20, 25, 30, 35 °C. Let them reach temperature.
3. Add the **same number of eggs** (e.g. 40, counted under a hand lens, or the same mass) to each container.
4. Keep light, salinity, pH, volume, aeration and egg batch (age) the same.
5. Count the hatched larvae at set intervals (e.g. every 12 h for 48 h). Use a lamp briefly to attract larvae for counting (they swim towards light), keeping it away from the containers between counts so it doesn't heat them.
6. Repeat at least 3 times at each temperature; calculate mean % hatched.

Seedling growth

1. Use seeds of the **same species and variety**, germinated together; choose seedlings of the same starting height.
2. Grow them in incubators at different temperatures with the same light, water, soil and humidity.
3. Measure the height (or dry mass of a sample) every day for, e.g., 10 days.
4. Growth rate = mean change in height ÷ number of days.

5.22 Results and processing

WORKED EXAMPLE: BRINE SHRIMP HATCH RATE

40 eggs per container, counted after 48 h (mean of 3 repeats):

Temperature / °C	15	20	25	30	35
Mean number hatched	6	13	25	31	18
% hatched	15	33	63	78	45
Hatch rate / larvae h^{-1}	0.13	0.27	0.52	0.65	0.38

% hatched at 25 °C = 25 ÷ 40 × 100 = **63%** (62.5 rounded). Rate = 25 ÷ 48 = **0.52 larvae h^{-1}**.
Q10 between 15 and 25 °C = 25 ÷ 6 = **4.2** (use the unrounded counts: the time is the same, so it cancels).
Fewer hatch at 35 °C: enzymes begin to denature, and warm water holds **less dissolved oxygen**.

5.22 Variables, evaluation, ethics

- IV: temperature. DV: number hatched per unit time, or growth in height per day. Controls: listed in the method.
- Errors: temperature drift in water baths (check with a thermometer); eggs are hard to count accurately (use mass and a calibration); hatching is hard to judge; seedlings differ genetically. Repeats, means and **standard deviation** (error bars) show whether differences are significant.
- **Ethics**: invertebrates and plants raise fewer ethical concerns than vertebrates. Use the **smallest number** that gives reliable results; avoid temperatures known to kill; handle gently; dispose of or release organisms responsibly (don't release brine shrimp into local waters).

MARK-SCHEME LANGUAGE

- Choosing organisms for research (nematodes): "lower-order organisms / fewer ethical issues", "**short life cycle**, so several generations can be studied", "lay **large numbers of eggs**, so large sample size / repeatability", "live in a range of environments, so controls are possible".
- Design answers: name the IV and its range, how the DV is measured, at least three controlled variables and **how** each is controlled, repeats and a mean.
- A "range of sizes" (or temperatures) must be **used** when that is the independent variable, not "controlled".

Often asked as: the method itself hasn't been set in WBI14 2020–2026. Related questions: development or size against temperature in insects and lizards, fertility against temperature, why a species suits laboratory research, designing a preference investigation.

COMMON MISTAKE

Controlling the variable the question asks you to vary.
Ignoring dissolved oxygen: higher temperature lowers it, which also affects hatching.

BEYOND THE SPEC, BUT EXAMINED

- Q10 as the natural way to process development data (WBI14 Oct 2023, Oct 2025).
- The ethical case for invertebrates and their practical advantages (WBI14 Oct 2025).

SPEC 5.23 Evolution by mutation and natural selection

WHAT THIS SPEC POINT ASKS

Understand how evolution (a change in allele frequency) can happen through gene mutation and natural selection.

5.23 Evolution is a change in allele frequency

- A **gene** is a length of DNA coding for a polypeptide; **alleles** are different versions of the gene.
- The **gene pool** is all the alleles in a population. **Allele frequency** is how common one allele is in the gene pool.
- **Evolution** is a **change in allele frequency** in a population over generations.

5.23 Where new alleles come from

- **Gene mutation**: a change in the DNA base sequence (substitution, insertion, deletion). Most are neutral or harmful; a few give an advantage in particular conditions.
- Mutations are **random**: they happen whether or not they are useful. They must occur in cells that make **gametes** (or in a bacterium itself) to be inherited.
- Meiosis and random fertilisation shuffle alleles into new combinations, adding variation.

5.23 Natural selection: the five-step chain

1. **Mutation** produces a **new allele**, so there is variation in the population.
2. A change in the environment (a new chemical, predator, disease, food source, climate) acts as a **selection pressure**.
3. Individuals with the advantageous allele are **more likely to survive and reproduce**.
4. They **pass the allele on to their offspring**.

5. Over many generations, the **frequency of the allele increases** in the population.

WORKED EXAMPLE: ANTIBIOTIC RESISTANCE

A mutation in one bacterium's DNA produces an allele for an enzyme that breaks down an antibiotic. The antibiotic is a **selection pressure**: susceptible bacteria die, the resistant one survives and divides by **binary fission**, forming a **clone** of resistant bacteria. The resistance allele can also be passed to other bacteria on a **plasmid** (conjugation). The frequency of the resistance allele rises.

- Resistance spreads fastest in organisms with **short life cycles**, **high mutation rates** and **many offspring**: bacteria, insects, viruses.
- Many features (size, horn length, neck length) are **polygenic**: controlled by many genes, so they show **continuous variation**, and selection shifts the mean.

EXAM TIP

Write the chain **in the context** of the question, naming the organism and the selection pressure. Avoid purpose words: organisms don't mutate "in order to" survive. The mutation was already there before the selection pressure arrived.

MARK-SCHEME LANGUAGE

- "**Mutation** (in the DNA)" produces a new **allele**; "acts as a **selection pressure**"; "**survive and reproduce**"; "**pass the allele on** to offspring"; "**increasing the allele frequency**".
- Bacteria: "by asexual reproduction / binary fission / conjugation / plasmids", "forming a clone of resistant bacteria".
- Polygenic: "variation in (size) / polygenic", "continuous variation".
- **Not accepted**: "the selection pressure causes the mutation", "gene" used for "allele", "immune" for "resistant".

Often asked as: explain how a trait evolved, in almost every paper (microbes digesting plastic, insecticide resistance, antibiotic resistance, squirrel pouches, apple maggot flies, divers' larger spleens); why short life cycles speed it up; 6-marks combining data with natural selection.

COMMON MISTAKE

"The pesticide caused the insects to mutate." The mutation happens **randomly first**; the pesticide only selects.
Generic answers that never mention the context.
Saying bacteria become "immune". They become **resistant**.
Mixing up gene and allele.

BEYOND THE SPEC, BUT EXAMINED

- Short life cycle, high mutation rate and many offspring explain fast evolution of resistance (WBI14 Jan 2021).
- Horizontal transfer of resistance plasmids between bacteria (WBI14 Oct 2020, Jun 2024).
- Evolution of a human population (larger spleens in divers) in a 6-mark (WBI14 Jun 2024).

SPEC 5.24 Isolation and speciation

WHAT THIS SPEC POINT ASKS

Understand how isolation reduces gene flow between populations, leading to allopatric or sympatric speciation.

5.24 Species and gene flow

- A **species** is a group of organisms that can **interbreed to produce fertile offspring**.
- **Gene flow** is the movement of alleles between populations when individuals interbreed. It keeps populations genetically similar.
- **Speciation** is the formation of a new species. It needs **reproductive isolation**: gene flow between two populations stops.

5.24 How speciation happens

1. A population is split into groups that **can't interbreed** (**isolation**), so **gene flow stops**.
2. Each group has different **mutations**, and faces different **selection pressures** (and chance effects in small populations: genetic drift).
3. **Allele frequencies change differently** in each group.
4. Over many generations the groups become so different that, even if they meet again, they **can't interbreed to produce fertile offspring**: they are separate species.

5.24 Two types

	Allopatric	Sympatric
Where	populations in **different places**	populations in the **same place**
Isolation by	a **geographical barrier**: sea, mountain, river, road, habitat fragmentation	a **reproductive barrier** with no physical separation
Example	lemurs in Madagascar isolated from ancestors in Africa	apple maggot flies switching to a new host plant; species with different diets or breeding times in one habitat; polyploid plants

5.24 Reproductive isolating mechanisms (often the cause in sympatric speciation)

- **Temporal**: breed or are active at **different times** of year or day.
- **Behavioural**: different **courtship** displays, songs or calls.
- **Mechanical**: incompatible **genitalia** or flower structure.
- **Ecological / habitat**: use different parts of the habitat or different host plants.
- **Gametic**: sperm can't fertilise the egg (incompatible gametes).
- After mating: hybrids die early or are **sterile** (e.g. mules).

MARK-SCHEME LANGUAGE

- Sympatric speciation: "formation of **one new** species (from a pre-existing one)" "while both continue to live in the **same location**" (or "without a physical barrier"). **Not** "two new species" or "subspecies".
- Allopatric: "isolated (from a common ancestor) by a geographical barrier".
- Habitat fragmentation: "genetic diversity decreases because **gene flow is reduced**", "inbreeding".
- Wildlife corridors: "populations can mix / mate", "increases the gene pool / gene flow", "prevents inbreeding", "prevents extinction".
- Isolation examples: "eggs laid at different times of year", "different breeding behaviours", "incompatible gametes / genitalia".

Often asked as: which type of speciation happened (lemurs); define sympatric speciation; fragmentation and genetic diversity; a corridor between tiger populations; isolating mechanisms between two fly species; endemic species.

COMMON MISTAKE

"One species becomes two new species." The original species usually continues; **one** new species forms.
"Not due to a geographical barrier" instead of the positive "in the **same area**".
Writing a full speciation account when the question only asks about genetic diversity in one species.
Describing allopatric speciation when the context is clearly sympatric (or the reverse).

BEYOND THE SPEC, BUT EXAMINED

- **Endemic**: found only in one place (e.g. one country), not "its habitat" (WBI14 Jan 2023).
- Gene pool, gene flow and allele combinations in a wildlife-corridor question (WBI14 Jun 2025).
- Learned behaviour (birdsong) lost when a population becomes too small (WBI14 Jun 2022).
- **Polyploidy** in plants: an error in meiosis doubles the chromosome number; the polyploid can't make fertile offspring with the parent species, so a new species forms in one generation (a classic sympatric example).

SPEC 5.25 Who reaches the conclusion: controversial issues

WHAT THIS SPEC POINT ASKS

Understand that scientific conclusions about controversial issues (such as how much humans affect climate, and what should be done) can depend on who is reaching them.

5.25 Why people reach different conclusions

- **Data**: has the whole data set been used, or only selected parts? Is the sample big enough? Has it been **repeated** by others? Is the result **statistically significant**? Is it a correlation being presented as a cause?
- **Source**: who **funded** the work, and who benefits from the conclusion? Fossil-fuel companies, governments facing elections, campaign groups and businesses may each have **financial, political or commercial interests**.
- **Publication**: has it been **peer reviewed** in a scientific journal, or is it a press release, blog or advert?
- **Expertise**: is the author qualified in this field?
- **Complexity**: climate has natural variation too; a single hot summer or a single fire can't be blamed on climate change alone.

- **Values**: people weigh costs and benefits differently (jobs v environment), and education, media, religion and trust in scientists all affect what people accept.

5.25 The scientific consensus

- The large majority of climate scientists, through bodies such as the IPCC, conclude that the climate is warming and **human greenhouse gas emissions are the main cause**. Disagreement is mainly about **how fast**, **how much**, and **what to do**.

EXAM TIP

"Discuss" or "evaluate" questions need **both sides**. A good structure: claim → evidence for → evidence against or limitation → who is making the claim and why it might matter → a balanced conclusion.

MARK-SCHEME LANGUAGE

- "There is no evidence that climate change actually caused the forest fires / it is **only a correlation**."
- "Certain {parties / politicians / business people} have different opinions / want to continue their activities."
- "Forest fires have other causes / other (non-human) factors cause climate change"; "lack of education"; "lack of trust in scientists".
- Evaluating a proposal (e.g. fuel from faeces): at least one **advantage** and one **disadvantage**, each explained.
- Local people and conservation plans: they "rely on the land" and "need to understand the issues / feel their livelihoods are safe".

Often asked as: why not everyone accepts that climate change caused something; trust in scientists; 6-mark evaluation of a new technology; local people's views on reforestation; conflicting land uses.

COMMON MISTAKE

Giving one reason when the question has two marks.
One-sided answers to "discuss": examiners say most answers give only the positives.

BEYOND THE SPEC, BUT EXAMINED

- The same skill applied to other controversies: vaccine safety and trust in scientists (WBI14 Jan 2021); turning waste into fuel (WBI14 Jan 2026).

SPEC 5.26 Reforestation, sustainable resources and biofuels

WHAT THIS SPEC POINT ASKS

Understand how reforestation and the use of sustainable resources, including biofuels, manage the conflict between human needs and conservation.

5.26 The conflict

- People need land, food, fuel, timber, jobs and income. Conservation needs habitats, biodiversity and carbon stores. **Sustainable** use meets today's needs **without reducing the ability of future generations** to meet theirs.

5.26 Reforestation

For	Against / limitations
more photosynthesis, so CO_2 **removed** and carbon **stored** in wood and soil	takes decades for trees to mature; young trees store little carbon at first
restores habitats and **biodiversity**; can **join fragments** into corridors	plantations of one fast-growing species (monoculture) support little biodiversity
reduces soil erosion and flooding; protects water supplies	uses land that local people may need for farming
sustainable timber if felled trees are replaced	a planted forest is not the same as old-growth forest

- **Protecting existing forest** matters more than replanting: destroying it releases stored carbon, and established native forest is more resistant to fire, storm, drought and disease.
- Fast-growing trees have a **high NPP**, so they fix CO_2 quickly; but natural woodland stores more carbon overall and holds more species.
- Choosing species: **native** species (adapted to local conditions; non-natives may outcompete them), a **genetically diverse** mix, and endangered species where possible.
- Choosing sites: avoid destroying other ecosystems or local people's land; link fragments; make sure conditions suit the trees.

5.26 Biofuels

- Fuels made from recently grown biomass: **bioethanol** (fermenting sugar cane, maize) and **biodiesel** (plant oils, algae).

For	Against
renewable: crops can be regrown	land used for fuel isn't used for **food**: food prices rise
CO_2 released on burning was **recently absorbed** by the crop for photosynthesis (Calvin cycle), so less **net** CO_2 added than fossil fuels	forests cleared to grow biofuel crops release stored carbon and destroy habitats
reduces dependence on fossil fuels	growing, fertilising, harvesting, transporting and processing all release CO_2 (fertiliser production also releases N_2O)
algae and waste materials avoid using farmland	monocultures reduce biodiversity; pesticides harm food chains

5.26 Other sustainable resources

- **Sustainable forestry**: fell only as much as regrows; replant; selective felling.
- **Fishing quotas** and minimum mesh sizes so young fish can breed.
- Recycling (e.g. textiles) reduces the land, water and energy needed for new materials.
- Seaweed farming: fixes CO_2 and provides food and biofuel without using land.

MARK-SCHEME LANGUAGE

- Protect forests: "if existing trees are destroyed, carbon (dioxide) will be released"; "established / native, so more resistant to fire, storms, drought, disease"; "homes / food for organisms".
- Planting sites: "avoid destroying other ecosystems", "avoid destroying homes or farmland of local people", "join together fragments", "conditions at the site must be suitable".
- Species: "native species, adapted to the conditions", "avoid non-native species outcompeting", "endangered species to avoid extinction", "genetically diverse species".
- Biofuel: "plants can be regrown, removing carbon dioxide for the Calvin cycle". The phrase "carbon neutral" alone earns nothing: explain it.
- 6-mark discussions are marked by **aspect** (e.g. local people, global warming, biodiversity). Each aspect needs a positive **and** a negative, explained. For food production, adding **energy loss along food chains** reaches the top mark.

Often asked as: 6-marks on sustainable food production, crops v forests, seaweed farming, biodegradable packaging, recycled textiles, sea otters and salt marshes; plan a reforestation scheme; biofuel from algae or waste.

COMMON MISTAKE

Describing the graph or table only. "If you are told to use your own knowledge, then you must do so."
Positives only in a "discuss" answer.
Writing about woodland loss when the scheme is at sea (seaweed farms).
Stopping at "carbon neutral" without the photosynthesis explanation.

BEYOND THE SPEC, BUT EXAMINED

- Land use per kilogram of different foods: a plant-based diet needs less land because fewer trophic levels lose energy (WBI14 Jan 2022).
- Monocultures reduce biodiversity; pesticides and fertilisers cost money and harm food chains (WBI14 Jan 2023).
- Evaluating a conservation study: one site, no error bars, correlation not cause (sea otters and salt marshes) (WBI14 Oct 2025).
- Island restoration judged with a diversity index (WBI14 Jun 2025).

Topic 6 – Microbiology, Immunity and Forensics

SPEC 6.1 Culturing microorganisms and aseptic technique

WHAT THIS SPEC POINT ASKS

Know how microorganisms are grown in the lab (media, conditions, inoculation, incubation) and how aseptic technique keeps cultures pure and people safe.

6.1 Why we culture microorganisms

- Single bacteria are far too small to see. Put them on a nutrient medium in the right conditions and each cell divides again and again. A visible **colony** is a clone of millions of cells grown from one original cell.
- Culturing lets you count microorganisms, identify them, test antibiotics on them, and study how fast they grow.

6.1 What a culture needs

Need	How it is supplied
Nutrients	a **culture medium** with a carbon source (e.g. glucose), a nitrogen source (amino acids, peptones), minerals, vitamins and water
Liquid or solid	**nutrient broth** (liquid, in a flask or bottle) or **nutrient agar** (solid, poured into Petri dishes). Agar is a gel made from seaweed that most microorganisms can't digest.
Temperature	an **incubator** at a set temperature, so enzymes work at a steady rate and are not denatured
pH	a **buffer** in the medium
Oxygen	aerobes need air; anaerobes need it excluded

- **Selective media** contain substances that let only one type of microorganism grow (e.g. an antibiotic, to pick out resistant bacteria). **Indicator media** change colour around colonies that have a particular feature, which helps identify them.

6.1 Aseptic technique: every step with its reason

Aseptic technique = methods that prevent contamination of the culture by other microorganisms, and of the people and lab by the culture.

Step	Reason
Wash hands; wipe the bench with **disinfectant** before and after	kills microorganisms on hands and surfaces
Sterilise media and glassware in an **autoclave** (121 °C, high-pressure steam, about 15 minutes)	kills all bacteria and spores before use
Work next to a lit **Bunsen burner**	the hot air rising makes an **updraft** that carries airborne microorganisms away from the culture
Heat the **inoculating loop** in the flame until red hot, then let it cool	kills microorganisms on the loop; cooling stops it killing the culture you pick up
Flame the neck of bottles and flasks when you open and close them	kills microorganisms at the opening; the updraft stops air entering
Lift the Petri dish lid only a little, for as short a time as possible	reduces exposure to microorganisms in the air
Close doors and windows	stops draughts carrying microorganisms in
Tape the lid with a few short strips, not all the way round	the lid can't fall off, but air can still get in, so conditions stay **aerobic** and **anaerobic bacteria** (which include many pathogens) can't grow
Incubate plates **upside down**	condensation can't drip onto the agar and spread the colonies
Incubate at **25 °C or below** in school labs	discourages human pathogens, which grow best at body temperature (37 °C)
Autoclave (or soak in disinfectant) all cultures and equipment before disposal	kills the microorganisms so none are released

6.1 Inoculating

- **Broth**: dip a sterile loop into the culture, transfer it into the sterile broth, swirl, replace the lid and flame the neck. A **shaking incubator** keeps the cells evenly suspended, supplies oxygen and nutrients to all of them, and keeps the temperature even.
- **Agar plate**: spread a small measured volume (e.g. 0.1 cm^3) of liquid culture over the surface with a sterile spreader (a **spread plate**), or streak a loopful across the surface (a **streak plate**, used to get separate colonies from a colony or a thick suspension).

MARK-SCHEME LANGUAGE

- Aseptic technique: "methods used to **prevent contamination** (of the person / the culture) with other microorganisms".
- Each technique needs its **reason** too: "work beside a Bunsen burner" + "to create an **updraft** / move microorganisms in the air away"; "autoclave the agar / equipment" + "to kill any bacteria present"; "flame the necks of bottles"; "reduce exposure of cultures to air"; "close doors and windows".
- Why it matters: other microorganisms "compete for nutrients", "produce toxins" or "give an over-estimate of the number"; the culture may contain **pathogens** that could infect you.
- Sealing a dish: tape the lid "so it does not fall off / no contamination" **and** "not completely sealed, so conditions remain **aerobic** / to prevent the growth of **anaerobic** bacteria". You need both.
- Incubation: a set temperature "so that enzymes function / are not denatured"; a long time (e.g. 72 hours) "so that growth becomes **visible**".

Often asked as: define aseptic technique; describe two techniques and explain why each matters; how to seal a Petri dish; how to inoculate a broth; why use a shaking incubator; plan a culture experiment (6-mark).

COMMON MISTAKE

"To avoid contamination" on its own scores 1 mark at most. Say **what** would be contaminated (the culture, or you) and **what would go wrong** (competition, toxins, wrong counts, infection).
Listing four techniques when the question asks you to explain two. Pick two and give a reason for each.
Sealing the lid all the way round "to stop contamination". It must stay aerobic.
Incubating at 37 °C in a school lab. That is the temperature human pathogens like best.

BEYOND THE SPEC, BUT EXAMINED

- Selective and indicator media count as ways to make sure you are growing (and identifying) the right microorganism (WBI14 Jan 2023).
- Streaking is for picking colonies off a plate. Don't suggest it for sampling a liquid suspension: a mark scheme rejected it (WBI14 Jan 2026).
- Soap removes skin bacteria physically **and** its chemicals and high pH kill them; water alone only washes them off (WBI14 Jan 2024).

SPEC 6.2

Measuring the growth of microorganisms

WHAT THIS SPEC POINT ASKS

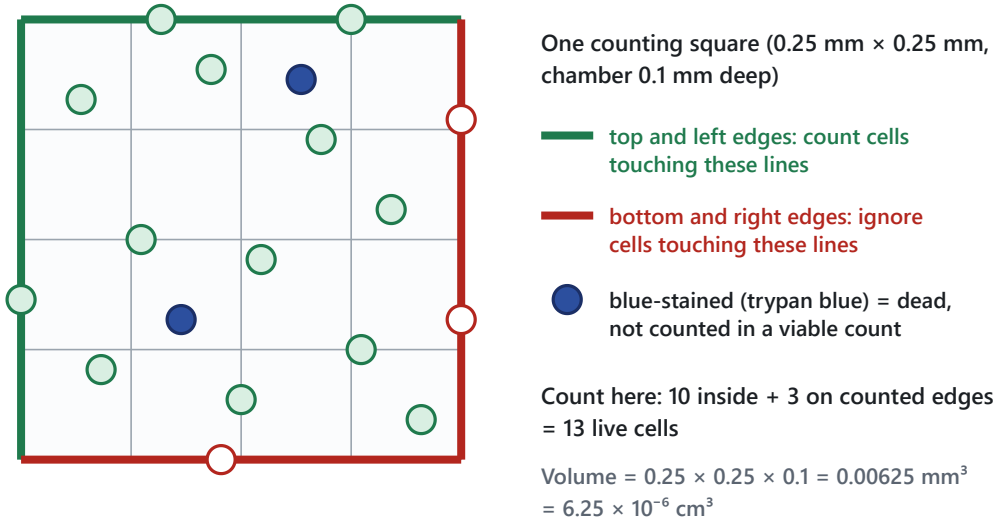
Know four ways to measure microbial growth (cell counts, dilution plating, mass, turbidity), what each one actually measures, and how to calculate numbers from them.

6.2 Overview: which method counts what?

Method	What it measures	Counts dead cells?	Notes
Cell count (haemocytometer)	total cells in a known tiny volume	yes, unless you use a stain that marks dead cells	fast; you need a microscope; the count is tedious
Dilution plating	**viable** (living) cells: each one grows into a colony	no	takes 1–3 days; assumes one colony = one cell
Mass (dry mass)	biomass	yes	used for fungi (mycelium) and large cultures; the sample is destroyed
Turbidity (optical density)	how cloudy the culture is	yes	quick and repeatable; needs a **calibration curve** to convert to numbers

6.2 Cell counts: the haemocytometer

- A haemocytometer is a thick microscope slide with a counting chamber of known depth (usually **0.1 mm**) and a grid of squares of known area etched on it. So every square holds a **known volume**.
- Mix the culture with an equal volume of **trypan blue** (an exclusion dye). Dead cells take up the dye and look blue; live cells keep it out and stay clear. Count only the clear cells to get a **viable** count. Remember that mixing 1 : 1 dilutes the sample by 2.
- Fill the chamber under the coverslip, count the cells in several squares and take the **mean**.
- **Edge rule**: count cells touching two sides of the square (e.g. top and left) and ignore those touching the other two (bottom and right). Otherwise you count the same cell twice in neighbouring squares.
- If the cells overlap or are too many to count, dilute the culture first and multiply by the dilution factor at the end.



Edge rule: cells on the top and left lines count, cells on the bottom and right lines don't. Blue (dead) cells are left out of a viable count.

WORKED EXAMPLE: HAEMOCYTOMETER COUNT

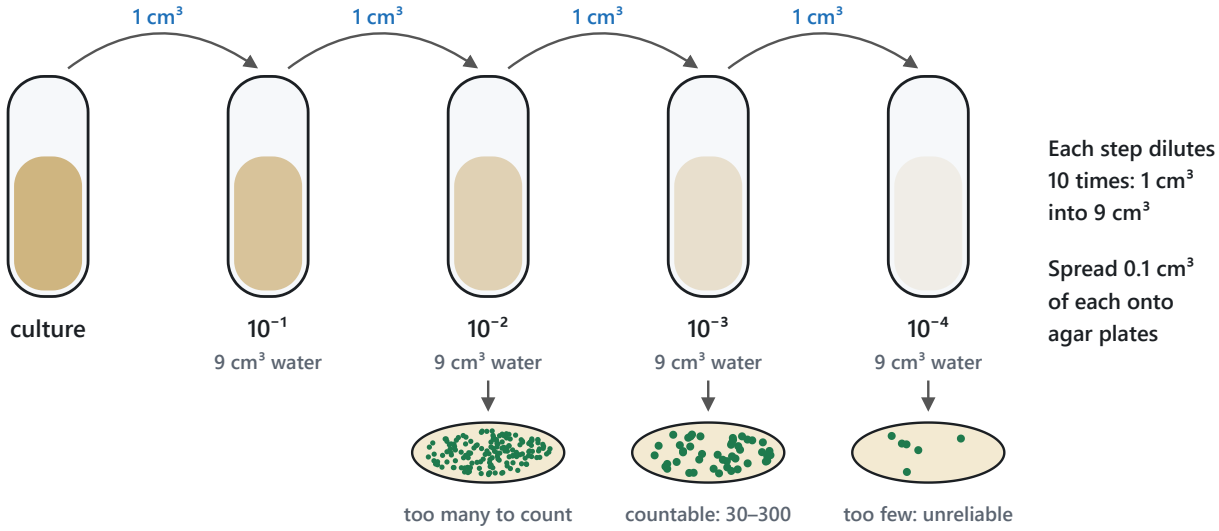
Yeast culture mixed 1 : 1 with trypan blue (dilution factor 2). Each square is 0.25 mm × 0.25 mm, chamber 0.1 mm deep. Live cells in five squares: 12, 15, 11, 14, 13.

1. Mean = 65 ÷ 5 = **13 cells per square**.
2. Volume of one square = 0.25 × 0.25 × 0.1 = 0.00625 mm^3. 1 cm^3 = 1000 mm^3, so this is **6.25×10^{-6} cm^3**.
3. Cells per cm^3 in the chamber = 13 ÷ (6.25×10^{-6}) = 2.08×10^6.
4. Undo the dilution: $2.08 \times 10^6 \times 2$ = **4.16×10^6 cells cm^{-3}**.

Shortcut for the large 1 mm × 1 mm squares: volume = 0.1 mm^3 = 10^{-4} cm^3, so cells per cm^3 = mean per square × 10^4 × dilution factor.

6.2 Dilution plating (viable counts)

- A culture contains far too many cells to give separate colonies, so it is **diluted** first, in steps (a **serial dilution**). The usual step is 1 cm^3 of culture into 9 cm^3 of sterile water: a **1 in 10** dilution. Repeat with a fresh sterile pipette each time to get 10^{-1}, 10^{-2}, 10^{-3} …
- Spread a known volume (e.g. 0.1 cm^3) of each dilution onto an agar plate, incubate, then count the colonies on a plate with a **countable** number (roughly 30–300 well separated colonies).
- Each colony grew from one living cell, so: cells per cm^3 in the original = colonies ÷ volume spread × dilution factor.
- Weaknesses: two cells close together give one colony (an underestimate); some species won't grow in the conditions used; it takes days.



A 1 in 10 serial dilution. Only the plate with a countable number of separate colonies is used.

WORKED EXAMPLE: DILUTION PLATING

0.1 cm^3 of the 10^{-3} dilution was spread on a plate. 46 colonies grew.

1. Cells per cm^3 of the 10^{-3} dilution = 46 ÷ 0.1 = 460.
2. The original is 10^3 times more concentrated: 460×10^3 = **4.6×10^5 viable cells cm^{-3}**.

Check your answer is sensible: the 10^{-2} plate should have about 10 times as many colonies (too many to count), the 10^{-4} plate about 10 times fewer.

WORKED EXAMPLE: 1 IN 2 SERIAL DILUTION

To halve a concentration each step: mix **equal volumes** of the solution and water (or broth), then repeat using the **previous** dilution each time. 1 in 2, 1 in 4, 1 in 8 … after *n* steps the dilution is 1 in 2^n (e.g. 1 in 256 after 8 steps).

6.2 Mass

- For fungi that grow as a mat of hyphae (and for large cultures), measure biomass: filter or centrifuge a sample, dry it in an oven until the mass stops changing, and weigh it (**dry mass**). The sample is killed, so take separate samples at each time.
- On agar, fungal growth can also be followed by the **diameter** (or area) of the colony.

6.2 Turbidity (optical methods)

- The more cells in a broth, the **cloudier** (more turbid) it is. A **colorimeter** shines light through a sample in a cuvette: more cells → more light scattered and absorbed → higher **absorbance** (optical density), lower transmission.
- Zero the colorimeter with sterile broth (a **blank**) first.
- Absorbance isn't a cell number. Make a **calibration curve**: measure the absorbance of cultures whose cell numbers are known (from haemocytometer counts), plot absorbance against number, and read unknown samples off the curve.
- Turbidity counts **dead cells too** (and any other particles), so it measures **total** cells, not living ones.

MARK-SCHEME LANGUAGE

- "Optical methods / turbidity measure **total** cell counts (dead and alive cells)"; "dilution plating measures **living** bacteria".
- Dilution plating: "use **serial dilutions**" **and** "sample added to agar and **spread**", then count colonies. Describe both steps.
- Haemocytometer: count cells touching two sides, ignore the others; use an **exclusion dye** (trypan blue, eosin) to count living cells only.
- Calibration curves: one curve can't be used for every organism because "some organisms are **larger** than others" and a larger cell "is more likely to be hit by light and **reflect / scatter** it"; at high concentrations the curve "**levels off**" (light scattered by several cells), so a new curve may be needed.
- Limits of culturing: "some bacteria may **not grow** in the culture conditions used"; "one colony may come from more than one bacterium".

Often asked as: name the method from a description; total v viable count on a graph; haemocytometer calculation with edge cells; explain a calibration curve (6-mark); plan how to compare the growth of two bacteria.

COMMON MISTAKE

Ignoring every cell on the edge, or counting them all. Use the two-sides rule.
Forgetting the dilution factor, or the 1 : 1 dilution with the dye.
Unit slips: 1 cm^3 = 1000 mm^3, so 1 mm^3 = 10^{-3} cm^3. Write standard form with **one** non-zero digit before the decimal point (4.6×10^5, not 46×10^4).
Describing the serial dilution and stopping. You must spread a known volume on agar, incubate and count.
The dye is **trypan** blue, not "tryptophan blue" (tryptophan is an amino acid).

BEYOND THE SPEC, BUT EXAMINED

- Exclusion dyes and the edge rule for haemocytometers (WBI14 Jan 2024).
- The reasons a calibration curve differs between species and levels off at high cell densities (WBI14 Jan 2020).
- Results expressed as **relative abundance** (% of each group of bacteria) in gut-flora studies (WBI14 Jun 2023).

SPEC 6.3

The bacterial growth curve and the growth rate constant

WHAT THIS SPEC POINT ASKS

Describe and explain the four phases of a bacterial growth curve (lag, exponential, stationary, death) and calculate the exponential growth rate constant, k.

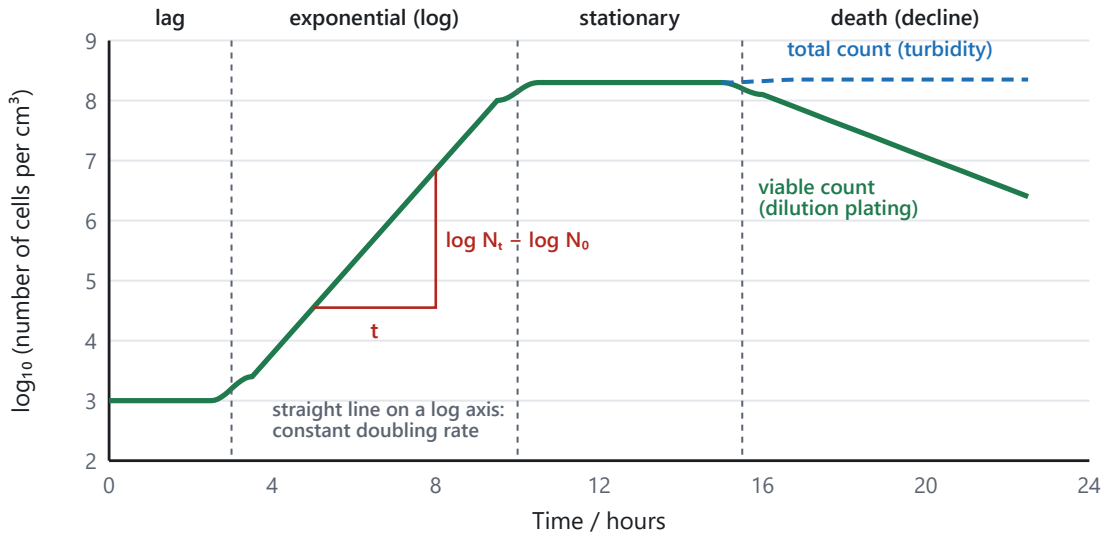
6.3 How bacteria multiply

- Bacteria divide by **binary fission**: the circular DNA is copied (plasmids too), the cell grows, and it splits into two genetically identical cells. It is **not** mitosis.
- In good conditions the number **doubles** every generation (as often as every 20 minutes for some species). Doubling gives **exponential** growth: 1 → 2 → 4 → 8 → 16 …

6.3 Why the y-axis is a log scale

- Numbers rise from thousands to billions. On an ordinary axis the early part would be squashed flat. A **$\log_{10}$** axis fits the whole range: each step of 1 is **ten times** more cells.
- On a log axis, exponential growth plots as a **straight line**. The gradient of that line tells you the growth rate.
- Values read off a log axis are **already logs**. Don't take the log of them again.

6.3 The four phases



Growth curve in a closed (batch) culture. Turbidity keeps counting dead cells, so it stays level when the viable count falls.

Phase	What happens	Why
Lag	little or no increase	cells are **adjusting** to the new medium: taking up water and nutrients, growing, and making the **enzymes** (switching on genes) needed for the new nutrients before they can divide
Exponential (log)	number doubles at a constant rate; straight line on a log axis	plenty of **nutrients** and space, little waste; nothing limits division, so births far outnumber deaths
Stationary	number stays constant	nutrients running out and toxic **waste** building up (and less oxygen); rate of new cells = rate of cell death
Death (decline)	number of living cells falls	nutrients used up and waste (e.g. acids, toxins) at toxic levels; death rate > rate of division

6.3 The growth rate constant, k

The number of cells at time t:

$N_t = N_0 \times 2^{kt}$

- N_0 = number at the start; N_t = number after time t; k = growth rate constant = **number of generations (doublings) per unit time** (e.g. h^{-1}).
- Take logs and rearrange (the paper prints this version, so you don't need to learn it):

$k = (\log_{10} N_t - \log_{10} N_0) \div (0.301 \times t)$

- 0.301 is $\log_{10} 2$. **Generation (doubling) time** = 1 ÷ k.
- Only use points from the **exponential phase**.

WORKED EXAMPLE 1: K FROM CELL NUMBERS

$N_0 = 2.0 \times 10^3$ cells cm^{-3}; after 4 hours $N_t = 6.4 \times 10^6$ cells cm^{-3}.
log N_t = 6.806; log N_0 = 3.301; difference = 3.505.
k = 3.505 ÷ (0.301 × 4) = 3.505 ÷ 1.204 = **2.91 h^{-1}** (about 3 generations per hour).
Generation time = 1 ÷ 2.91 = 0.34 h ≈ **21 minutes**.

WORKED EXAMPLE 2: K FROM A LOG GRAPH

The y-axis is $\log_{10}$ (cells cm^{-3}). In the exponential phase, the line reads 4.2 at 4 h and 7.4 at 8 h.
These are already logs: difference = 7.4 − 4.2 = 3.2. Time = 8 − 4 = 4 h.
k = 3.2 ÷ (0.301 × 4) = **2.66 h^{-1}**.
Say where the numbers came from: "two points read from the straight (exponential) part of the line".

WORKED EXAMPLE 3: REARRANGING FOR TIME

$N_0 = 5.0 \times 10^2$, k = 1.5 h^{-1}. How long to reach $N_t = 1.0 \times 10^6$?
t = (log N_t − log N_0) ÷ (0.301 × k) = (6.000 − 2.699) ÷ (0.301 × 1.5) = 3.301 ÷ 0.4515 = **7.31 hours**.
Prediction the other way: N_0 = 100, k = 2 h^{-1}, t = 3 h gives $N_t = 100 \times 2^6$ = **6400** cells.

MARK-SCHEME LANGUAGE

- Lag: cells "adjust to the new environment / **synthesise (new) enzymes**". Exponential: "conditions suitable / **nutrients available**", so cells can divide. Stationary: "**birth rate equals death rate**" / "number of new cells equals cells dying". Death: "nutrients run out", "**toxins** build up".
- Drawing the curve: both axes labelled ("log (number of living) cells" and "time / **hours**", with units), correct shape (flat lag, **straight** exponential section), all four phases labelled.
- Finding k from a graph: show **where the log values come from**, use the **exponential** phase, subtract the times to get t, then use the equation.
- Calculations: give the answer to the stated decimal places. "Taking the log of logs" loses the mark (error carried forward only).

Often asked as: order the phases (MCQ); explain the lag and exponential phases; sketch and label the curve; calculate k or t; explain why the y-axis is $\log_{10}$; link a product (e.g. a pigment) to the phases.

COMMON MISTAKE

Logging values that came off a log axis. If the axis says $\log_{10}$, the numbers are already logs.
Missing the time unit on the x-axis, or drawing a curved exponential phase on a log graph.
Explaining the lag phase but not saying **why** cells can divide quickly in the exponential phase (nutrients available).
Saying bacteria divide by mitosis. It's **binary fission**.
Reading the y-axis without checking the heading: "$\times 10^6$" or "$\log_{10}$" changes the number.

BEYOND THE SPEC, BUT EXAMINED

- A product made by the cells (e.g. a pigment) lags behind cell numbers: none in the lag phase while the enzymes are made, a rise in the exponential phase, levelling off in the stationary phase (WBI14 Jun 2025).
- Changing pH or temperature could raise the growth rate either way if the current conditions are not optimal (WBI14 Jun 2022).

SPEC 6.4 Core Practical 13: rate of growth of microorganisms in a liquid culture

WHAT THIS SPEC POINT ASKS

Investigate how fast a microorganism grows in a liquid culture, using safe, aseptic methods and treating the organisms ethically.

6.4 Principle

Yeast (a single-celled fungus) grows in a sugar broth. As the cells multiply, the broth gets cloudier. Measure the **absorbance** (turbidity) at regular intervals, convert to cell numbers with a calibration curve (or count samples directly), and plot a growth curve.

CORE PRACTICAL 13: METHOD IN BRIEF

1. Disinfect the bench. Work next to a lit Bunsen burner.
2. Put a measured volume of sterile glucose broth (e.g. 250 cm^3) in a sterile conical flask.
3. Add a measured mass of dried yeast (e.g. 1 g) using aseptic technique. Plug the flask with **cotton wool** (keeps microorganisms out but lets air in) and cover loosely with foil.

4. Keep it at a constant temperature (e.g. 20–25 °C water bath or incubator), stirred or shaken so the cells stay suspended and get oxygen.
5. Zero the colorimeter with sterile broth (the **blank**).
6. At time 0, and then at regular intervals (e.g. every 30 minutes at first, then every few hours, for 24–48 hours), aseptically remove a sample, mix it, put it in a cuvette and read the absorbance.
7. Optional check: count cells in a sample with a haemocytometer (or plate a dilution) at some time points, to build the calibration curve.
8. Plot absorbance (or $\log_{10}$ cell number) against time.
   Extension: repeat at different temperatures, glucose concentrations or pH values to compare k.

6.4 Variables

Independent	time (or, in an extension, temperature / glucose concentration)
Dependent	absorbance (turbidity), converted to number of cells
Control	temperature, volume and concentration of glucose broth, mass of yeast, stirring speed, colorimeter wavelength/filter, same cuvette, time samples are left before reading

6.4 Processing the results

- Convert absorbances to cell numbers with the calibration curve.
- Plot **$\log_{10}$ (cell number)** against time. Identify the phases. Pick two points on the straight part and calculate **k** (see 6.3).
- Comparing conditions: compare k values, or the time to reach the stationary phase. Repeat each condition (at least three cultures) and use means.

6.4 Safety and ethics

- Use a **non-pathogenic** organism (baker's yeast), keep incubation at **25 °C or below** (not 37 °C), and use aseptic technique so the culture isn't contaminated with something harmful.
- Wear eye protection; take care with the Bunsen; wash hands.
- **Ethical use**: grow only as much culture as you need, and kill all cultures (autoclave or disinfectant) before disposal so nothing is released into the environment.

6.4 Sources of error and improvements

Problem	Fix
Cells settle in the cuvette, so readings fall	mix the sample just before reading; read quickly
Turbidity includes **dead** cells and debris	back up with a viable count (dilution plating or trypan blue)
Fingerprints or scratches on the cuvette	handle by the ridged sides; use the same cuvette
Contamination by other microbes	strict aseptic technique; a flask of sterile broth (no yeast) as a control
Colorimeter drift	re-zero with the blank before each reading

MARK-SCHEME LANGUAGE

- Comparing the growth of two bacteria: "both cultured in the **same conditions**", "aseptic technique", "**calibration curve** used to determine concentration", "describe how growth rate is determined" (or a named statistics test).
- Shaking: "keep bacteria **in suspension**", "supply of **oxygen / nutrients**", "keep temperature constant".
- Results language from 6.2 and 6.3 applies: total v viable counts, log graphs, k from the exponential phase.

Often asked as: the method itself has not been set as a full question (2020–2026); instead you get data from liquid cultures (optical density, samples plated at intervals, pH and temperature effects) and planning questions that need this method.

COMMON MISTAKE

Saying "use a control" without saying what it is. Here: a flask of sterile broth with no yeast, treated the same way, to show any change is caused by the yeast.
Measuring at irregular intervals, or too rarely in the first few hours, so the lag and exponential phases are missed.

BEYOND THE SPEC, BUT EXAMINED

- Serial 1 in 2 dilutions: "equal volumes of the substance and water / broth", "repeated using the **previous** solution" (WBI14 Oct 2021).

SPEC 6.5

Bacteria and viruses: structure, the lytic cycle and latency

WHAT THIS SPEC POINT ASKS

(i) Compare the structure of bacteria and viruses (nucleic acid, capsid, envelope), using Ebola, TMV, HIV and λ phage as examples. (ii) Know what the lytic cycle and latency mean.

6.5 Bacteria compared with viruses

Feature	Bacteria	Viruses
Cellular?	yes: prokaryotic cells	**no**: particles, not cells
Size	about 0.5–5 μm	about 20–300 nm (Ebola threads can be longer)
Genetic material	**DNA** only: one circular chromosome, free in the cytoplasm (no nucleus, no histones), often plus **plasmids**	**DNA or RNA** (never both), single- or double-stranded
Outer layer	cell surface membrane inside a **peptidoglycan (murein)** cell wall; some have a capsule	protein coat: the **capsid**, made of protein units (**capsomeres**); some have a lipid **envelope** outside it
Ribosomes / metabolism	**70S** ribosomes; own enzymes and respiration	none: they use the host cell's ribosomes, enzymes and ATP
Other structures	flagella, pili	attachment proteins (often glycoproteins in the envelope); some carry enzymes (e.g. reverse transcriptase)
Reproduction	binary fission	only inside a living host cell
Antibiotics?	affected	**not** affected

6.5 The four named viruses

Virus	Nucleic acid	Capsid	Envelope?	Host
λ (lambda) phage	**DNA** (double-stranded)	**complex**: polyhedral head plus a tail with tail fibres	**no**	bacteria (*E. coli*)
TMV (tobacco mosaic virus)	RNA (single-stranded)	**helical**: a rigid rod	**no**	plants (tobacco, tomato)
Ebola	RNA (single-stranded)	**helical**, inside a long thread-shaped particle	**yes**, with glycoprotein spikes	humans and other primates
HIV	RNA (two single strands); a **retrovirus** with **reverse transcriptase**	polyhedral (Edexcel papers call it icosahedral; in reality it is cone-shaped)	**yes**, with glycoproteins (gp120)	humans (T helper cells)

Memory aid: only **λ phage** has DNA; only λ phage has a **complex** capsid; the **helical** pair is TMV and Ebola; the **enveloped** pair is Ebola and HIV.

6.5 How viruses get in and reproduce

- **Attachment**: proteins on the virus bind to complementary **receptors** on the host cell surface (λ phage's tail protein binds a sugar-transport protein, maltoporin, on *E. coli*; HIV's gp120 binds **CD4** on T helper cells).
- **Entry**: phages inject only their nucleic acid; enveloped viruses fuse with the membrane or are taken in by endocytosis; plant viruses get in through wounds (often made by insects).

- **Replication**: the host's enzymes, ribosomes, nucleotides, amino acids and ATP make new viral nucleic acid and proteins. Then new particles are **assembled**.
- **Release**: by bursting the cell (lysis) or by **budding** out through the membrane (taking a piece as an envelope).

6.5 The lytic cycle and latency

- **Lytic cycle**: the virus replicates straight away. New virus particles are made and assembled, then the host cell **bursts (lyses)** and releases them to infect other cells. In λ phage, proteins called lysins break down the peptidoglycan wall so the cell bursts.
- **Latency**: the viral nucleic acid stays in the host cell **without** making new virus particles, so there are no symptoms. It can stay hidden for a long time, then enter the lytic cycle later (often when the host is stressed or weakened).
  - **λ phage**: its DNA is inserted into the bacterial chromosome (it is now a **prophage**; this is the **lysogenic** pathway). It is copied every time the bacterium divides, so all daughter cells carry it.
  - **HIV**: reverse transcriptase makes a DNA copy of the RNA; integrase inserts it into the T helper cell's DNA as a **provirus**. It can stay latent for years.

MARK-SCHEME LANGUAGE

- MCQ facts: Ebola = RNA, helical capsid, envelope; HIV = RNA, polyhedral capsid, envelope, reverse transcriptase; TMV = RNA, helical, no envelope, plants; λ phage = DNA, complex capsid, no envelope, bacteria.
- Viral genome: "**genetic information / code for (viral) proteins**", with a named protein if one is given.
- Attachment: glycoproteins "attach to the host cell by binding to **specific receptors**". λ phage: "attaches to the maltoporin so that the **DNA** can enter the cell".
- Lysis: lysins "break bonds in the **peptidoglycan / murein**" so the cell bursts and the particles are released.
- Provirus to lytic: drugs don't remove all the virus, so the "**provirus is activated**" and "new viral particles burst out of the T helper cells".

Often asked as: which-virus MCQs (most papers); label or explain the parts of an unfamiliar virus (6 marks on Ebola); phage therapy graph (6-mark); fungi v bacteria v plants v animals.

COMMON MISTAKE

Treating every virus like HIV. **Only retroviruses** (HIV) have reverse transcriptase and make a provirus. Ebola and TMV are RNA viruses that copy RNA directly (Ebola brings its own RNA polymerase, because host cells don't have one that copies RNA). λ phage is a DNA virus.
"Viruses are killed / die." They aren't alive in the usual sense: say they are **destroyed** or **inactivated**.
Viruses and bacteria "divide by mitosis". Bacteria use binary fission; viruses are assembled.
"The DNA is transcribed and translated." DNA is **transcribed**; mRNA is **translated**.
Bacterial DNA as a "single strand". It is one circular **double-stranded** molecule.

BEYOND THE SPEC, BUT EXAMINED

- How holins and lysins burst a bacterium: holins form a hydrophilic channel through the membrane so lysins reach the wall (WBI14 Oct 2020).
- Positive- and negative-strand RNA viruses (rabies): the positive strand has the codons for the proteins and is the template for new negative strands (WBI14 Oct 2021).
- Phage therapy graph: bacteria rise at first (the lytic cycle takes time), then fall as cells burst; phage numbers rise after the bacteria, and the pattern is cyclical because not all bacteria are infected (WBI14 Jun 2021).
- Fungi: not plants (glycogen, chitin wall), not animals (cell wall), not bacteria (nucleus, mitochondria) (WBI14 Oct 2020).
- Protease inhibitors stop HIV's protease cutting a long viral protein into parts, so new particles can't be assembled (WBI14 Jan 2023).

SPEC 6.6

How TB and HIV infect human cells and cause death

WHAT THIS SPEC POINT ASKS

Know how Mycobacterium tuberculosis and HIV infect human cells and cause the symptoms that can lead to death.

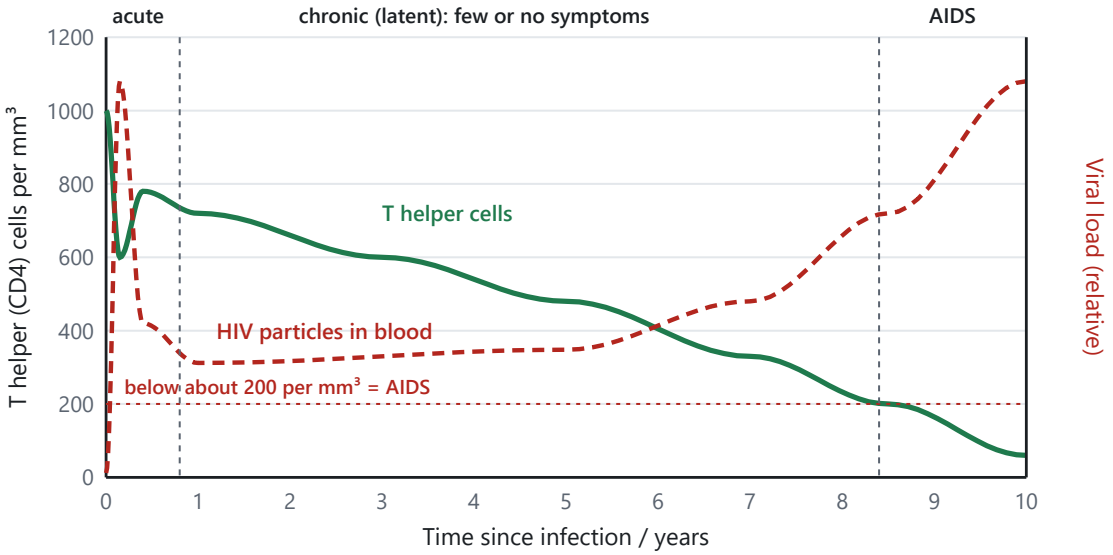
6.6 Tuberculosis (TB)

- **Pathogen**: the bacterium *Mycobacterium tuberculosis*. It has a thick, **waxy** cell wall that resists digestion and drying.
- **Spread**: inhaled in **droplets** from an infected person's coughs and sneezes. Overcrowding, poor ventilation, poor nutrition and HIV infection all raise the risk.
- **Infection**:
  1. Bacteria reach the alveoli and are engulfed by **macrophages**.
  2. They **survive inside the macrophages**: they stop the lysosomes fusing with the phagocytic vacuole, so they aren't digested. They multiply slowly.
  3. In most healthy people the immune system walls off the infected macrophages in a mass of cells called a **tubercle** (granuloma). The bacteria stay **dormant** (latent TB): no symptoms, not infectious.
  4. If the immune system weakens (HIV, malnutrition, old age, drugs), the bacteria become active, multiply and break out (**active TB**).
- **Symptoms**: persistent cough (sometimes with blood), fever, night sweats, weight loss, tiredness.
- **Why it can kill**: the bacteria and the inflammation they cause **destroy lung tissue**, so the surface area for gas exchange falls → not enough oxygen for aerobic respiration → **respiratory failure**. Bacteria can spread in the blood to other organs (**organ failure**).
- **Treatment**: several antibiotics together, for **months**: the bacteria grow slowly and hide in tubercles and macrophages, and using several drugs means a mutant resistant to one is killed by another.

6.6 HIV (human immunodeficiency virus)

- **Spread**: in body fluids: unprotected sex, shared needles, infected blood or blood products, and from mother to child (across the placenta, at birth, or in breast milk).
- **Infection and replication**:
  1. **gp120** on the envelope binds to the **CD4 receptor** on a **T helper cell** (also on macrophages).

2. The envelope fuses with the cell membrane; the capsid enters and releases the two RNA strands and the enzymes.
3. **Reverse transcriptase** makes a single DNA strand complementary to the viral RNA, then a double-stranded DNA copy.
4. **Integrase** inserts this DNA into the host's DNA as a **provirus**. It may stay latent for years.
5. When active, the provirus is transcribed; the host's ribosomes make viral proteins; new particles are assembled.
6. New HIV particles **bud** out of the cell, taking some of its membrane as their envelope. The T helper cell is destroyed (by the virus, or by T killer cells that recognise it as infected).



Typical course without treatment. After the first flu-like illness, T helper numbers fall slowly over years until opportunistic infections take hold.

6.6 How HIV leads to AIDS and death

- **Acute stage** (first weeks): flu-like symptoms; viral numbers rise sharply, T helper count drops, then partly recovers as the immune system responds.
- **Chronic (latent) stage**: few or no symptoms, often for **years**, while T helper numbers fall steadily.
- **AIDS** (acquired immune deficiency syndrome): T helper count falls very low (below about 200 per mm^3). Without T helper cells, **B cells aren't activated** (few plasma cells, few antibodies) and **T killer cells aren't activated**. The immune system can't fight infections.
- Death comes from **opportunistic infections** (e.g. TB, pneumonia) and some cancers that a healthy immune system would control. HIV does not kill directly.
- How fast this happens depends on the strain, other infections, age, nutrition and access to **antiretroviral drugs**.

6.6 Why TB and HIV go together

HIV destroys T helper cells, so fewer B cells are activated → fewer antibodies → bacteria aren't **opsonised** (marked for phagocytes) → macrophages destroy fewer *Mycobacterium*. Dormant TB becomes active, and new infections take hold. TB is the leading cause of death in people with HIV.

MARK-SCHEME LANGUAGE

- HIV: "**infects / targets T helper cells**" (ACCEPT CD4 cells); "number of T helper cells reduced"; "without T helper cells the humoral response is **not initiated**"; "fewer antibodies, so Mtb are **not opsonised**"; "phagocytosis less effective".
- Drugs: "if **reverse transcriptase** is inhibited, a **DNA copy** of the viral RNA can't be made"; "if **integrase** is inhibited, the DNA can't be **inserted into the host DNA** / no provirus". Link each effect to its enzyme.
- Combination therapy: "the virus may have **mutated** and become resistant to one drug"; other drugs act at other stages.
- TB death: "**tubercles** in lungs", "destroys lung tissue, so **reduced gas exchange**", "insufficient oxygen for aerobic respiration", "organ failure", "immune system weakened so other infections".

Often asked as: why HIV is an "immunodeficiency" virus; how HIV drugs work; why people with HIV get TB (4 marks); why TB kills; reading graphs of HIV infections and deaths.

COMMON MISTAKE

"HIV virus" (V already means virus). HIV is the **virus**; AIDS is the **syndrome**.
"RNA is converted into DNA." Reverse transcriptase **makes a DNA copy** using the RNA as a template.
"*M. tuberculosis* targets T cells." It infects **macrophages**; HIV is the one that targets T helper cells.
Writing a long account of HIV replication when the question asks why it causes immunodeficiency. Answer the question: fewer T helper cells → B cells and T killer cells not activated → no effective response.
TB "tubercules": the word is **tubercles**.

BEYOND THE SPEC, BUT EXAMINED

- Tumour necrosis factor (TNF) drives inflammation and helps macrophages; anti-TNF drugs raise the risk of TB because fewer bacteria are engulfed and destroyed (WBI14 Oct 2020).
- Drugs that stop DNA synthesis stop mitosis, so there is **no clonal expansion** of T and B cells (WBI14 Jun 2021).
- Why HIV numbers rise when new infections fall: **better treatment** means infected people survive longer (WBI14 Jun 2022).
- TB case numbers are underestimates: not everyone is tested, false negatives, people hide their HIV status (WBI14 Jan 2024).

SPEC 6.7 Routes of entry and the body's barriers

WHAT THIS SPEC POINT ASKS

(i) Know the main ways pathogens get into the body. (ii) Understand how barriers (skin, stomach acid, gut and skin flora) protect against infection.

6.7 Routes of entry

Route	How	Examples
Respiratory system (inhalation)	droplets from coughs and sneezes	TB, influenza, measles
Digestive system (ingestion)	contaminated food and water	cholera, *Salmonella*
Skin (inoculation)	cuts, wounds, needles, animal bites	tetanus, hepatitis B, rabies
Vectors	an animal (often an insect) carries the pathogen into the blood	malaria (mosquito)
Direct contact / body fluids	sex, blood, blood transfusions	HIV, Ebola
Indirect contact	touching contaminated surfaces (fomites) then the mouth, nose or eyes	many bacteria and viruses
Other openings	urinogenital tract, eyes, ears	

6.7 Barriers

Barrier	How it protects
Skin	a tough outer layer of dead, **keratinised** cells that pathogens can't easily cross. **Sebum** (oil) contains fatty acids that slow microbial growth. Cuts are sealed quickly by **blood clots**.
Stomach acid	**hydrochloric acid** (pH about 1–2) denatures the enzymes of most microorganisms swallowed in food and drink, killing them
Skin flora and gut flora	harmless (**commensal**) bacteria that live on the skin and in the gut. They **compete** with pathogens for **space** (attachment sites) and **nutrients**, and some make substances (e.g. lactic acid) that stop pathogens growing. Pathogens can't easily colonise.
Lysozyme	an enzyme in tears, saliva and mucus that hydrolyses bacterial cell walls, so bacteria burst
Mucus and cilia	mucus in the airways traps microorganisms; cilia sweep it up to the throat, where it is swallowed
Others	ear wax, nasal hairs, eyelashes, coughing and sneezing reflexes

- Broad-spectrum antibiotics kill gut flora as well as pathogens. With less competition, harmful bacteria (e.g. *Clostridium difficile*) can multiply.

MARK-SCHEME LANGUAGE

- Routes: "respiratory system / inhalation", "urinogenital tract / sexual transmission", "damaged skin / open wound", "vector / injections / bites / transfusions", "eyes", "ears".
- Barriers: "skin / keratin / sebum", "skin flora", "gut flora", "stomach acid / HCl", "mucus / cilia", "blood clots", "lysozyme in tears / saliva". "Immune system" is ignored as a barrier.
- Flora: "**compete with pathogens** for {space / nutrients / a named nutrient}"; "produce toxins / lactic acid"; "prevent **colonisation** by pathogens". Write **nutrients**, never "food".
- Antibiotics and gut flora: "antibiotics are **not specific** to one type of bacteria".
- Fairness in flora studies: sample "the same part of the body", "same washing regime", "same environment", people "not taking antibiotics".

Often asked as: two barriers; how skin flora protect; why antibiotics upset the gut flora; conclusions and design points from skin or gut bacteria data.

COMMON MISTAKE

"Flora compete for food." Say **nutrients** (or space).
Not saying the bacteria that are kept out are **pathogenic**. Skin flora are bacteria too.
"All antibiotics affect gut flora." Say broad-spectrum antibiotics are not specific, so they kill many types.
Giving "sweat" or "the immune system" as a barrier.

BEYOND THE SPEC, BUT EXAMINED

- Gut flora make **vitamin K** for us (WBI14 Oct 2020).
- Skin flora data by group (Firmicutes, Actinobacteria, Proteobacteria): draw conclusions about which group dominates at each age (WBI14 Jun 2022).

SPEC 6.8

The non-specific response: inflammation, lysozyme, interferon and phagocytosis

WHAT THIS SPEC POINT ASKS

Understand the body's non-specific responses to infection: inflammation, lysozyme action, interferon and phagocytosis.

Non-specific responses happen straight away and are the **same whatever the pathogen**. The specific immune response (6.9–6.10) takes days and targets one antigen.

6.8 Inflammation

1. Damaged tissue and **mast cells** release **histamine** (and other chemicals, e.g. cytokines).
2. Histamine makes arterioles **dilate** (vasodilation) → more blood flows to the area → it becomes **red** and **warm**.
3. Histamine makes capillary walls **more permeable** → plasma, proteins and white blood cells leak out into the tissue → **swelling** (oedema). The swelling and chemicals cause **pain**.
4. Phagocytes are attracted to the site (by chemicals released there) and leave the capillaries to destroy pathogens.

Signs and their roles:

Sign	Role
Redness / swelling	more blood, phagocytes, antibodies and tissue fluid reach the site
Heat	speeds up enzyme activity, so phagocytes (and lysozyme) work faster; may slow pathogens
Pain	warns you of damage, so you protect the area

6.8 Lysozyme

- An enzyme in tears, saliva, mucus and inside phagocytes (in lysosomes).
- It **hydrolyses the peptidoglycan** in bacterial cell walls (especially Gram-positive bacteria). The weakened wall can't resist the osmotic entry of water, so the bacterium bursts (**lysis**).

6.8 Interferon

- Proteins **secreted by virus-infected cells**.
- They bind to nearby uninfected cells and make them **stop viral replication** (they inhibit viral protein synthesis), so the virus spreads more slowly.
- They also **activate** macrophages and T killer cells and boost inflammation.

6.8 Phagocytosis

1. Phagocytes (**neutrophils** and **macrophages**) are attracted to the pathogen by chemicals it releases or by chemicals from damaged cells (**chemotaxis**).
2. Receptors on the phagocyte bind to **non-self** molecules on the pathogen (binding is easier when the pathogen is covered in antibodies, see 6.9).
3. The phagocyte **engulfs** the pathogen by endocytosis into a **phagocytic vacuole** (phagosome).
4. **Lysosomes** fuse with the vacuole and release **hydrolytic enzymes** (including lysozyme) that **digest** the pathogen.
5. Macrophages then **display** fragments of the pathogen's antigens on their surface: they become **antigen-presenting cells** and set off the specific response.

	Neutrophils	Macrophages
Nucleus	lobed	large, round or bean-shaped
Timing	first to arrive; short-lived	arrive later; live longer
Antigen presentation	no	**yes**: they link non-specific and specific responses

MARK-SCHEME LANGUAGE

- Signs: "pain", "redness", "heat", "swelling", "loss of function" (rubor, calor, dolor accepted). Fever, itching and rash are ignored.
- Roles: pain "**alerts** the person to an injury"; heat "**speeds up** enzyme activity / phagocytosis / chemical reactions"; swelling and redness bring more "blood / white blood cells / phagocytes / antibodies / tissue fluid". Raised temperature may also "denature the pathogen's enzymes".
- Interferon: "**secreted by infected cells**", "**anti-viral** proteins", "prevent **viral replication** / synthesis of viral proteins / spread of virus", "activate macrophages / T killer cells". It is **not** part of the specific response.
- Phagocytosis: "engulf the bacteria" and "destroy them with **hydrolytic enzymes / lysozyme**".

Often asked as: name two signs of inflammation and give the role of each; why a raised temperature helps; how interferon works (two ideas for 2 marks); the role of macrophages.

COMMON MISTAKE

Giving vasodilation as a **sign** of inflammation. It's a cause; the signs are redness, heat, swelling and pain.
Explaining what **causes** each sign when the question asks for its **role**.
Confusing **lysosome** (the organelle) with **lysozyme** (the enzyme).
Phagocytes "eat" bacteria. Write **engulf** and **digest with hydrolytic enzymes**.

BEYOND THE SPEC, BUT EXAMINED

- Cytokines such as **TNF** (tumour necrosis factor) trigger inflammation and phagocytosis; anti-TNF drugs treat inflammatory disease but weaken defence against infection (WBI14 Oct 2020).
- Comparing anti-inflammatory treatments from data: describe both, then say which is more effective (WBI14 Jan 2021).
- Ebola's protein VP35 blocks interferon, so neighbouring cells are not protected (WBI14 Jan 2020).

SPEC 6.9 Antigens, antibodies and antigen-presenting cells

WHAT THIS SPEC POINT ASKS

Understand how antigens and antibodies work in the immune response, including the roles of plasma cells, macrophages and antigen-presenting cells.

6.9 Antigens

- An **antigen** is a molecule (usually a protein or glycoprotein) that triggers an immune response. Antigens are on the surface of pathogens, infected cells, cancer cells, transplanted cells and toxins.
- Your own cells carry **self** antigens, which your lymphocytes tolerate. Antigens on anything else are **non-self** (foreign).
- Each antigen has a specific 3D shape, which depends on its amino acid sequence.

6.9 Antigen-presenting cells (APCs)

- A **macrophage** engulfs a pathogen and digests it (6.8). It then places fragments of the pathogen's antigens on its cell surface membrane (held by proteins called MHC). It is now an **antigen-presenting cell**.
- The APC presents the antigen to **T helper cells**. Only the T helper cell with a **complementary** receptor binds it and is activated (6.10).

6.9 Antibodies

- Antibodies (immunoglobulins) are **proteins** secreted by **plasma cells** (which develop from B cells).
- Structure: **four polypeptide chains** (two **heavy**, two **light**) held together by **disulfide bridges**, forming a Y shape.
  - **Variable region** at the tip of each arm: a different amino acid sequence for each antibody, so a different shape. It forms the **antigen-binding site**, which is **complementary** to one specific antigen. Two binding sites per antibody.
  - **Constant region**: the same within a class of antibodies; binds to receptors on phagocytes.
  - **Hinge region**: flexible, so the arms can bind antigens different distances apart.

6.9 What antibodies do

Antibodies don't destroy pathogens themselves. They bind to antigens to form **antigen–antibody complexes**, which:

- **Agglutinate** pathogens: each antibody has two binding sites, so it links pathogens together into clumps. Phagocytes can engulf many at once, and the pathogens can't spread.
- **Neutralise**: antibodies bind to **toxins** (antitoxins) so they can't bind their target cells, or block the sites a pathogen uses to attach to host cells.
- **Opsonise**: antibodies coat (mark) the pathogen so phagocytes bind and engulf it more easily.

MARK-SCHEME LANGUAGE

- Antigen presentation: "**macrophages** present antigen (on their surface / on MHC) to **T helper** cells". Presentation to T killer cells or B cells is not accepted.
- Antibody action: antibodies "{bind to / neutralise / agglutinate} the antigen"; "**opsonisation**" so macrophages engulf and digest; "antibodies bind the macrophage to the virus".
- Specificity: an antibody "can only bind to its antigen because of the **complementary shape** of the antigen-binding site (variable region)".
- B cells: "B cells **differentiate into plasma cells**, which produce antibodies".
- Wrong blood transfusion: "B antigens recognised as **foreign**", "humoral response", "antibodies released by plasma cells", "**agglutination** / destruction of red blood cells".

Often asked as: describe antigen presentation; how a vaccine gives immunity (4 marks); why an antibody is specific; how antibodies help destroy cancer cells or plaques; primary and secondary responses on a log graph (6-mark).

COMMON MISTAKE

The most repeated errors in every examiner report: "antibodies **kill / destroy** pathogens" (they bind, then phagocytes destroy); "B cells produce antibodies" (plasma cells do); "plasma cells become antibodies"; "the virus is killed".

Mixing up **opsonisation** (marking for phagocytes), **agglutination** (clumping) and **neutralisation** (blocking toxins or binding sites). Name the one that fits.

Calling the antigen-binding site an "active site". Antibodies aren't enzymes.

"The pathogen becomes an antigen-presenting cell." The **macrophage** does.

BEYOND THE SPEC, BUT EXAMINED

- Blood groups: antigens A and B on red blood cells; group O has neither, so it can be given to anyone (WBI14 Oct 2020).
- **ELISA**: an antibody fixed to a plate catches a specific antigen; a second labelled antibody and a colour reaction show it; unbound antibody is washed off (WBI14 Jan 2023).
- Monoclonal and bispecific antibodies (one arm binds a cancer cell, the other a T killer cell, bringing them together) (WBI14 Jun 2025).
- Comparing titres on a log axis: $10^{4.4} \div 10^{3.1} = 10^{1.3} \approx$ **20 times** higher (WBI14 Jan 2024).

SPEC 6.10 B cells and T cells

WHAT THIS SPEC POINT ASKS

Understand the different roles of B cells (B memory and B effector cells) and T cells (T helper, T killer and T memory cells) in the immune response.

6.10 Lymphocytes and clonal selection

- Lymphocytes are made in the bone marrow. **B cells** mature in the bone marrow, **T cells** in the thymus.
- Each lymphocyte has receptors of **one** shape on its surface. Among millions of lymphocytes, a few have receptors complementary to any particular antigen.
- When a pathogen arrives, only the lymphocytes with matching receptors are **selected** and activated (**clonal selection**). They divide by **mitosis** to form a large **clone** of genetically identical cells (**clonal expansion**).

6.10 Step by step

1. A macrophage engulfs the pathogen and presents its antigens (APC).
2. A **T helper cell** with a complementary receptor binds the presented antigen and is **activated**. It divides by mitosis into a clone of T helper cells and **T memory** cells.
3. Activated T helper cells release **cytokines** (chemical signals) that:
   - activate the **B cells** whose surface antibodies have bound the same antigen;
   - activate **T killer cells** with complementary receptors;
   - stimulate phagocytes.
4. **Humoral response (B cells)**: activated B cells divide by mitosis. Most become **B effector cells**, which differentiate into **plasma cells** that secrete large amounts of one antibody into the blood and lymph. Some become **B memory cells**.
5. **Cell-mediated response (T killer cells)**: activated T killer cells divide by mitosis. They bind to body cells displaying the antigen (virus-infected or cancer cells) and release **perforins**, which make pores in the cell membrane, so the cell dies (**lyses**). This releases the viruses inside, where antibodies and phagocytes can deal with them. Some become **T memory** cells.

Cell	Role
T helper	binds antigen on an APC; releases cytokines that activate B cells, T killer cells and phagocytes
T killer (cytotoxic)	destroys body cells that display the antigen (infected or cancer cells)
T memory	stay in the body for years; on re-infection, divide quickly into T helper and T killer cells
B effector	divide and differentiate into **plasma cells**, which secrete antibodies
B memory	stay in the body for years; on re-infection, divide quickly into plasma cells, giving the fast secondary response

6.10 Why both T helper and T killer cells are needed

- Without T helper cells, B cells and T killer cells aren't activated (the reason HIV causes immunodeficiency).
- Antibodies can't reach viruses hiding **inside** cells. T killer cells destroy the infected cells, so the virus can't keep replicating there.

MARK-SCHEME LANGUAGE

- T helper cells "**activate / stimulate** B cells (to divide)" and "T killer cells"; they "**release cytokines**".
- T killer cells "release **perforins**" that "cause the infected / cancer cells to **burst / lyse**"; they must be activated and "**divide** so there are enough T killer cells". ACCEPT cytotoxic, CD8.
- Destroying infected cells "releases the viruses so macrophages can destroy them". Not "kills the virus".
- Clones: "**genetically** identical cells"; the word "genetically" is required.
- Memory cells: on re-infection, a "**faster, higher** secondary response"; plasma cells make antibodies sooner.

Often asked as: which cell presents to which (MCQ); why both T helper and T killer cells are needed; how T killer cells are activated and act; the effect of a drug that stops cell division; an immune response to cancer cells (4 marks).

COMMON MISTAKE

"T helper cells turn into T killer cells." They are separate cell types; T helpers **activate** T killers.
Confusing the cell types, or listing everything you know about T killer cells instead of answering the "why" (e.g. why so many are needed: one cell can't destroy many infected cells).
Leaving out "genetically" when you describe the clone made by mitosis.

BEYOND THE SPEC, BUT EXAMINED

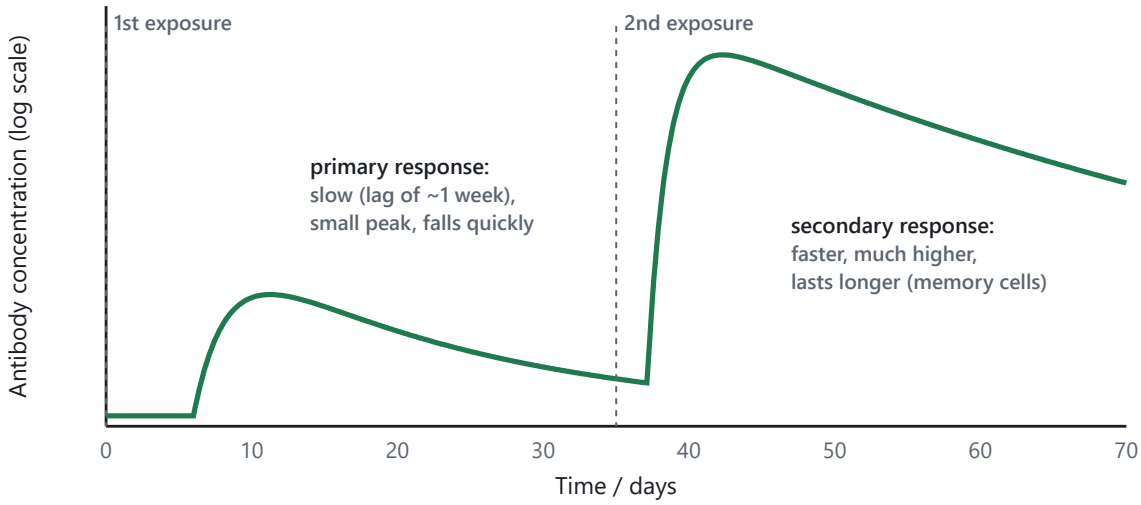
- Receptor names: **CD4** on T helper cells, **CD8** on T killer cells; **perforins** and **cytokines** by name (WBI14 Oct 2024).
- Immunosuppressive drugs that block DNA synthesis stop mitosis, so no clonal expansion of T or B cells (WBI14 Jun 2021).
- CAR-T therapy: a patient's own T cells are given a new receptor gene (using a virus vector), cloned and returned; they aren't rejected because they carry no foreign antigens (WBI14 Jan 2020).
- Cancer cells carry new antigens (neoantigens) made by mutations in their DNA (WBI14 Oct 2023).

SPEC 6.11 Developing immunity: natural, artificial, active and passive

WHAT THIS SPEC POINT ASKS

Understand the four ways a person can become immune: natural or artificial, and active or passive.

6.11 Primary and secondary responses



The secondary response is faster, larger and longer-lasting because memory cells are already there.

- **Primary response** (first exposure): only a few lymphocytes match the antigen. Selection, clonal expansion and plasma cell formation take days, so antibody levels rise slowly and stay fairly low. You usually get **ill** (symptoms) before the pathogen is cleared. **Memory cells** are made.
- **Secondary response** (same antigen again): many **memory cells** recognise it straight away and divide quickly into plasma cells. Antibodies are made **sooner, faster and in much larger amounts**, and stay high for longer. The pathogen is usually destroyed **before symptoms** appear. You are **immune**.

6.11 Active and passive immunity

- **Active**: your own immune system meets the antigen, makes antibodies and **memory cells**. Takes time to develop but **long-lasting**.
- **Passive**: you receive **ready-made antibodies** from another source. Protection is **immediate** but **short-term**: no memory cells are made, and the antibodies are broken down within weeks to months.

	Active (own antibodies + memory cells)	**Passive (antibodies given)**
Natural	catching the disease and recovering	antibodies from the mother cross the **placenta** to the fetus, or pass to the baby in **breast milk** (colostrum)
Artificial	**vaccination**	**injection of antibodies** (e.g. antitoxin for tetanus, antivenom)

6.11 Vaccines

- A vaccine contains **antigens** in a harmless form: an **attenuated** (weakened) or **inactivated** pathogen, a purified antigen or toxoid, or mRNA that makes your cells produce the antigen.
- It causes a primary response, so memory cells form without you getting the disease. A real infection later gets a fast secondary response.
- **Boosters** give a secondary response, raising antibody levels and memory cell numbers further.
- Vaccines may need updating when a pathogen's antigens change (6.12).

WORKED EXAMPLE: READING AN IMMUNITY QUESTION

A pregnant woman is vaccinated. Her baby has antibodies at birth that fall over the next 3 months. The mother's immune system made the antibodies (**artificial active** immunity for her). They crossed the **placenta** into the **fetus**, so the baby has **natural passive** immunity. No memory cells were made in the baby, and the antibodies are broken down, so the level falls.

MARK-SCHEME LANGUAGE

- Active: "B cells differentiate into plasma cells", "**memory cells** are formed", long-term. Passive: "antibodies injected / received", short-term, no memory cells.
- Vaccine: "an **inactive / attenuated** (weakened) form of the pathogen". "Dead virus" is not accepted.
- Maternal vaccination: "plasma cells released antibodies"; "antibodies **pass across the placenta** (to the fetus)" and "remain in the blood until after birth". In this question, antibodies "in the milk" or the baby's cells crossing are not accepted.
- Secondary response: similarity "antibody concentration rises (and falls)"; difference "rises **sooner / higher / faster**".
- Herd immunity: "vaccinated people won't develop the disease, so there are **fewer infected people to pass it on**", protecting those who can't be vaccinated.

Often asked as: tick-grids on types of immunity; how a vaccine works; herd immunity; maternal vaccines; compare primary and secondary response graphs; 6-mark on vaccine data for two groups of animals.

COMMON MISTAKE

"Dead viruses in the vaccine." Viruses aren't alive: write **inactivated** or **attenuated**.
"Baby" when you mean "fetus" (before birth); "the vaccine given to the mother" when the question says pregnant woman.
Confusing "**infected**" with "**develops the disease**". A vaccinated person may be infected but the pathogen is destroyed before symptoms appear.
Saying passive immunity gives memory cells. It doesn't.

BEYOND THE SPEC, BUT EXAMINED

- **Herd immunity** is not in the spec but is credited (WBI14 Jan 2021).
- Antibody levels fall because antibodies are broken down (and, in a pregnant animal, transferred to the young) (WBI14 Jan 2026).
- A vaccine made from one virus subtype can protect against a similar one if their antigens are similar in shape (WBI14 Oct 2024).
- Planning a vaccine trial: control the dose, timing, age and health of the subjects; use large groups (WBI14 Jan 2024).

SPEC 6.12

The evolutionary race between pathogens and hosts

WHAT THIS SPEC POINT ASKS

Understand how the ways pathogens evade the immune system support the idea of an "evolutionary race" between pathogens and their hosts.

6.12 The idea

- Hosts with better defences survive and reproduce more, so defences evolve. That puts a **selection pressure** on pathogens: any pathogen with a **mutation** that helps it evade the defences survives and reproduces, so evasion evolves. That in turn selects for new host defences, and so on.
- Neither side "wins" for long. The evasion mechanisms we see in pathogens today only make sense as the result of this repeated natural selection, so they are **evidence** for the race.

6.12 Evasion by HIV

- **Kills T helper cells**, the cells that coordinate the whole specific response.
- **High mutation rate** (reverse transcriptase makes many copying errors) → the genes for its surface proteins change → **antigens change shape** (antigenic variation). Memory cells and antibodies from earlier infection no longer fit, so each new variant needs a new primary response. This also makes a vaccine very hard to design.
- Hides as a **provirus** (latency) inside cells where the immune system can't see it.
- Reduces the display of its antigens on infected cells, so T killer cells are less likely to recognise them.

6.12 Evasion by *Mycobacterium tuberculosis*

- **Survives inside macrophages** by stopping lysosomes fusing with the phagocytic vacuole, so it isn't digested.
- Its thick **waxy cell wall** resists digestion.
- Hides **dormant** inside tubercles for years.
- Interferes with antigen presentation, so T cells are activated more slowly.

6.12 Other examples (beyond the spec)

- **Influenza**: small mutations change its antigens every year (antigenic drift), so new vaccines are needed each season.
- **λ phage and *E. coli***: a mutation in the *E. coli* gene for maltoporin changes its shape so the phage can't attach; a mutation in the phage's attachment protein lets it bind the new shape. The maltoporin must stay functional, because the bacterium needs it to take in sugars for respiration.
- **Ebola** blocks interferon (6.8). Hosts respond in turn, e.g. with interferon and with T killer cells that detect infected cells.

MARK-SCHEME LANGUAGE

- Antigen change: "**mutation** changes the amino acid sequence", "so the protein is a **different shape**", "**memory cells won't recognise** the new antigen", "another **primary response** is needed".
- λ phage 6-mark: bacteria — "mutation in the gene changing the **structure of the maltoporin** so the J protein can no longer bind"; phage — "mutation in the viral DNA coding for the J protein, able to bind the altered maltoporin"; "the virus acts as a **selection pressure**"; top mark: "maltoporin must **remain functional**, otherwise no sugars for respiration", or how mutations pass to the next generation.
- Why we need new antibiotics: "bacteria are becoming **resistant**", "so we won't be able to treat resistant infections"; "to win the evolutionary race" is accepted.

Often asked as: 6-mark on a phage–bacterium race in context; why new antibiotics are needed; why memory cells fail when an antigen mutates.

COMMON MISTAKE

Generic answers ("pathogens evolve") without the context's named molecules (e.g. J protein, maltoporin).
Saying pathogens mutate "in order to" evade. Mutations are **random**; evasion is **selected**.
Viruses and bacteria "dividing by mitosis".
Bringing in antibiotics as selection pressures when the question asks something else (e.g. why a new drug is needed).

BEYOND THE SPEC, BUT EXAMINED

- The λ phage / *E. coli* maltoporin race (WBI14 Jan 2022).
- HIV *env* gene mutations change envelope protein shape, so a vaccine based on one strain may not work (WBI14 Oct 2025).
- Note: examiners judged slime-capsule answers beyond the scope of the phage question (WBI14 Jan 2022).

SPEC 6.13

Bacteriostatic and bactericidal antibiotics

WHAT THIS SPEC POINT ASKS

Understand the difference between bacteriostatic antibiotics (which stop bacteria growing) and bactericidal antibiotics (which kill them).

6.13 Antibiotics

- An **antibiotic** is a substance that kills bacteria or stops them growing, without harming human cells.
- **Bactericidal**: **kills** bacteria. Example: **penicillin** stops the enzymes that form cross-links in the peptidoglycan cell wall. In growing bacteria the wall becomes weak; water enters by osmosis and the cell **bursts** (osmotic lysis). Vancomycin also acts on the wall.
- **Bacteriostatic**: **stops** bacteria **growing and dividing** but doesn't kill them. Example: **tetracycline** binds to bacterial ribosomes and stops protein synthesis. The population stays the same size (like a stationary phase), and the immune system destroys the bacteria. If the drug is stopped too early, the bacteria can start dividing again. High doses of some bacteriostatic drugs can kill.

	Bactericidal	**Bacteriostatic**
Effect on cells	kills them	stops growth and division
Viable count in a culture	**falls**	stays **constant**
Relies on the immune system?	less	yes, to remove the bacteria
Examples	penicillin, vancomycin	tetracycline

6.13 Why antibiotics don't harm human cells or viruses

- Antibiotics target structures human (eukaryotic) cells don't have, or have in a different form: the **peptidoglycan cell wall**, **70S ribosomes** (ours are 80S), particular bacterial enzymes (e.g. DNA gyrase).
- **Viruses** have no cell wall, ribosomes or metabolism of their own, so antibiotics have nothing to act on. Viral diseases (flu, colds) can't be treated with antibiotics.

MARK-SCHEME LANGUAGE

- Antibiotic / bactericidal: "a {chemical / substance / drug} that {kills / inhibits the growth of} bacteria". **Bactericidal must say "kills"**. Viruses or antibodies score nothing.
- Bacteriostatic: "**prevents cell division / growth**"; keeps bacteria in the stationary phase.
- Compare: a categorical similarity ("**both** target bacteria") and a paired difference ("bacteriostatic **prevent growth** whereas bactericidal **kill**").
- Why bacteria not humans: "bacteria are **prokaryotic** (humans are eukaryotic)", or a named target such as the cell wall or DNA gyrase ("human cells do not have DNA gyrase, therefore not harmed").
- Telling them apart: grow bacteria, add the antibiotic, then plate out samples (colonies grow from bacteriostatic but not bactericidal); or count living cells (constant = bacteriostatic, falls = bactericidal); or use an exclusion dye.

Often asked as: define bactericidal; compare two named antibiotics; identify the type from a growth graph; effects on a pie chart of throat bacteria; plan a test to tell them apart.

COMMON MISTAKE

Confusing **antibiotics** with **antibodies**: this scores zero.
Definitions that don't say "**substance**" or "chemical".
"Bacteria become **immune** to antibiotics." They become **resistant**.
Defining the types by mechanism ("bactericidal destroy the wall, bacteriostatic stop protein synthesis").
Define by the **effect**: kill v stop growth.

BEYOND THE SPEC, BUT EXAMINED

- Bacterial DNA is supercoiled by **DNA gyrase**; an antibiotic that blocks it stops bacteria dividing but doesn't affect human cells, which lack the enzyme (WBI14 Oct 2024).
- A helper drug that makes the membrane or wall more permeable lets an antibiotic reach its target inside the cell (WBI14 Jan 2025).
- Broad-spectrum antibiotics kill many types, including gut flora (WBI14 Jan 2021).

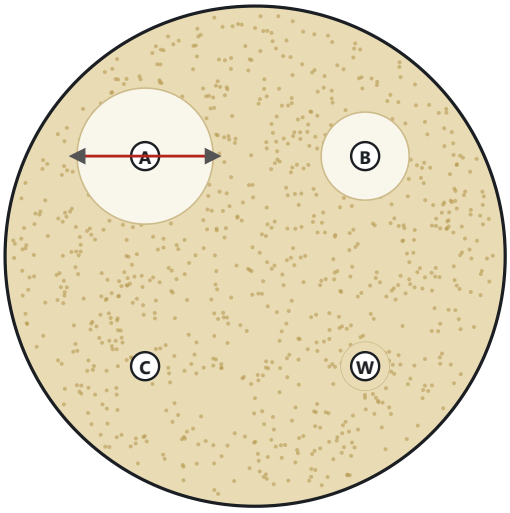
SPEC 6.14 Core Practical 14: the effect of different antibiotics on bacteria

WHAT THIS SPEC POINT ASKS

Investigate the effect of different antibiotics (or different concentrations) on bacteria, using the size of the clear zone they produce on an agar plate.

6.14 Principle

Bacteria are spread evenly over agar so they grow as a continuous **lawn**. Paper discs soaked in antibiotic are placed on top. The antibiotic **diffuses** out into the agar. Where it is concentrated enough to stop growth, a **clear zone** (zone of inhibition) forms around the disc. A bigger clear zone means the bacteria are more sensitive to that antibiotic.



Lawn of bacteria on agar, with antibiotic discs

A: large clear zone = most effective here
B: smaller clear zone = less effective
C: no clear zone = bacteria resistant
W: sterile water disc (negative control)

measure the diameter of the clear zone (mean of two at right angles)

Area of clear zone = πr^2
(r = diameter ÷ 2)

Clear zones round discs A and B; no zone round C (bacteria resistant) or the water control W.

CORE PRACTICAL 14: METHOD IN BRIEF

1. Disinfect the bench; work next to a lit Bunsen burner (updraft).
2. Use a sterile pipette to put a known volume (e.g. 0.1 cm^3) of a broth culture of a non-pathogenic bacterium onto a sterile nutrient agar plate. Spread it evenly with a sterile spreader (flamed metal, or a sterile plastic one) to give a lawn.
3. With **sterile forceps** (flamed), place discs soaked in different antibiotics (or one antibiotic at different concentrations) on the agar, well spaced so the zones don't overlap. Label the underside of the plate.
4. Include a **control** disc soaked in sterile distilled water (or the solvent used for the antibiotic).
5. Tape the lid with a few strips (not sealed all round), invert the plate and incubate at about **25 °C** for 24–48 hours.
6. Without opening the plate, measure the **diameter** of each clear zone with a ruler across two diameters at right angles, and take the mean.
7. Repeat with at least three plates.

6.14 Variables

Independent	type of antibiotic (or its concentration)
Dependent	diameter or area of the clear zone
Control	species and volume of bacterial culture (and its age), disc size, volume of antibiotic solution per disc, concentration (when comparing types), agar depth and type, incubation temperature and time

WORKED EXAMPLE: CLEAR-ZONE AREA

Antibiotic A: diameters 17 mm and 19 mm at right angles. Mean = 18 mm, so r = 9 mm.
Area = πr^2 = 3.14 × 9^2 = **254 mm²** (3 s.f.).
Antibiotic B: mean diameter 12 mm, r = 6 mm, area = π × 6^2 = **113 mm²**.
A's zone is 254 ÷ 113 = **2.25 times** the area of B's, so these bacteria are more sensitive to A at this concentration.
If you want only the cleared agar, subtract the disc: a 6 mm disc has area π × 3^2 = 28 mm², so A clears 254 − 28 = 226 mm².

6.14 Evaluation

- A larger zone doesn't always mean a better medicine: zone size also depends on how fast the antibiotic **diffuses** through agar (small molecules spread further) and how much was on the disc. And a clear zone doesn't show whether the drug is bactericidal or bacteriostatic (6.13).
- Results on agar may not match how the drug works in the body.
- Use the **same concentration** to compare types; repeat plates and calculate means (and SD) to judge whether differences are significant.

MARK-SCHEME LANGUAGE

- Choosing an antibiotic from a plate: "not A and D because **no zone of inhibition** / bacteria are **resistant**"; "B because it has the **largest zone** of inhibition"; a higher mass that gives no bigger zone is no more effective.
- Controls: a disc or well with no antibiotic "to show the bacteria were **viable** / any effect is due to the antibiotic"; solvent-only control "so that the effect of the **drug itself** can be determined". "Control" or "placebo" alone is too vague.
- Data collection in a planning question: zone of inhibition (larger = greater effect), turbidity (less turbid = more effective), cell or dilution counts; **repeats**; a **statistical test** for significance.
- Units: 10 µg = **0.01 mg**.

Often asked as: choose antibiotics and doses from a plate diagram; MIC plate with serial dilutions; controls; plan a 6-mark investigation of natural compounds against MRSA.

COMMON MISTAKE

Using the diameter in πr^2. Halve it first.
Describing in general how zones are used when the question gives a diagram: read the actual zones and quote them.
Opening the plate to measure the zones. Measure through the lid or base.
Incubating at 37 °C, or sealing the plate completely.

BEYOND THE SPEC, BUT EXAMINED

- **MIC** (minimum inhibitory concentration): the lowest concentration that stops visible growth, found with a 1 in 2 serial dilution in wells. Comparing MICs of 1 in 8 and 1 in 256 gives 256 ÷ 8 = **32 times** more effective (WBI14 Oct 2021).
- Species richness of gut bacteria as a measure of antibiotic damage (WBI14 Jan 2025).

SPEC 6.15 Hospital-acquired infections and codes of practice

WHAT THIS SPEC POINT ASKS

Know what causes hospital-acquired infections (HAIs) and how this has led to codes of practice on prescribing antibiotics and on infection prevention and control.

6.15 Why infections spread in hospitals

- Many patients are already ill, elderly, very young, or **immunosuppressed**, so they are easily infected.
- **Wounds**, surgery, catheters and drips give pathogens a route in.
- Staff and visitors move between patients; equipment and surfaces get contaminated.
- Antibiotics are used heavily, so there is strong **selection pressure** for resistant bacteria such as **MRSA** (methicillin-resistant *Staphylococcus aureus*) and *Clostridium difficile*.

6.15 How antibiotic resistance develops (natural selection)

1. In a population of bacteria, a random **mutation** produces an allele that gives resistance (e.g. a gene for an enzyme that breaks down the antibiotic).
2. When the antibiotic is used, non-resistant bacteria are killed. The resistant ones **survive** (the antibiotic is the selection pressure).
3. They **reproduce** by binary fission, passing the resistance allele to their offspring (**vertical** transmission).
4. Resistance genes on **plasmids** can also pass to other bacteria, even other species, by conjugation (**horizontal** transmission).
5. Over generations, the **frequency of the resistance allele rises** in the population.

6.15 Codes of practice

Antibiotic prescribing	Infection control in hospitals
don't prescribe for viral or minor infections	wash hands (or alcohol gel) before and after every patient contact
use **narrow-spectrum** antibiotics where possible	gloves, aprons, "bare below the elbows" (no long sleeves, watches, ties)
test which antibiotic the bacteria are sensitive to before prescribing	**screen** patients for MRSA on admission
prescribe the right dose for the right length of time	**isolate** infected patients
patients **finish the course**, take them as directed and don't share them	disinfect surfaces; sterilise equipment; deep-clean wards
rotate antibiotics; limit routine preventive use	dispose of waste (e.g. dressings, needles) safely

- These reduce the selection pressure for resistant bacteria (so existing antibiotics keep working longer) and stop pathogens spreading between patients.

MARK-SCHEME LANGUAGE

- Advice to **patients**: "**finish the course** of antibiotics", "take them as directed", "don't share your antibiotics", "don't take antibiotics not prescribed". Advice to doctors isn't credited here.
- Why reduce use: "antibiotic use is a **selection pressure**, so the number of resistant bacteria increases"; reducing use "decreases the **development / spread** of resistance" and "prolongs the time antibiotics can be used"; otherwise "antibiotics may become useless and people will remain ill / die".
- Prescriptions v resistance graph: "**correlation**"; "use of the antibiotic is a selection pressure"; "resistant bacteria reproduce and non-resistant ones die"; resistance falls when prescriptions fall because **non-resistant bacteria survive and compete**.

Often asked as: advice to patients; describe a code of practice and explain how it reduces resistance; why MRSA means new drugs are needed; interpret prescription and resistance data.

COMMON MISTAKE

Describing the graph without explaining it. When prescriptions fall, resistant bacteria lose their advantage: non-resistant bacteria survive and outcompete them.
Giving advice to doctors when the question asks for advice to patients. A "describe" question doesn't need a reason.
"The antibiotic causes the mutation." Mutations are random; the antibiotic **selects** resistant bacteria that already exist.

BEYOND THE SPEC, BUT EXAMINED

- Artificial intelligence can search for new antibiotic targets and doses (WBI14 Oct 2025).
- Surveys of doctors who feel pressured to prescribe: "codes of practice are being ignored", so resistance rises (WBI14 Oct 2020).

SPEC 6.16 Microorganisms, decomposition and the carbon cycle

WHAT THIS SPEC POINT ASKS

Know how microorganisms decompose organic matter and so recycle carbon.

6.16 How decomposers work

- **Decomposers** are bacteria and fungi that feed on dead organisms, faeces and other organic waste (**saprotrophic** nutrition).
- They **secrete enzymes** onto the dead material (**extracellular digestion**). These **hydrolytic** enzymes break large insoluble molecules into small **soluble** ones:
  - carbohydrases (e.g. cellulase, amylase) hydrolyse **glycosidic bonds** → monosaccharides (glucose);
  - proteases → amino acids;
  - lipases → fatty acids and glycerol;
  - nucleases → nucleotides.
- The soluble products are **absorbed** into the microorganisms (by diffusion and active transport).
- They use them in **respiration**, which releases **CO_2** into the air (or into soil and water), and to make their own molecules (proteins, lipids, glycogen) for growth.
- Detritivores (earthworms, woodlice) break dead material into small pieces first, giving a larger **surface area** for the enzymes.

6.16 Recycling carbon

- Plants (producers) take in CO_2 in **photosynthesis** and make organic compounds. Animals eat them. When organisms die, the carbon is locked in their bodies.
- Decomposers release that carbon as **CO_2** by respiration, so plants can fix it again. Without decomposers, carbon would stay locked in dead material.
- In **anaerobic** conditions (waterlogged soils, landfill, animal guts), some microorganisms release **methane** (CH_4) instead.
- Decomposition also returns minerals (e.g. nitrogen as ammonium ions from amino acids) to the soil.

6.16 What controls the rate of decomposition

Factor	Effect
Temperature	warmer → faster enzyme activity (more enzyme–substrate collisions); too hot denatures enzymes
Water	needed for enzyme reactions and for microbes to grow; very dry → slow
Oxygen	aerobic decomposers are faster; waterlogged soil is anaerobic, so decay is slow
pH	acidic conditions slow most decomposers
Surface area	small pieces decay faster

When decay is very slow (cold, acidic, waterlogged peat bogs), carbon builds up as peat and, over millions of years, fossil fuels (Topic 5).

MARK-SCHEME LANGUAGE

- "Bacteria and fungi are **decomposers**", "**release (hydrolytic) enzymes onto** dead organisms" (2 marks), "break down organic matter", "digested molecules are **soluble**", "**absorbed / taken up** by the microorganisms", "**respired**, releasing **carbon dioxide**", "used by plants for **photosynthesis**".
- 6-mark on carbohydrates: Level 1 = general decomposition; Level 2 = **carbohydrases hydrolyse glycosidic bonds** to monosaccharides, absorbed by diffusion and respired; Level 3 = detail from the data (e.g. one bond broken in a disaccharide, two in a trisaccharide; monosaccharides need no digestion) **and** CO_2 from respiration.
- Faster at 25 °C: "enzymes work faster", "more enzyme–substrate collisions", "mass lost because CO_2 is released by **respiration** of decomposers".
- Products also used to build "**other organic molecules** / glycogen / protein / lipid". Starch or chlorophyll in bacteria isn't accepted.

Often asked as: role of microorganisms in recycling carbon; a 6-mark on decomposing carbohydrates; teabag or leaf mass loss data; plastic breakdown.

COMMON MISTAKE

A generic decomposition answer when the question is about specific molecules (e.g. carbohydrates): name the enzymes, the bonds, and the products.
Stopping at "enzymes break it down": say what happens to the soluble products (absorbed, respired, CO_2 released).
Not linking **mass loss** to CO_2 released by respiration.
"Food" for decomposers: say sugars or nutrients.

BEYOND THE SPEC, BUT EXAMINED

- Rate of mass loss from a **tangent** on a decomposition curve (WBI14 Oct 2021).
- Leaves with more sugar decompose faster; bacterial numbers rise then fall as sugars run out (WBI14 Jun 2024).
- Proteins are converted into amino acids and nucleic acids into nucleotides (WBI14 Jan 2026).

SPEC 6.17 PCR: amplifying DNA

WHAT THIS SPEC POINT ASKS

Know how the polymerase chain reaction (PCR) makes many copies of a piece of DNA.

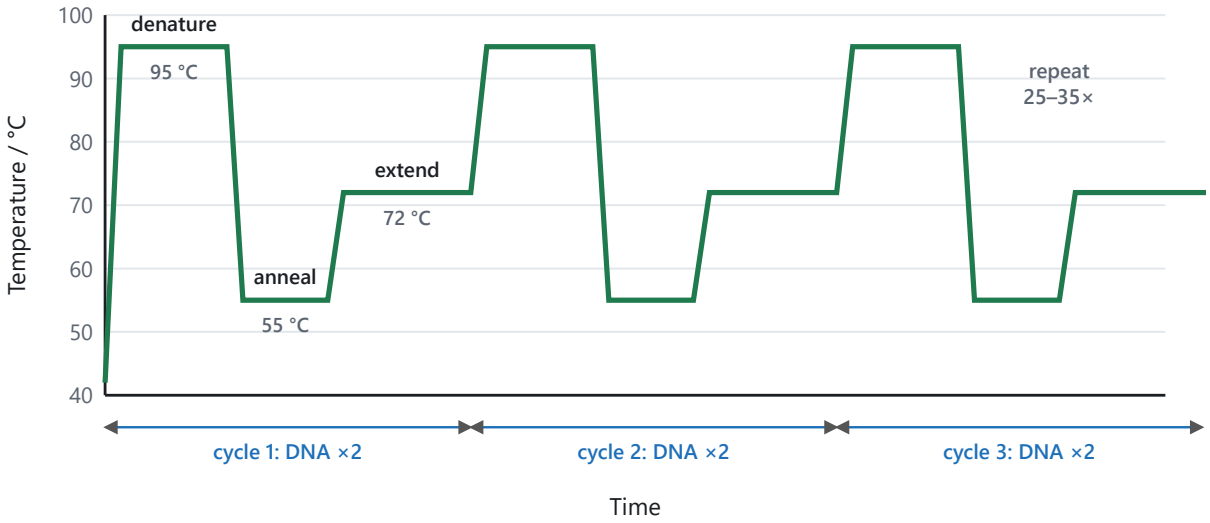
6.17 Why PCR is needed

Samples from a crime scene, a fossil, a feather or a hair contain only a **tiny amount of DNA**. PCR copies (amplifies) a chosen section millions of times, so there is **enough to analyse** (e.g. by gel electrophoresis). It happens in a test tube, in a machine called a **thermal cycler**.

6.17 What goes in the tube

Ingredient	Job
Template DNA	the sample to be copied
Primers	short, single-stranded pieces of DNA complementary to the start of the target section. **One primer for each strand** (the strands are antiparallel). DNA polymerase can only add nucleotides onto an existing strand, so it needs a primer to start from; primers also choose which section is copied.
DNA nucleotides (A, T, C, G)	the raw material for the new strands
Taq DNA polymerase	joins nucleotides to make new strands (forms **phosphodiester bonds**). From the bacterium *Thermus aquaticus*, which lives in hot springs, so it is **not denatured** at 95 °C.
Buffer	keeps the pH optimal

6.17 One cycle



Each cycle runs through three temperatures and doubles the amount of target DNA.

1. **Denaturation (about 95 °C)**: the high temperature breaks the **hydrogen bonds** between the bases, so the two DNA strands separate. (In cells, helicase does this; here heat does.)
2. **Annealing (about 50–60 °C)**: cooling lets the **primers bind** (by complementary base pairing) to the ends of the target section on each single strand.
3. **Extension (about 72 °C)**: Taq polymerase binds at the primer and adds free nucleotides that pair with the template, making a new complementary strand. 72 °C is Taq's optimum.

Repeat **25–35 times**. Each cycle **doubles** the number of copies.

WORKED EXAMPLE: COPIES AND TIME

Copies after n cycles = starting copies × 2^n.

50 molecules after 25 cycles: $50 \times 2^{25} = 50 \times 33\,554\,432 =$ **1.68×10^9** copies.

Time: 30 cycles of (30 s at 95 °C + 30 s at 55 °C + 60 s at 72 °C), plus 2 min at the start and 5 min at the end.

30 × 120 s = 3600 s; 3600 + 120 + 300 = 4020 s = 67 min = **1.12 h** (2 d.p.).

MARK-SCHEME LANGUAGE

- Why: "only **small quantities** of DNA can be collected", so we need to "**amplify** / increase the number of copies of DNA" so there is "enough to run on gel electrophoresis / analyse". "To increase the length of DNA" is not accepted.
- Ingredients: "(DNA) primers", "(DNA) nucleotides", "Taq / DNA polymerase", "buffer". RNA nucleotides or RNA polymerase aren't accepted.
- Cycle, **using the temperatures given**: "94 °C to **break the H bonds** / separate the strands"; "temperature lowered (54 °C) so **primers can attach**"; "temperature raised (72 °C) so nucleotides **join** to the bases"; "many cycles so that enough DNA is made". Maximum 2 marks if you never mention a temperature or temperature change.
- Taq polymerase: "binds to the DNA / primers", "lines up nucleotides along the strands", "forms **phosphodiester bonds**".
- Primers: "so that DNA polymerase can **bind**"; "**one primer for each strand**" because "the strands are **antiparallel** / complementary".

Often asked as: why PCR is needed; list the molecules needed; time for a PCR programme; explain a temperature diagram; why primers are needed; the role of Taq polymerase.

COMMON MISTAKE

Writing everything you know about PCR and electrophoresis. Use the marks available and the diagram you are given.
Writing about Taq's heat tolerance when asked for its **role**: it joins nucleotides into a new strand.
"One primer for each end of each strand." One primer per strand.
Leaving out the temperatures when a diagram gives them.

BEYOND THE SPEC, BUT EXAMINED

- Reading a PCR temperature programme and calculating its total time (WBI14 Jan 2022).
- Mitochondrial DNA is inherited only from the mother (through the egg), so it traces maternal lines (WBI14 Jan 2026).

SPEC 6.18

Gel electrophoresis

WHAT THIS SPEC POINT ASKS

Know how gel electrophoresis separates DNA fragments according to their length.

6.18 How it works



Wells at the negative end; DNA (negatively charged) moves to the positive electrode. Short fragments travel furthest.

(a) Suspect 2's bands match the crime-scene sample exactly. (b) Every band in the child comes from either the mother or man A.

1. **Get DNA fragments**: PCR products, or DNA cut with **restriction enzymes**. Restriction enzymes cut (hydrolyse the phosphodiester bonds) at specific base sequences called recognition sites. These sites are spaced unevenly, so the fragments are different lengths, and the lengths differ between individuals.
2. **Load**: put the DNA into **wells** cut in a slab of **agarose gel**, covered in a buffer solution. Put a **DNA ladder** (fragments of known lengths) in one well.
3. **Run**: apply a current. The wells are at the **negative** electrode (cathode). DNA is **negatively charged** (its **phosphate** groups), so it moves towards the **positive electrode (anode)**.
4. **Separate**: the gel is a mesh of fibres. **Shorter fragments** slip through the pores more easily, so they travel **further** in the same time. Longer fragments are held back more and stay nearer the wells.
5. **See the bands**: DNA is invisible, so it is stained with a dye that shows under UV light, or labelled with fluorescent or radioactive markers. Each band is many copies of a fragment of one length.

- A **higher gel concentration** has smaller pores (more molecules, more resistance), so all fragments move more slowly; it separates small fragments better.

WORKED EXAMPLE: ESTIMATING FRAGMENT SIZE FROM A LADDER

Ladder bands: 1000 bp at 20 mm, 500 bp at 34 mm, 250 bp at 48 mm from the well. Each halving of size moves the band 14 mm further: distance is linear with **log** of size.
An unknown band is at 41 mm: halfway between 500 bp and 250 bp on the log scale.
log size = log 500 − (7 ÷ 14) × log 2 = 2.699 − 0.151 = 2.548 → size = $10^{2.548}$ ≈ **350 bp**.
(Not 375 bp: halfway on a log scale isn't halfway in base pairs.)

MARK-SCHEME LANGUAGE

- Different fragment sizes: "restriction enzymes recognise **specific (base) sequences**"; "the recognition sites are **not equally spaced**"; "the enzyme **hydrolyses phosphodiester bonds**"; "different-sized fragments **move different distances** through the gel".
- Higher gel concentration: "more molecules / **smaller spaces**, so higher resistance / friction".
- Method details credited: "agarose gel", "apply a current", "use of a dye", "use of restriction enzymes".
- Write "**bands**", not "blocks".

Often asked as: name the enzyme that cuts DNA (MCQ); why fragments differ in size; read sizes off a ladder on a log scale; effect of gel concentration; which lanes are needed to check a GM plant.

COMMON MISTAKE

DNA moves to the "positive cathode". The positive electrode is the **anode**; the wells sit at the negative cathode.
Saying larger fragments move further, or explaining by charge alone. All DNA has a similar charge per length; it is the **size** that separates them, because of the gel mesh.
Writing about introns or hypervariable regions when asked why restriction fragments differ: the answer is unevenly spaced **recognition sites**.

BEYOND THE SPEC, BUT EXAMINED

- Reading fragment sizes from a ladder plotted on a log scale (WBI14 Jan 2021).
- Gel separation of proteins: thickest band = most concentrated; band that moved furthest = smallest (or most charged) (WBI14 Oct 2022).
- To show a GM broccoli has a gene from linseed, run original broccoli, GM broccoli and linseed side by side; the GM plant must share the linseed band (WBI14 Oct 2024).

SPEC 6.19

DNA profiling: identification and relationships

WHAT THIS SPEC POINT ASKS

Understand how DNA profiling identifies individuals and shows genetic relationships between organisms (plants and animals).

6.19 What a DNA profile is

- Much of the genome is **non-coding**. Some non-coding regions contain short base sequences repeated end to end (**short tandem repeats**, STRs; longer ones are VNTRs).
- The **number of repeats** at each of these places (loci) varies a lot between individuals. So the length of the DNA at each locus varies.
- A DNA profile is the pattern of **bands** made by these fragments on a gel. Apart from identical twins, no two people have the same profile when enough loci are used. You inherit **half your bands from each parent**.

6.19 Making a profile: the order of the steps

1. **Collect** a sample containing cells (blood, saliva, hair root, skin, feathers, leaf, droppings).
2. **Extract** the DNA.
3. **Amplify** it with **PCR**: the primers are chosen to flank the repeat regions, so those sections are copied.

4. **Cut** the DNA with **restriction enzymes** (if the method uses them) to give fragments.
5. **Separate** the fragments by **gel electrophoresis**.
6. **Visualise** the bands (dye under UV, or fluorescent/radioactive labels) and **compare** the profiles.

Mark schemes credit "PCR, followed by (gel) electrophoresis" and give credit for restriction enzymes as a detail of the electrophoresis stage. Modern STR profiling often skips the restriction step, because the PCR primers already cut out the right sections. Either way, PCR comes **before** electrophoresis.

6.19 Uses

Use	How the profile is used
Forensics	compare crime-scene DNA with suspects and the victim. All bands must match to link a person; one mismatch rules them out.
Paternity / maternity	every band in the child must come from one parent or the other. Bands not from the mother must all be in the father's profile.
Relationships between animals	the more bands two individuals (or species) share, the more **closely related** they are. Used in conservation **breeding programmes** to choose the **least related** mates, avoiding inbreeding (which makes harmful recessive alleles show up and reduces genetic diversity).
Plants	identify varieties, check that a GM plant carries the new gene, confirm the parentage of crosses, detect illegal timber or fake products
Origins of populations	e.g. show that animals on an island came from a particular mainland population, by comparing profiles

READING A PATERNITY GEL (FIGURE (B) ABOVE)

The child has four bands. Two match the mother (top band and third band).
The other two must come from the father. Man A has both; man B has neither.
So man A is the likely father. Man B is excluded.

MARK-SCHEME LANGUAGE

- Relationships: "DNA profiling so that **banding patterns** can be compared; the **more similar** the profiles, the **more closely related**" (or compare base sequences). "Common ancestor" on its own is ignored.
- Method: "take DNA samples from {feather / dropping / skin / blood}", "use of **PCR**", "(followed by) gel **electrophoresis**", "the {number / position / pattern} of **bands** shows similarities".
- Sex and identity from profiles: "bands specific to females on two samples but not on the male's"; "all the animals have **different banding patterns**". "Females will have the same bands" isn't accepted.

Often asked as: show two populations are related; identify individuals and their sex from a profile; compare a museum specimen with a living one; check a GM plant; describe how a profile is made in context.

COMMON MISTAKE

Describing the method in huge detail and forgetting the context. Name the sample source and say what the bands would show.
"Blocks" or "lines": write **bands**.
Saying a child has the same bands as its father. It shares **about half** its bands with each parent.

BEYOND THE SPEC, BUT EXAMINED

- Mitochondrial DNA (circular, inherited from the mother) compared with nuclear DNA (linear, with histones) (WBI14 Jan 2022).
- Profiling nematodes from a polluted zone and an unaffected area to look for genetic change (WBI14 Oct 2025).
- The chance of two unrelated people matching falls as more loci are used: if 1 in 10 people share a band pattern at each locus, matching at 10 independent loci is 1 in 10^{10}.

SPEC 6.20

Estimating time of death

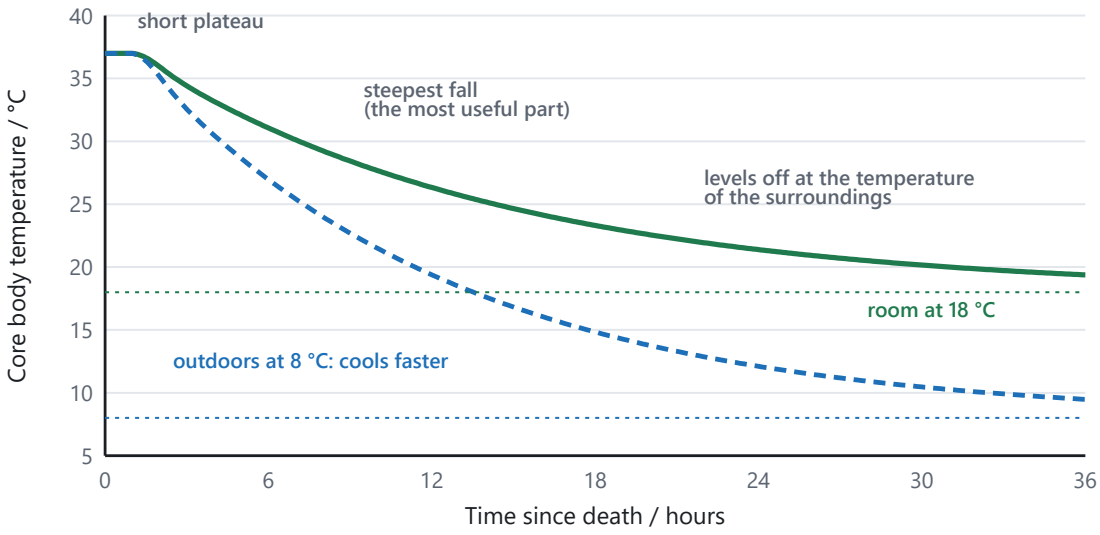
WHAT THIS SPEC POINT ASKS

Understand how time of death of a mammal is estimated from the extent of decomposition, the stage of succession, forensic entomology, body temperature and the degree of muscle contraction (rigor mortis).

Every method gives an **estimate with a range**, and each is affected by the conditions around the body. Forensic scientists use several methods together.

6.20 Body temperature (algor mortis)

- After death, respiration stops, so no more heat is made. The body loses heat to its surroundings until it reaches the surrounding (ambient) temperature.
- Measure **core** temperature with a probe in the **rectum** or **liver** (not the mouth or armpit). Assume it was about **37 °C** at death.
- Cooling is roughly steady (often quoted as about 1–1.5 °C per hour) for the first part, then slows as the body nears ambient temperature.
- Faster cooling (so a *shorter* time since death for the same temperature): cold surroundings, wind, wet clothes or water, a thin body (larger SA : V), no clothes.
- Slower cooling: warm surroundings, clothing, more body fat (an insulator), large body.
- Useful only for about the first **24 hours**: after that the body is at ambient temperature.



Body temperature falls towards that of the surroundings, faster when the surroundings are colder.

WORKED EXAMPLE: TIME OF DEATH FROM COOLING

One rate. Core temperature 30.4 °C; assume 37.0 °C at death; cooling 1.5 °C per hour.
Fall = 37.0 − 30.4 = 6.6 °C. Time = 6.6 ÷ 1.5 = **4.4 hours** before the body was found.

Two rates (use the table in the question). A body cools 0.8 °C per hour for the first 10 hours, then 0.5 °C per hour. It is found at 25.0 °C.
1. First 10 hours: 10 × 0.8 = 8.0 °C, so 37.0 − 8.0 = 29.0 °C after 10 h.
2. Remaining fall: 29.0 − 25.0 = 4.0 °C; 4.0 ÷ 0.5 = 8 hours.
3. Total = 10 + 8 = **18 hours**.

6.20 Muscle contraction (rigor mortis)

- After death, oxygen runs out, so aerobic respiration stops. Cells respire **anaerobically** for a short time: **lactic acid** builds up and the **pH falls**, which inhibits enzymes, so **ATP** can't be made.
- **ATP is needed to detach myosin heads from actin**. Without it the cross-bridges stay locked and the muscles become **stiff**. (Calcium ions leaking from the sarcoplasmic reticulum also let myosin bind.)
- Timing (at room temperature, approximate): starts in small muscles (face, jaw) after about **2–6 hours**; whole body rigid by about **12 hours**; wears off after about **24–36 hours** as enzymes and bacteria break down the muscle proteins.
- Warmth speeds up both onset and loss; cold slows them. Muscle mass and activity just before death also change the timing.

6.20 Extent of decomposition

Stage	What happens	Roughly when (temperate conditions)
Fresh	cells' own enzymes start digesting them (**autolysis**)	first day or two
Bloat	gut bacteria decompose tissue **anaerobically**, producing gases (e.g. methane, hydrogen sulfide) that swell the body; skin turns green then purple-black	from about 2–3 days
Active decay	tissues liquefy; gases escape; large **loss of mass**; strong smell	about 1–3 weeks
Advanced decay / dry remains	soft tissue gone; skin, cartilage, then bones	weeks to months; years for a skeleton

- Faster in **warm**, moist, aerobic conditions; slower in cold, dry, buried or underwater bodies.

6.20 Succession on a body

- A body is a new habitat. Different organisms arrive in a **predictable order**, and each group changes the conditions, making the body suitable for the next (like succession in 5.15).
- Bacteria first; then **blowflies**, which lay eggs within hours (attracted by the smell); then other flies; then beetles that eat fly larvae and, later, the dry tissue; mites and moths on the dry remains.
- The community present indicates how long the body has been there: useful from days to months and longer.

6.20 Forensic entomology

- Identify the **species** of the insects and the **stage** in the life cycle of the oldest ones (egg, larva of a given length or stage, pupa, empty pupal case).

- Each species develops at a known rate **at a given temperature**. Work back from the stage reached to when the eggs were laid. This gives a **minimum** time since death.
- Collect eggs or larvae and **rear them in the lab** at a known temperature to identify the species and check the timing.
- Things that affect it: **temperature** (most important), drugs in the body, humidity, whether flies could reach the body (indoors, buried, wrapped), and that flies lay eggs in daylight. Some species live only in certain habitats, so insects can show that a body was **moved**.
- Most useful after about **3 days**, when temperature and rigor no longer help.

WORKED EXAMPLE: LARVAE

Blowfly eggs hatch about 24 hours after laying (at this temperature). Larvae are 2.0 mm at hatching and grow 0.20 mm per hour. The longest larva found is 9.6 mm.
Growth time = (9.6 − 2.0) ÷ 0.20 = 38 hours. Add 24 hours as eggs: **62 hours** (minimum time since death).

6.20 Which method when?

Time since death	Best methods
up to about 24 h	body temperature; rigor mortis starting
about 3–36 h	rigor mortis (present = between about 3 and 36 h)
days to weeks	forensic entomology; stage of decomposition
weeks to years	succession of organisms; decomposition; skeleton

MARK-SCHEME LANGUAGE

- Measuring: "a thermometer / temperature probe in the **liver / rectum**" (deep in the body). Mouth or armpit is ignored.
- Cooling calculations: a mark for each stage; give the answer to the nearest hour if asked.
- Colder place: the estimate would be **too short** (an underestimate) "because the body **loses heat faster**". Clothing keeps heat in, so the real time is longer.
- Fat: "heat loss would be less / slower as **fat is an insulator**", so the cooling line is shallower.
- Rigor: "**ATP** falls (no aerobic respiration)", "lactic acid rises, pH falls", "myosin heads can't detach from actin"; stiffness alone is limited because "temperature affects rigor", "judging stiffness is subjective", and a stiff body only shows "**between 3 and 36 hours**".
- Body temperature as a method: "readily available", "formulae available", "**but** the change depends on several factors (ambient temperature, clothing, fat, size)", "so limited unless used with other methods".
- Entomology: most accurate after 72 h because "insects colonise a body **in a specific order** / show succession" and their life cycles are known; insects can show the **body was moved** because some species live only in certain **habitats**; collect eggs and rear them "to **identify the species**".
- A cage over a body in a study is "to prevent animals eating / moving it".

Often asked as: where to take the temperature; time-of-death calculations from cooling rates or a cooling graph; why cold surroundings change the estimate; why rigor alone is not enough; forensic entomology and insect succession; evaluate a method (6-mark with chemical markers).

COMMON MISTAKE

"The body cools faster" when the mark is for "**heat is lost** faster".
Not using the cooling rates in the table, or using one rate for the whole time when two are given.
Describing microbial decomposition when the question asks about forensic entomology. Always mention the **species** and the life-cycle **stage**.
Answering from memory of a similar past question instead of reading the graph you are given.

BEYOND THE SPEC, BUT EXAMINED

- Fitting a straight line to cooling data ($y = mx + c$), e.g. a gradient of −1.3 °C per hour (WBI14 Jun 2025).
- Mass of a carcass over time: little change at first; a drop at the end of bloat; a large drop in active decay as gases escape and insects eat the flesh (WBI14 Jan 2022).
- Chemical markers of decomposition (pH, ammonia, hypoxanthine, lactate) in animal models such as pigs; using humans is unethical because the dead can't consent (WBI14 Jun 2023).
- Larval growth rate from data (e.g. 0.12 mm per hour) and the limits of using it (WBI14 Jun 2021).